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Abstract. Silicon photonics is a technology based on fabricating integrated optical circuits by using the same paradigms of the dominating electronics industry. After twenty years of impetuous development, silicon photonics is entering the market with low cost, high performance and mass manufacturable optical devices. Up to now, most of the silicon photonic devices are based on linear optical effects, despite the many phenomenologies associated to nonlinear optics in both bulk materials and integrated waveguides. Silicon and silicon based materials have strong optical nonlinearities which are enhanced in integrated devices by the small cross section of the high index contrast silicon waveguides or photonic crystals. Here the photons are made to strongly interact with the medium where they propagate. This is the central argument of nonlinear silicon photonics. It is the aim of this review article to make the point of the state of the art in the field. Starting from the basic nonlinearities in a silicon waveguide or in optical resonator geometries, many phenomena and applications are described: from frequency generation, frequency conversion, frequency comb generation, supercontinuum generation, soliton formation, temporal imaging and time lensing, Raman lasing, up to comb spectroscopy. Emerging quantum photonics applications, as entangled photon sources, heralded single photon sources and integrated quantum photonic circuits, are also addressed at the end of the review.

1. Introduction to nonlinear silicon photonics

1.1. Silicon photonics

Silicon is the material which provides the backbone to the ICT (Information and Communication Technology) revolution whereas silica optical fibers provide the infrastructure for the information highway. Many billions of transistors, each one few tenths of nanometer in size, are integrated in a small $1{\times}1$ cm^2 silicon chip to realize multifunctional chips which power all the various applications of the present knowledge society. Millions of servers are running in dedicated data centers to provide the intelligence and the data which are continuously accessed by 1 over 4 world inhabitants. Several tens of exabytes per month are flowing among the different terminals (computers, smartphones, laptops, tablets, televisions, ...) since we want to be always connected and ready to tweet on the net. The Zettabyte Era is approaching and in 2020 there will be more than three networked devices per capita world wide [1]. The Internet of Everything will significantly change the way we consider internet and our devices, making the hardware disappearing (i.e. the hardware as we know it today will be ultimately integrated with real world objects and everyday environments) so that the users will be only concentrated on the services (the concept of the cloud). This evolution requires that photonics is brought to the heart of the nodes where the information is generated, elaborated and stored in order to satisfy the need of realtime worldwide networking. The only way to do this is to use the same technology platform for electronics and telecommunications as much as possible. The internet infrastructure, being smart, fast, and cheap, has to be transparent to the data flow and generation with a minimum loss between photonics and electronics. This is made possible by silicon photonics which is photonics in and on silicon. Silicon photonics is the natural extension to optics of the microelectronic paradigm and allows merging both intelligence and communication on the same chip. Silicon photonics is the technology to fabricate integrated photonic circuits by using the same fabrication process which is dominating in electronics, i.e. complementary metal-oxide-semiconductor (CMOS) technology. The motivation is simple: leverage on the know-how, the infrastructure, the economy of electronics to make mass-manufacturing of photonic circuits possible.

After twenty years of impetuous research, we are witnessing the penetration of silicon photonics in several devices both as the interconnect technology as well as the enabling technology. The applications of silicon photonics span from security to biosensing, from data centers to remote networking, from the car industry to medical devices. Still most of the silicon photonics applications are enabled by linear optic concepts. Only now, more elaborated devices are emerging which use the rich phenomenology of nonlinear optics. In fact, silicon is characterized by a very high refractive index contrast with respect to silicon dioxide, which is the most commonly used cladding material (at the wavelength of 1550 nm the refractive index of silicon is about 3.48, the refractive index of silica is about 1.44). This allows manufacturing very compact photonic devices which can be integrated in huge numbers in a single chip (record to date is more than thousands per cm^2 [2]). In addition, the small core section of silicon photonic devices (typically 250×500 nm^2) allows reaching easily the high power density which is needed for the onset of nonlinear optical effects. Other important characteristics of silicon are the high thermal conductivity (149 $Wm^{-1}K^{-1}$) and the mechanical hardness (Young's Modulus of 150 GPa, and a Poisson's ratio of 0.17), which make easy the cooling and the handling of the photonic devices. Nonlinearities in silicon are not as large as in other semiconductor materials such as III-V semiconductors, polymers, organic materials, chalcogenide compounds, etc. [3]. However the CMOS compatibility of silicon photonics is a strong motivation to develop nonlinear devices within this technological platform, in addition to the already developed heavily integrated linear optical silicon photonics circuitries.

It is the aim of this review to present the state-of-the-art of nonlinear silicon photonics with a special emphasis on the physics of the processes. Let us remark here that, as it is common in the silicon photonics community, the emphasis will be on optical devices realized with silicon based materials and not only with pure crystalline silicon. This means that, when needed, experimental data about silicon nitride or silicon oxide will be discussed to describe relevant nonlinear phenomena. After a general introduction to nonlinear optics (section 1.2), we will discuss the equations and phenomenologies of pulse propagation in nonlinear waveguides (section 2). Here all the different first order, second order, third order optical processes occurring in nonlinear silicon photonics will be described. For the sake of conciseness, fifth order nonlinear processes, such as 3-photon absorption which is mostly relevant in the mid infrared, are not discussed. In addition, since silicon is a semiconductor, free-carrier effects such as free-carrier absorption and free-carrier dispersion will be discussed. Section 3 deals with nonlinear optical microresonators. A brief introduction to the microresonator physics is followed by the discussion of dispersion engineering, frequency comb generation and harmonic generation. Section 4 reports on nonlinear optical devices fabricated in silicon photonics. Here we will present temporal imaging and lenses, frequency comb applications, ultra-short pulse generation and wavelength conversion. Finally, section 5 presents the use of nonlinear silicon photonics to quantum optics applications. Starting from deterministic and probabilistic sources, we will report on the generation of entangled photon states, on the quantum description of nonlinear wave mixing and on the steps towards an integrated quantum photonic chip.

1.2. Nonlinear optical processes

In an ideal linear optical material the linear polarization vector $\mathbf{P}_0$ and the applied optical field $\mathbf{E}$ are related by [4, 5]:

$$\mathbf{P}_0 = \varepsilon_0 \chi^{(1)} \cdot \mathbf{E} = \varepsilon_0 \sum_{ij} \chi^{(1)}_{ij} E_i \hat{\mathbf{u}}_j, \tag{1}$$

being ε_0 the vacuum permittivity, $\chi^{(1)}$ the first order susceptibility (which is a second rank tensor whose elements are labeled by $\chi^{(1)}_{ij}$), E_i the $i-$th component of the field $\mathbf{E}$, while $\hat{\mathbf{u}}_j$ is the unitary versor. In a semiconducting material, free carriers and nonlinear optical effects can be important, inducing a modification $\delta\mathbf{P}$ in the polarization vector. Therefore, the polarization vector $\mathbf{P}$ can be written in general as:

$$\mathbf{P} = \mathbf{P}_0 + \delta\mathbf{P} = \mathbf{P}_0 + \delta\mathbf{P}_L + \mathbf{P}_{NL}, \tag{2}$$

where $\delta\mathbf{P}_L$ is due to linear first order processes associated with free carriers, while $\mathbf{P}_{NL}$ is due to nonlinear optical effects and is defined as:

$$\mathbf{P}_{NL} = \mathbf{P}^{(2)} + \mathbf{P}^{(3)} + ... = \varepsilon_0 \left[\chi^{(2)} : \mathbf{E}^2 + \chi^{(3)} \vdots \mathbf{E}^3 + ... \right] =$$

$$= \varepsilon_0 \left[\sum_{ijk} \chi^{(2)}_{ijk} E_i E_j \hat{\mathbf{u}}_k + \sum_{ijkl} \chi^{(3)}_{ijkl} E_i E_j E_k \hat{\mathbf{u}}_l + ... \right] \tag{3}$$

being $\mathbf{P}^{(2)}$ and $\mathbf{P}^{(3)}$ the second and third order nonlinear polarization vectors, while $\chi^{(2)}$ and $\chi^{(3)}$ are the second and the third order susceptibilities. In general, the $j-$th order nonlinear susceptibility is a tensor of rank $(j+1)$: for isotropic or amorphous materials (such as silica-based optical fibers) it can be considered as a scalar quantity, while for crystalline structures (such as silicon) it must be considered in its tensorial form. Higher order susceptibility terms are in smaller and smaller and they are effective only for large field intensities.

1.2.1. First order processes The first order susceptibility coefficient $\chi^{(1)}$ is related to the electronic excitations that can be induced in the material by single photons, and is related to the refractive index n by: $\chi^{(1)}+1=n^2$. Therefore, the real part of $\chi^{(1)}$ is associated to the real part of the refractive index, while its imaginary part is related to losses or gain inside the material [6].

When dealing with semiconductors, such as silicon, the first order susceptibility coefficient must take into account also the effect of free-carriers. Free-carriers introduce a modification in both the real and the imaginary part of the refractive index, giving rise to the Free-Carrier Absorption (FCA) and Free-Carrier Dispersion (FCD) effects. By taking into account these effects, the complex silicon refractive index can be written as [6, 7, 8]:

$$n(\lambda, N_e, N_h) = n_0(\lambda) + \delta n_{FC}(N_e, N_h) + i\frac{\lambda}{4\pi}\left[\alpha_0(\lambda) + \delta\alpha_{FC}(N_e, N_h)\right]. \quad (4)$$

The coefficients $n_0(\lambda)$ and $\alpha_0(\lambda)$ express the wavelength dependence of refractive index and absorption. δn_{FC} and $\delta\alpha_{FC}$ are respectively the free-carrier induced index change and free-carrier absorption coefficient, which both depend on the free electron and hole densities N_e and N_h. Silicon is highly transparent from the band edge (which at room temperature is at 1.1071 μm) up to 5 μm, where lattice absorption bands due to lattice vibration appear [9, 10]. In the transparent region, the absorption coefficient $\alpha_0(\lambda)$ vanishes, while $n_0(\lambda)$ is usually described by the Sellmeier equation [8, 9]:

$$n_0(\lambda) = \varepsilon + \frac{A}{\lambda^2} + \frac{B\lambda_1^2}{\lambda^2 - \lambda_1^2}, \quad (5)$$

where $\varepsilon = 11.6858$, $A = 0.939816$ μm^2, $B = 8.10461\times10^{-3}$ and $\lambda_1 = 1.1071$ μm. The FCA and FCD coefficients at a wavelength of 1550 nm are expressed by [11, 12]:

$$\delta\alpha_{FC} = 8.5\times10^{-18}N_e + 6.0\times10^{-18}N_h, \quad (6)$$

$$\delta n_{FC} = -8.8\times10^{-22}N_e - 8.5\times10^{-18}N_h^{0.8}. \quad (7)$$

In the case of nonlinear optical processes, the free-carriers are mainly introduced by the Two-Photon Absorption (TPA), a third order nonlinear process which we will introduce in section 1.2.3. In this situation, free-carriers are created with an equal density, i.e. $N_e = N_h = N$.

1.2.2. Second order processes The inversion symmetry inhibits the existence of second order nonlinear processes in materials with a centrosymmetric crystalline structure, such as silicon [4]. Nevertheless, recent experiments showed that it is possible to induce second order effects in silicon by breaking its centrosymmetry [13].

The second order effects can be evaluated by representing the propagating electric field $\mathbf{E}$ in the material as the superposition of two waves $\mathbf{E}_1$ and $\mathbf{E}_2$:

$$\mathbf{E}(\mathbf{r},t) = \Re\left[\sum_{n=1}^{2} \mathbf{E}_n(\mathbf{r},\omega_n)e^{-i\omega_n t}\right] = \sum_{n=1}^{2}\left[\mathbf{E}_n(\mathbf{r},\omega_n)e^{-i\omega_n t} + c.c.\right], \tag{8}$$

where $\Re$ means the real part and *c.c.* stands for "complex conjugate". From Eq. (3), the second order nonlinear polarization vector can be derived [4]:

$$\begin{aligned}\mathbf{P}^{(2)}(\mathbf{r},t) = \varepsilon_0\chi^{(2)} : \Big[&\mathbf{E}_1^2(\mathbf{r},\omega_1)e^{-i2\omega_1 t} + \mathbf{E}_2^2(\mathbf{r},\omega_2)e^{-i2\omega_2 t} + \\ &+2\mathbf{E}_1(\mathbf{r},\omega_1)\mathbf{E}_2(\mathbf{r},\omega_2)e^{-i(\omega_1+\omega_2)t} + 2\mathbf{E}_1(\mathbf{r},\omega_1)\mathbf{E}_2^*(\mathbf{r},\omega_2)e^{-i(\omega_1-\omega_2)t} + \\ &+\mathbf{E}_1(\mathbf{r},\omega_1)\mathbf{E}_1^*(\mathbf{r},\omega_1) + \mathbf{E}_2(\mathbf{r},\omega_2)\mathbf{E}_2^*(\mathbf{r},\omega_2)\Big] + c.c. \end{aligned} \tag{9}$$

The first two terms in Eq. (9) correspond to Second Harmonic Generation (SHG), a process where a photon at frequency $2\omega_1$ is generated from two photons at frequency ω_1. A schematic representation of this process is shown on the left side of Fig. 1. The third and the fourth terms in Eq. (9) correspond to Sum Frequency Generation (SFG) and to Difference Frequency Generation (DFG). As it is clear from Fig. 1, in these processes a photon is generated (respectively) at frequencies $\omega_1 + \omega_2$ and $\omega_1 - \omega_2$. The last terms of Eq. (9) give rise to the generation of a DC component of the polarization vector. This process is called Optical Rectification (OR). An interesting process occurs if one of the two frequencies in (9) is much lower than the other, or is set to DC. In this situation, the SFG and the DFG terms consist in a polarization vector oscillating at the high frequency input signal, determining a change in the material refractive index which linearly depends on the low-frequency wave amplitude. This phenomenon is known as the linear electro-optic effect or the Pockels effect.

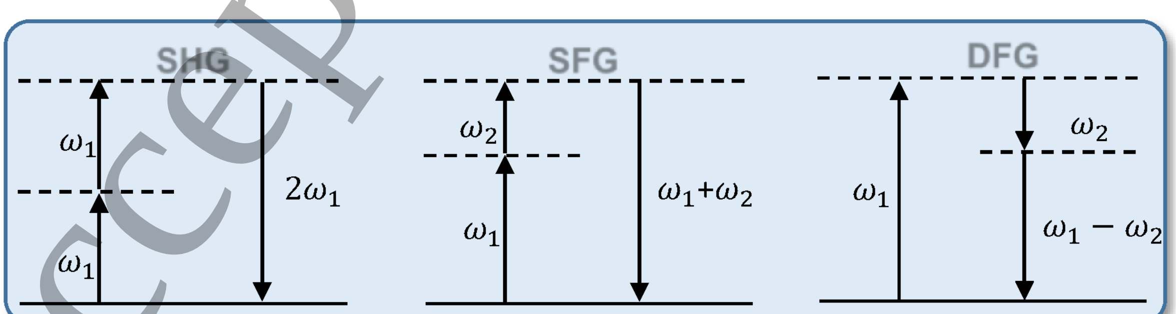

Figure 1. Schematic representation of the main second order nonlinear process: Second Harmonic Generation (on the left), Sum Frequency Generation (in the center) and Difference Frequency Generation (on the right).

1.2.3. Third order processes Following the approach used for the second order case, the third order nonlinear polarization vector $\mathbf{P}^{(3)}$ can be written by expanding the total electric field as the sum of three waves [6]:

$$\mathbf{P}^{(3)}(\mathbf{r},t) = \varepsilon_0\chi^{(3)}\vdots\Big[\mathbf{E}_1^3(\mathbf{r},\omega_1)e^{-i3\omega_1 t} + 3\mathbf{E}_1^2(\mathbf{r},\omega_1)\mathbf{E}_2(\mathbf{r},\omega_2)e^{-i(2\omega_1+\omega_2)t} + 3\mathbf{E}_1^2(\mathbf{r},\omega_1)\mathbf{E}_2^*(\mathbf{r},\omega_2)e^{-i(2\omega_1-\omega_2)t} + 6\mathbf{E}_1(\mathbf{r},\omega_1)\mathbf{E}_2(\mathbf{r},\omega_2)\mathbf{E}_3(\mathbf{r},\omega_3)e^{-i(\omega_1+\omega_2+\omega_3)t} + 6\mathbf{E}_1(\mathbf{r},\omega_1)\mathbf{E}_2(\mathbf{r},\omega_2)\mathbf{E}_3^*(\mathbf{r},\omega_3)e^{-i(\omega_1+\omega_2-\omega_3)t} + 3\left|\mathbf{E}_1(\mathbf{r},\omega_1)\right|^2\mathbf{E}_1(\mathbf{r},\omega_1)e^{-i\omega_1 t} + 6\left|\mathbf{E}_2(\mathbf{r},\omega_2)\right|^2\mathbf{E}_1(\mathbf{r},\omega_1)e^{-i\omega_1 t}\Big] + c.c. \tag{10}$$

In this formula, for reasons of simplicity, all the possible permutations of the photon indices (i.e. of the waves) have been omitted. The first term in the Eq. (10) is related to Third Harmonic Generation (THG). In this process, as it is represented in Fig. 2, three photons with identical frequencies ω_1 generate a photon at triple frequency $3\omega_1$. The second to fifth elements of the summation are responsible of Four Wave Mixing (FWM), where the annihilation of two incident photons, the pump photons, generate two other frequencies, the idler and the signal photons. A possible scheme for this process is shown in Fig. 2. The sixth term of the summation is responsible of both Self Phase

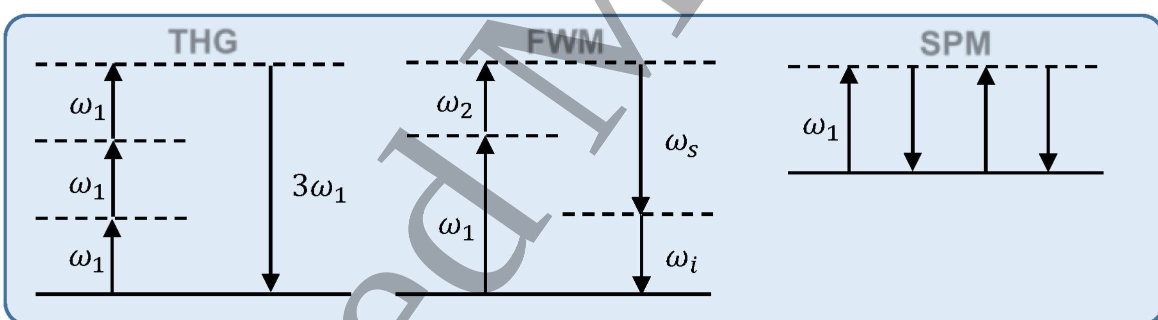

Figure 2. Schematic representation of some third order non linear processes: Third Harmonic Generation, Four Wave Mixing and Self Phase Modulation.

Modulation (SPM) and Two Photon Absorption (TPA). SPM is related to the real part of the third order susceptibility $\chi^{(3)}$, while TPA is related to the imaginary part of $\chi^{(3)}$. When this term is inserted into the nonlinear wave equation, it gives intensity-dependent perturbations of the refractive index and of the absorption coefficient in the form [6]:

$$n = n_0 + n_2 I + i\frac{\lambda}{4\pi}\left[\alpha_0 + \beta_{TPA} I\right], \tag{11}$$

where I is the field intensity, n_2 the Kerr coefficient and β_{TPA} the TPA absorption coefficient. The coefficients n_2 and β_{TPA} are defined as:

$$n_2 = \frac{3}{4\varepsilon_0 c n^2}\Re\left(\chi^{(3)}_{eff}\right) \quad \beta_{TPA} = \frac{-3\omega}{2\varepsilon_0 c^2 n^2}\Im\left(\chi^{(3)}_{eff}\right), \tag{12}$$

where $\Im$ means the imaginary part and $\chi^{(3)}_{eff}$ is an effective third order nonlinear coefficient which takes into account the independent terms of the 4−th rank tensor $\chi^{(3)}$

[8]. The physical origin of the TPA process is related to the absorption of two photons whose energies sum up to the energy required for the excitation of an electron from the valence to the conduction band. In this process, the initial and the final quantum mechanical states are not identical, since an electron is removed from the valence band and is promoted to the conduction band. For this reason the TPA effect is a first example of a non-parametric process, which distinguishes from parametric processes where the initial and the final quantum mechanical states are the same, and where the population can be moved from the ground state to a virtual level only for short time intervals [4]. The nonlinear index variation and the nonlinear absorption are usually compared by means of the figure of merit FOM, which is defined as the number of 2π nonlinear phase accumulations felt by the pulse propagating in the material for a length equal to the nonlinear absorption length. In the case of third order nonlinearities, one can write $FOM = n_2/\lambda\beta_{TPA}$ [6]. Since both n_2 and β_{TPA} vary with wavelength, the FOM has a strong wavelength dependence. As it is shown in Fig. 3, for silicon n_2 increases with wavelength up to $1.8-1.9$ μm, while β_{TPA} strongly decreases after 2 μm, approaching a frequency halved with respect to the silicon band gap. Therefore, for silicon the mid infrared shows an interestingly large FOM, which is in turn limited by higher order nonlinear optical processes, such as the three photon absorption.

The last term in the summation in Eq. (10) is responsible for the cross Phase Modulation (XPM). XPM is still related to an intensity dependent refractive index, but here it is a signal at frequency ω_2 that influences the propagation of a signal at frequency ω_1. The coefficients appearing in Eq. (10) show that XPM causes a modification of the refractive index that is doubled with respect to the modification induced by SPM.

Up to now, only electronic contributions to the third order nonlinear susceptibility have been considered. In general, one should also consider the contribution of phonons to $\chi^{(3)}$. In this case, one can write $\chi^{(3)} = \chi_e^{(3)} + \chi_R^{(3)}$, being $\chi_e^{(3)}$ the electronic contribution to the third order nonlinearity which has been considered so far and $\chi_R^{(3)}$ the Raman contribution which involves optical phonons [7]. In this framework additional third order processes are present, related for example to the enhancement of the Raman scattering [8, 14, 15]. In the Stimulated Raman Scattering (SRS) Stokes photons are launched together with the pump signal, increasing the scattering rate and enhancing the generation of additional Stokes photons. Another effect related to Raman scattering is the Coherent Anti-Stokes Raman Scattering (CARS), where the interaction of two pump photons at frequency ω_p with a signal photon at frequency ω_s produces an output photon at frequency $\omega_{CARS} = 2\omega_p - \omega_s$. Similarly to SRS, also the stimulated Brillouin scattering (SBS) can occur. In this situation the scattering process involves acoustic phonons, and it is enhanced by the presence of a stimulating signal [14].

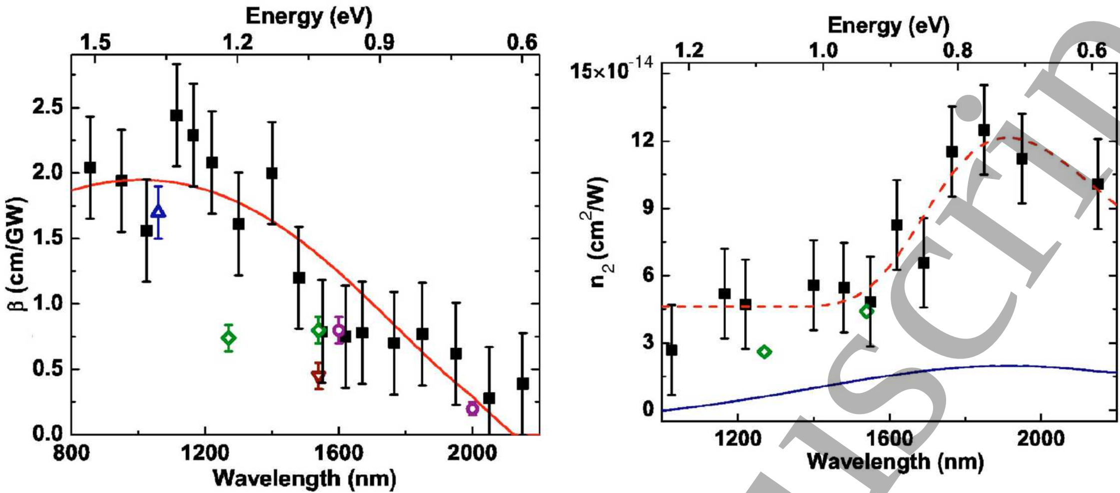

Figure 3. Measured data of TPA and Kerr coefficients for silicon. This figure is reprinted with permission from Ref. [16].

2. Optical Pulses in Nonlinear Waveguides

2.1. Theory of pulse propagation in waveguides

2.1.1. Basic concepts A channel waveguide is formed by a high refractive index medium, which is called core and has typically a rectangular shape. The core is surrounded by a lower refractive medium, called cladding. A sketch of a typical rectangular waveguide is given on the left side of Fig. 4. Let us divide the modes propagating in a waveguide in Transverse Electric (TE) modes, when the electric field is in the horizontal plane, and Transverse Magnetic (TM) modes, when the electric field is in the vertical direction. Each mode is characterized by a field profile $\mathbf{E}(\mathbf{r}, \omega) \sim \mathbf{e}(\mathbf{r}_\perp, \omega)e^{i\beta z}$, where z is the propagation direction, β the propagation constant, ω the frequency, $\mathbf{r}_\perp$ a vector in the plane orthogonal to z and $\mathbf{e}(\mathbf{r}_\perp, \omega)$ the modal profile. In the right side of Fig. 4, the planar profiles of some modes supported by a rectangular waveguide are shown. It is possible to define the effective refractive index n_{eff} of each mode as:

$$n_{eff} = \frac{c}{v} = \frac{c\beta}{\omega}, \tag{13}$$

being c the speed of light in vacuum and v the phase velocity of the propagating wave. The propagation constant β depends on the frequency, and it can be expanded as a Taylor series around a central frequency ω_0:

$$\beta(\omega) = \sum_m \frac{\beta_m}{m!}(\omega - \omega_0)^m. \tag{14}$$

The first term of the expansion, β_1, is related to the group velocity of the mode v_g and to the group index of the mode n_g by the relation $v_g = 1/\beta_1 = c/n_g$. The second term of the expansion β_2 is called the Group Velocity Dispersion (GVD), while β_3 is the Third Order Dispersion (TOD), and so on. When $\beta_2 > 0$ the dispersion is said to be normal, while it is anomalous if $\beta_2 < 0$.

Especially for small structures, the values assumed by β_m are dominated by the waveguide geometry, and can be tailored by suitable choice of the waveguide dimensions.

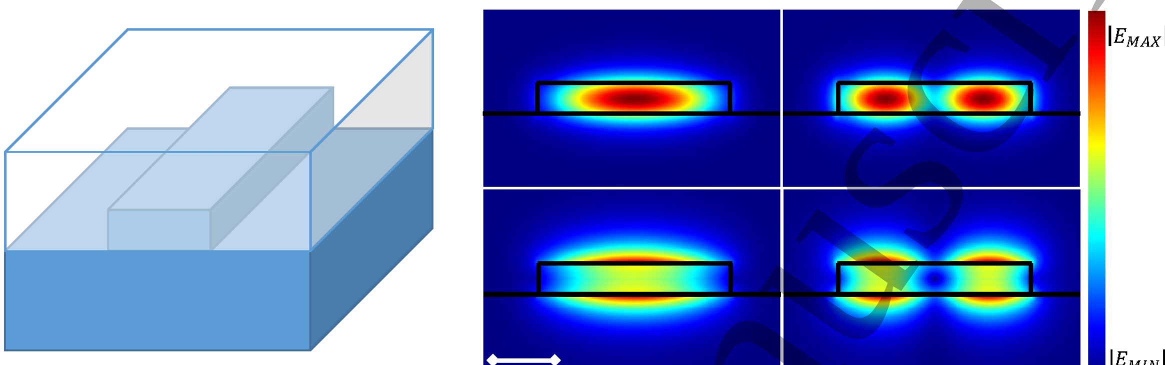

Figure 4. On the left: sketch of a rectangular waveguide fabricated on a Silicon On Insulator (SOI) wafer. On the right: modal profiles for the two lowest order TE and TM modes supported by a (1.5×0.25) µm^2 silicon waveguide with a silica cladding. The profiles have been evaluated using a finite element simulation software for a wavelength of 1.55 μm.

2.1.2. General remarks on the pulse propagation As it is shown in Eq. (2), free-carriers and nonlinear optical processes induce a perturbation $\delta\mathbf{P}$ of the polarization vector. To describe the propagation of optical pulses in a waveguide under the effect of this perturbation, let us consider $(\mathbf{E_0}, \mathbf{H_0})$ as the electromagnetic fields propagating in the unperturbed waveguide ($\delta\mathbf{P} = 0$). Due to the perturbation, the electromagnetic fields are modified to $(\mathbf{E_1}, \mathbf{H_1})$. From Lorentz's reciprocity theorem, it follows that [8]:

$$\frac{\partial}{\partial z}\int_{A_\infty}(\mathbf{E_0^*}\times\mathbf{H_1}+\mathbf{E_1}\times\mathbf{H_0^*})\cdot\hat{\mathbf{z}}\ \ dA = i\omega\int_{A_\infty}\mathbf{E_0^*}\cdot\delta\mathbf{P}\ \ dA. \tag{15}$$

In both terms, the integration is carried over all the plane A_∞ transverse to the propagation direction $\hat{\mathbf{z}}$. If the electromagnetic field is linearly polarized, the perturbation can be introduced as a modulation $u(z,\omega)$. The unperturbed and the perturbed fields can be then rewritten as:

$$\mathbf{E_0}(\mathbf{r},\omega_0) = \frac{1}{2}\mathbf{e}(\mathbf{r}_\perp,\omega_0)e^{i\beta_0 z} \qquad \mathbf{E_1}(\mathbf{r},\omega) = \frac{1}{2}\mathbf{e}(\mathbf{r}_\perp,\omega)u(z,\omega)e^{i\beta z}, \tag{16}$$

$$\mathbf{H_0}(\mathbf{r},\omega_0) = \frac{1}{2}\mathbf{h}(\mathbf{r}_\perp,\omega_0)e^{i\beta_0 z} \qquad \mathbf{H_1}(\mathbf{r},\omega) = \frac{1}{2}\mathbf{h}(\mathbf{r}_\perp,\omega)u(z,\omega)e^{i\beta z}. \tag{17}$$

Here ω_0 is the carrier frequency and $\beta_0 = \beta(\omega_0)$. The quantities $\mathbf{e}(\mathbf{r}_\perp,\omega)$ and $\mathbf{h}(\mathbf{r}_\perp,\omega)$ express the field profiles in the plane orthogonal to the main waveguide axis. They are not modified by the perturbation, so that $\mathbf{e}(\mathbf{r}_\perp,\omega)\sim\mathbf{e}(\mathbf{r}_\perp,\omega_0)$ and $\mathbf{h}(\mathbf{r}_\perp,\omega)\sim\mathbf{h}(\mathbf{r}_\perp,\omega_0)$.

In this framework, the total optical power P_0 carried by the unperturbed electromagnetic field is given by:

$$P_0 = \frac{1}{4}\int_{A_\infty} [\mathbf{e}(\mathbf{r}_\perp,\omega)\times\mathbf{h}^*(\mathbf{r}_\perp,\omega)+\mathbf{e}^*(\mathbf{r}_\perp,\omega)\times\mathbf{h}]\cdot\hat{\mathbf{z}}\ \ dA, \tag{18}$$

while in the case of the perturbed field the total optical power P it is given by $P = P_0\,|u(z,\omega)|^2$. Inserting the definitions given in Eq. (16), (17) and (18) into Eq. (15), one can derive the following equation:

$$\frac{du(z,\omega)}{dz} + i(\beta-\beta_0)u(z,\omega) = \frac{i\omega}{2P_0}e^{-i\beta z}\int_{A_\infty}\mathbf{e}^*(\mathbf{r}_\perp,\omega)\cdot\delta\mathbf{P}(\mathbf{r},\omega)\ \ dA. \tag{19}$$

Optical pulses propagating in the same waveguide are characterized by their frequency spectrum. If their spectra do not overlap, a separate equation of the form of Eq. (19) can be written for each pulse. Otherwise, a single equation approach must be used.

2.2. Pulse propagation in cubic media

In this section, the theory describing the propagation of optical pulses in waveguides under the effect of third order nonlinearities will be treated. First of all, the propagation of single pulses will be analyzed, passing then to the description of the propagation of multiple pulses and to the analysis of the generation processes, which will be treated considering the example of the FWM.

2.2.1. Propagation of single pulses In order to simplify the treatment for the propagation of a single optical pulse in a waveguide under the effect of nonlinear phenomena, the different nonlinear contributions are considered separately. In this way, the physical origin of each term will appear more clear.

Third order effects The modification of the polarization vector describing SPM and TPA is expressed in Eq. (10) as:

$$\delta\mathbf{P}_{\chi^{(3)}}(\mathbf{r},\omega) = 3\varepsilon_0\chi^{(3)}\vdots\,|\mathbf{E}(\mathbf{r},\omega)|^2\,\mathbf{E}(\mathbf{r},\omega). \tag{20}$$

Inserting this expression into Eq. (19) and by taking the Fourier transform, it is possible to derive the following partial differential equation describing the spatial and the temporal evolution of the mode amplitude $u(z,t)$ under the effect of the SPM and TPA [8]:

$$\frac{du}{dz} + \sum_{m\geq 1}\frac{(i)^{m-1}\beta_m}{m!}\frac{\partial^m u}{\partial t^m} = i\gamma^{(3)}P_0\,|u|^2\,u. \tag{21}$$

This is referred as the nonlinear Schrödinger equation. In the derivation of this equation, the propagation constant $\beta(\omega)$ has been expanded as in Eq. (14). Moreover, deriving Eq. (21) the unperturbed pulse power P_0 has ben written as $P_0 = v_g W$, where W is the energy density for unitn length that is given by [17]:

$$W = \frac{\varepsilon_0}{2}\left(\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega)|^2\ \ dA\right)^2, \tag{22}$$

where the quantity $n(\mathbf{r}_\perp)$ denotes the local refractive index distribution. The summation on the left-hand side of Eq. (21) is usually truncated depending on the strength of the coefficients β_m. The first term describes the propagation of the pulse at a group velocity $v_g = 1/\beta_1$ without any temporal or spectral distortion. The modification of the pulse shape due to dispersion effects is described by the successive terms of the summation. The complex factor $\gamma^{(3)}$ in Eq. (21) is the nonlinear waveguide parameter, defined as [8]:

$$\gamma^{(3)} = \frac{3\omega n_g^2}{4\varepsilon_0 A_0 c^2}\Gamma^{(3)}, \tag{23}$$

where A_0 is the waveguide cross section, while the parameter $\Gamma^{(3)}$ is given by:

$$\Gamma^{(3)} = \frac{A_0 \int_{A_\infty} \mathbf{e}^*(\mathbf{r}_\perp,\omega)\chi^{(3)}\vdots\mathbf{e}(\mathbf{r}_\perp,\omega)\mathbf{e}^*(\mathbf{r}_\perp,\omega)\mathbf{e}(\mathbf{r}_\perp,\omega)\ dA}{\left(\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega)|^2\ dA\right)^2}. \tag{24}$$

If the $\chi^{(3)}$ tensor inside the waveguide is much larger than outside, the integral at the numerator of Eq. (24) can be taken over the waveguide transverse section A_0 despite of the whole transverse plane A_∞. From Eq. (23) it follows that the parameter $\gamma^{(3)}$ strongly depends on the wavelength and on the waveguide geometry. This dependence is shown in Fig. 5.

The strengths of the various effects described by Eq. (21) are usually compared by introducing several characteristic lengths. From one side, the SPM strength is evaluated by means of the nonlinear length $L_{NL} = \left[P_0\Re(\gamma^{(3)})\right]^{-1}$. On the other side, the strength of the terms in the summation on the left-hand side of Eq. (21) are compared by means of the $n-$th order dispersion length $L_D^{(n)} = T_0^n/|\beta_n|$, being T_0 the pulse temporal width. The first order dispersion length $L_D^{(1)} = T_0/|\beta_1| = v_g T_0$ simply defines the spatial extension of the pulse in the waveguide. The major effects on the pulse shape are described by the second and the third order dispersion lengths. For example, if $L_{NL} \gg L_D^{(2)}, L_D^{(3)}$, the evolution of the pulse inside the waveguide is dominated by SPM. Otherwise, if the parameters L_{NL}, $L_D^{(2)}$ and $L_D^{(3)}$ are comparable, some interesting effects appear, such as the creation of temporal solitons. To this purpose, it is useful to introduce the soliton number $N_s = (L_D^{(2)}/L_{NL})^{1/2}$ as a measure of the strength of solitonic effects [18].

Neglecting the effect of TPA, i.e. assuming a purely real $\chi^{(3)}$, a simple solution to Eq. (21) can be obtained by considering the propagation of a continuous wave or a pulse with $L_{NL} \gg L_D^{(2)}, L_D^{(3)}$. In this situation $u(z) = u_0 e^{i\gamma^{(3)}P_0|u_0|^2 z}$, where $u_0 = u(z=0)$. From this solution it is clear that the effect of the nonlinear perturbation is to induce a modulation of the phase of the propagating signal which is proportional to the optical power $P_0|u_0|^2$ carried by the signal itself. This effect corresponds to SPM.

Free-carrier effects The free-carrier induced modification δn of the silicon refractive index shown in Eq. (11) perturbs the first order susceptibility as:

$$\delta\chi^{(1)} = 2n\delta n = 2n\left[\delta n_{FC} + \frac{ic}{2\omega}\delta\alpha_{FC}\right]. \tag{25}$$

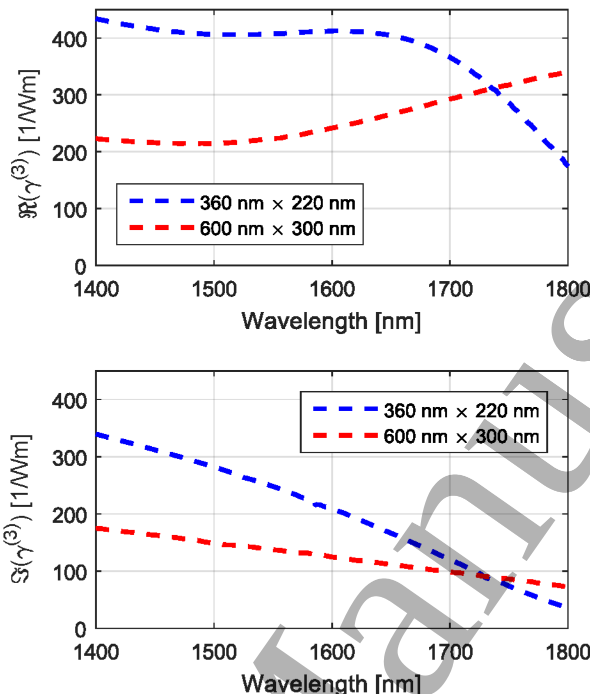

Figure 5. Real and imaginary parts of the nonlinear waveguide parameter $\gamma^{(3)}$ against wavelength for different silicon waveguide geometries. Data taken from Ref. [8].

Therefore also the polarization changes as $\delta\mathbf{P} = \varepsilon_0 \delta\chi^{(1)}\mathbf{E}$. The nonlinear Schrödinger equation results then in [8]:

$$\frac{du}{dz} + \sum_{m\geq 1} \frac{(i)^{m-1}\beta_m}{m!}\frac{\partial^m u}{\partial t^m} = -\frac{\sigma}{2}\left[1 + i\mu\right] Nu - \frac{\alpha}{2}u, \tag{26}$$

where:

$$\sigma = \frac{1}{N}\frac{c\kappa}{nv_g}\alpha_{FC} \quad \mu = -\frac{1}{\sigma N}\frac{2\omega\kappa}{nv_g}\delta n_{FC}. \tag{27}$$

and:

$$\kappa = \frac{n^2 \int_{A_0} |e(\mathbf{r}_\perp, \omega)|^2 \; dA}{\int_{A_\infty} n(\mathbf{r}_\perp)^2 |e(\mathbf{r}_\perp, \omega)|^2 \; dA}. \tag{28}$$

σ and μ describe the Free-Carrier Absorption (FCA) and the Free-Carrier Dispersion (FCD) effects, while α is associated to the waveguide linear losses due to the scattering effects induced by the surface roughness [19]. Eq. (26) must be coupled to an equation describing the free carrier density (N) evolution. If the optical wave frequency is smaller than the silicon band gap frequency, the free carriers are mainly generated by TPA, and in this situation it is possible to write [8]:

$$\frac{\partial N}{\partial t} = -\frac{N}{\tau_c} + \frac{3\beta_1^2 P_0^2 \Im(\Gamma^{(3)})}{4\varepsilon_0 \hbar A_0^2}|u|^4. \tag{29}$$

τ_c is the characteristic free-carrier lifetime, which is assumed here to be the same for electrons and holes. Due to the fast diffusion of free carriers to the edges of the waveguide, it depends on the waveguide geometry and, in the case of silicon small core waveguides, τ_c is of the order of few nanoseconds [8].

General equation for the propagation of single pulses The propagation of a single optical pulse in a silicon waveguide taking into account the effect of propagation losses, free-carriers, TPA and SPM is described by the nonlinear Schrödinger equation:

$$\frac{du}{dz} + \sum_{n\geq 1} \frac{(i)^{n-1}\beta_n}{n!}\frac{\partial^{(n)}u}{\partial t^n} = -\frac{\sigma}{2}\left[1 + i\mu(N)\right] Nu - \frac{\alpha}{2}u + i\gamma^{(3)} P_0 \left|u\right|^2 u. \quad (30)$$

This equation is coupled to Eq. (29), which describes the evolution of the free carrier density N.

2.2.2. Propagation of multiple pulses When multiple pulses of different frequencies are co-propagating in the same waveguide, one has to write a nonlinear Schrödinger equation for each of them. In this situation, one has to consider also the presence of XPM. For reasons of simplicity, in the following equations we will neglect the presence of generation processes, such as FWM. Assuming to have two optical pulses u_a and u_b co-propagating in the waveguide, they will be described by the following equations [8]:

$$\frac{du_a}{dz} + \sum_{m\geq 1} \frac{(i)^{m-1}\beta_{m,a}}{m!}\frac{\partial^m u_a}{\partial t^m} = -\frac{\sigma_a}{2}\left[1 + i\mu_a\right] Nu_a - \frac{\alpha_a}{2}u_a + i\left(\gamma_a^{(3)} P_a \left|u_a\right|^2 + 2\gamma_{ab}^{(3)} P_b \left|u_b\right|^2\right) u_a \quad (31)$$

$$\frac{du_b}{dz} + \sum_{m\geq 1} \frac{(i)^{m-1}\beta_{m,b}}{m!}\frac{\partial^m u_b}{\partial t^m} = -\frac{\sigma_b}{2}\left[1 + i\mu_b\right] Nu_b - \frac{\alpha_b}{2}u_b + i\left(\gamma_b^{(3)} P_b \left|u_b\right|^2 + 2\gamma_{ba}^{(3)} P_a \left|u_a\right|^2\right) u_b \quad (32)$$

$$\frac{\partial N}{\partial t} = -\frac{N}{\tau_c} + \frac{3}{4\varepsilon_0\hbar A_0^2}\left[\frac{P_a^2\Im\left(\Gamma_a^{(3)}\right)}{v_{g,a}^2}|u_a|^4 + \frac{P_b^2\Im\left(\Gamma_b^{(3)}\right)}{v_{g,b}^2}|u_b|^4 + \right.$$
$$\left. + \frac{4\left[\omega_a\Im\left(\Gamma_{ba}^{(3)}\right) + \omega_b\Im\left(\Gamma_{ab}^{(3)}\right)\right] P_a P_b}{(\omega_a + \omega_b)v_{g,a}v_{g,b}}|u_a u_b|^2\right] \quad (33)$$

$\gamma_i^{(3)}$ $(i = a, b)$ is the nonlinear waveguide parameter defined in Eq. (23), while $\gamma_{ij}^{(3)}$ and $\Gamma_{ij}^{(3)}$ are defined as follows:

$$\gamma_{ij}^{(3)} = \frac{3\omega_i n_{g,i} n_{g,j}}{4\varepsilon_0 A_0 c^2}\Gamma_{ij}^{(3)}, \quad (34)$$

$$\Gamma_{ij}^{(3)} = A_0 \frac{\int_{A_0} \mathbf{e}^*(\mathbf{r}_\perp,\omega_j)\chi^{(3)}\vdots\mathbf{e}(\mathbf{r}_\perp,\omega_i)\mathbf{e}^*(\mathbf{r}_\perp,\omega_i)\mathbf{e}(\mathbf{r}_\perp,\omega_j)\ \ dA}{\left(\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega_i)|^2\ \ dA\right)\left(\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega_j)|^2\ \ dA\right)}. \quad (35)$$

2.2.3. Generation processes in waveguides In this section, the generation processes will be introduced into the formalism describing the propagation of optical pulses in a waveguide. The case of FWM process will be considered as an example. The same theory can be generalized also for other generation processes, such for example SHG or THG. The concept of phase-matching, which will be formally derived here for the FWM, is a general notion that is still valid also for all the other generation processes. For reasons of simplicity, in this derivation the TPA and the free-carrier effects are neglected. Also, the time dependence of $u(z,t)$ will be neglected, assuming continuous signals. The formalism can be generalized to account for these terms as in the previous sections.

Theoretical description of the generation process: the case of FWM In the FWM process, two pump waves at frequencies ω_1 and ω_2 and a signal wave at frequency ω_s propagate in the nonlinear waveguide, generating an idler wave at a frequency ω_i. These waves must satisfy the energy conservation relationship $\omega_1 + \omega_2 = \omega_i + \omega_s$. The equation describing the propagation of each of the four pulses can be derived from Eq. (19). For each pulse the polarization vector $\delta\mathbf{P}$ must be written considering the SPM, the XPM and the FWM terms. As an example, for the first pump wave $\delta\mathbf{P}$ is given by:

$$\delta\mathbf{P} = \varepsilon_0 \chi^{(3)} \vdots \left\{ 6\mathbf{E}_i\mathbf{E}_s\mathbf{E}_2^* + 3\left[|\mathbf{E}_1|^2 + 2\left(|\mathbf{E}_2|^2 + |\mathbf{E}_s|^2 + |\mathbf{E}_i|^2 \right) \right] \mathbf{E}_1 \right\}. \quad (36)$$

For reasons of compactness, in this expression we omitted the dependency of each field on position and frequency. The equations for the propagation of the four pulses are then given by:

$$\frac{du_1}{dz} = -\frac{\alpha_1}{2}u_1 + i\left[\gamma_1^{(3)}P_1|u_1|^2 + 2\left(\gamma_{12}^{(3)}P_2|u_2|^2 + \gamma_{1s}^{(3)}P_s|u_s|^2 + \gamma_{1i}^{(3)}P_i|u_i|^2 \right) \right] u_1$$
$$+2i\gamma_{1is2}^{(3)}\sqrt{\frac{P_2P_iP_s}{P_1}}u_iu_su_2^*e^{i\Delta\beta z}, \quad (37)$$

$$\frac{du_2}{dz} = -\frac{\alpha_2}{2}u_2 + i\left[\gamma_2^{(3)}P_2|u_2|^2 + 2\left(\gamma_{21}^{(3)}P_1|u_1|^2 + \gamma_{2s}^{(3)}P_s|u_s|^2 + \gamma_{2i}^{(3)}P_i|u_i|^2 \right) \right] u_2$$
$$+2i\gamma_{2is1}^{(3)}\sqrt{\frac{P_1P_iP_s}{P_2}}u_iu_su_1^*e^{i\Delta\beta z}, \quad (38)$$

$$\frac{du_i}{dz} = -\frac{\alpha_i}{2}u_i + i\left[\gamma_i^{(3)}P_i|u_i|^2 + 2\left(\gamma_{i1}^{(3)}P_1|u_1|^2 + \gamma_{i2}^{(3)}P_2|u_2|^2 + \gamma_{is}^{(3)}P_s|u_s|^2 \right) \right] u_i$$
$$+2i\gamma_{i12s}^{(3)}\sqrt{\frac{P_1P_2P_s}{P_i}}u_1u_2u_s^*e^{-i\Delta\beta z}, \quad (39)$$

$$\frac{du_s}{dz} = -\frac{\alpha_s}{2}u_s + i\left[\gamma_s^{(3)}P_s|u_s|^2 + 2\left(\gamma_{s1}^{(3)}P_1|u_1|^2 + \gamma_{s2}^{(3)}P_2|u_2|^2 + \gamma_{si}^{(3)}P_i|u_i|^2 \right) \right] u_s$$
$$+2i\gamma_{s12i}^{(3)}\sqrt{\frac{P_1P_2P_i}{P_s}}u_1u_2u_i^*e^{-i\Delta\beta z}. \quad (40)$$

The parameter $\gamma^{(3)}_{ijkl}$ is defined as follows:

$$\gamma^{(3)}_{ijkl} = \frac{3\omega_i\sqrt{n_{g,i}n_{g,j}n_{g,k}n_{g,l}}}{4\varepsilon_0 A_0 c^2}\Gamma^{(3)}_{ijkl}, \tag{41}$$

with

$$\Gamma^{(3)}_{ijkl} = A_0\frac{\int_{A_0}\mathbf{e}^*(\mathbf{r}_\perp,\omega_i)\chi^{(3)}\vdots\mathbf{e}(\mathbf{r}_\perp,\omega_j)\mathbf{e}^*(\mathbf{r}_\perp,\omega_k)\mathbf{e}(\mathbf{r}_\perp,\omega_l)\ \ dA}{\prod_{n=ijkl}\left(\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega_n)|^2\ \ dA\right)^{1/2}}. \tag{42}$$

On the right-hand side of the Eq. (37) the first term is due to losses, the second to SPM and XPM while the last one is due to FWM. In this last term, the phase mismatch parameter $\Delta\beta = \beta_i + \beta_s - \beta_1 - \beta_2$ has been introduced. The strength of the FWM terms depends on the coefficients $\gamma^{(3)}_{ijkl}$, which are related to the confinement of the optical modes. Therefore, FWM is stronger in waveguides where the electromagnetic field is more confined. If the pulses propagate on different modal orders of the waveguide some of the coefficient $\gamma^{(3)}_{ijkl}$ can vanish for symmetry reasons. This fact strongly affects the efficiency of FWM processes in multimodal waveguides.
A particular case of interest is when the pump photons have the same frequency $\omega_1 = \omega_2 = \omega_p$. In this situation, if one expresses the signal frequency as $\omega_s = \omega_p + \Delta\omega$, the idler frequency is $\omega_i = \omega_p - \Delta\omega$, and the signal and the idler are symmetrically separated with respect to the pump.

Efficiency of the FWM process A simplified expression for the efficiency of FWM can be derived for the idler in the so called *undepleted pump approximation* [20], which considers the pump power $P_p|u_p|^2$ much stronger than the signal power, which in turn can be considered stronger than the idler power. In this scenario, the FWM generation efficiency is given by:

$$|u_i(L)|^2 = \frac{4|\gamma^{(3)}_{s12i}|^2}{P_i}P_p^2|u_p(0)|^4 P_s|u_s(0)|^2 L^2 sinc^2\left(\frac{\Delta\beta' L}{2\pi}\right), \tag{43}$$

where $\Delta\beta'$ is the general phase mismatch parameter that takes into account the additional phase mismatch introduced by the SPM and the XPM processes, and is defined as :

$$\Delta\beta' = \Delta\beta + \gamma^{(3)}_s P_s|u_s(0)|^2 + (2\gamma^{(3)}_{s1} - \gamma^{(3)}_1)P_1|u_1(0)|^2 + (2\gamma^{(3)}_{s2} - \gamma^{(3)}_2)P_2|u_2(0)|^2. \tag{44}$$

Equation (43) shows that the efficiency depends quadratically on the pump intensity and linearly on the signal intensity. In Fig. 6 the dependence of the generation efficiency on the phase mismatch parameter $\Delta\beta'$ is reported for different values of L. It is clear that, independently on L, the efficiency is maximum for $\Delta\beta' = 0$. This condition is called perfect phase matching, and can be interpreted in the quantum mechanical description as the momentum conservation between the two pump photons that are destroyed and the two generated signal and idler photons. As $\Delta\beta'$ increases, the efficiency reduces and vanishes at $L_c = 2\pi/\Delta\beta'$, which is called the coherence length of the FWM process. Its physical interpretation is the length over which the generation process is efficient.

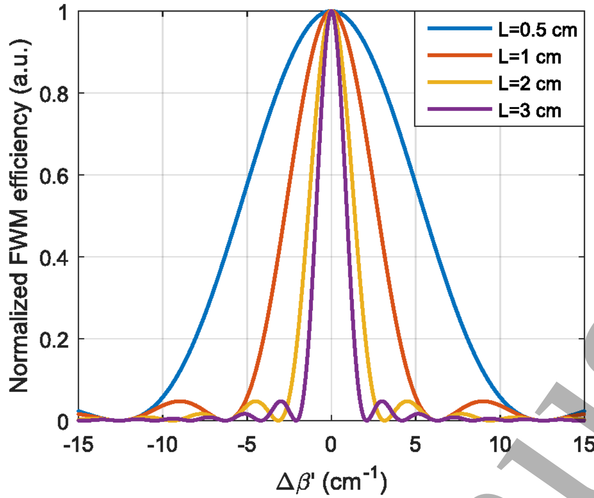

Figure 6. FWM efficiency dependence on the phase mismatch parameter $\Delta\beta'$ for various values of the waveguide length L.

The phase-matching issue In the previous paragraph the phase-matching condition has been formally introduced. In order to achieve an efficient FWM generation, the phase mismatch $\Delta\beta$ must be equal to zero. This concept is quite general and all the generation processes require phase matching. Here we introduce few methods to achieve perfect phase-matching. For reasons of simplicity, these concepts will be applied to FWM of degenerate pump photons, but they can be easily generalized to other processes.

Neglecting the terms due to SPM and the XPM processes, the phase mismatch parameter can be written as:

$$\Delta\beta = 2\omega_p n_{eff}(\omega_p) - [\omega_s n_{eff}(\omega_s) + \omega_i n_{eff}(\omega_i)] , \qquad (45)$$

where $n_{eff}(\omega)$ is the modal index of the considered optical modes. Since the frequency dependence of n_{eff} results from material and waveguide dispersion, to achieve phase matching the dispersion of the waveguide has to be engineered acting on its geometric dimensions. Expanding the phase mismatch parameter as a Taylor series around the pump frequency, the odd power terms in $\Delta\omega$ are canceled out for symmetry reasons. So, at the second order in $\Delta\omega$, $\Delta\beta$ is given by:

$$\Delta\beta \simeq \beta_2(\Delta\omega)^2. \qquad (46)$$

Therefore, the phase mismatch grows quadratically with the frequency mismatch $\Delta\omega$, reducing the spectral region where phase matching can be achieved. This spectral region is called the bandwidth of the process. In general, from the efficiency expression given in Eq. (43) it follows that the maximum wavevector mismatch interval that can be tolerated is given by $\Delta\beta_{max} = 4\pi/L$. For this reason, when the approximation done in Eq. (46) is valid, the process bandwidth BW can be written as:

$$BW \simeq \sqrt{\frac{\Delta\beta_{max}}{\beta_2}} = \sqrt{\frac{4\pi}{\beta_2 L}}. \qquad (47)$$

Therefore, the bandwidth can be enlarged through an engineering process of the waveguide which can guarantee the achievement of $\beta_2 \sim 0$ at the pump frequency. Another possibility to achieve phase matching is suggested by expanding $\Delta\beta$ to the following order in $\Delta\omega$:

$$\Delta\beta \simeq \beta_2(\Delta\omega)^2 + \frac{\beta_4}{12}(\Delta\omega)^4. \tag{48}$$

The waveguide dimensions can be thus engineered to achieve a GVD parameter β_2 and a fourth order dispersion parameter β_4 with opposite sign and with the proper magnitude to have $\Delta\beta \sim 0$ [21].

Another opportunity is offered by the use of multimodal waveguides. In this approach, the signals propagate on different modal orders with different dispersion relations. In this way, by expanding the propagation constants of the three different modes around the pump frequency, it is possible to write:

$$\Delta\beta = \Delta\beta_0 + \Delta\omega(\beta_{1,i} - \beta_{1,s}) \tag{49}$$

where $\Delta\beta_0 = 2\beta_p(\omega_p) - \beta_i(\omega_p) - \beta_s(\omega_p)$ is the phase mismatch at the pump frequency. The basic idea of the multi-modal phase matching technique is then to balance the $\Delta\beta_0$ parameter by means of the difference between the signal and idler group velocities. Since the balance is between the propagation constants β_i, it is less sensitive to fabrication defects than the method based on balancing the second and the fourth order dispersion parameters [20].

2.3. Third order processes in silicon waveguides

Third order nonlinearities in silicon are particularly important due to the wide range of applications that they open in the field of integrated photonics.

2.3.1. Parametric mixing The most studied process related to parametric wave mixing in silicon is FWM, that is ruled by the $\chi^{(3)}$ tensor. When the two pump frequencies are equal, the FWM process is called Degenerate FWM (DFWM). Moreover, when at the input of the system only the pump waves are provided, the FWM can spontaneously generate both the idler and the signal waves, and the process is called Spontaneous FWM (SFWM).

FWM can be used for a wide range of applications, such as entangled photons generation [22, 23], wavelength conversion [24], optical phase conjugation [25], optical phase erasure [26] and optical isolation [27]. After the first experimental measurement of FWM in a silicon waveguide in 2005 [28], with an efficiency of -35 dB and a 3 dB bandwidth lower than 30 nm in a simple 1.58-cm-long SOI waveguide, several strategies have been adopted to optimize the process. The main limitations to efficient FWM are due to nonlinear losses, in particular TPA and FCA, while bandwidth is mostly limited by GVD.

By tailoring the GVD of the waveguide structure, a -9.6 dB efficiency with 150 nm bandwidth in a 1.8-cm-long SOI waveguide was achieved [29]. TPA and FCA can be significantly reduced by using a pump wavelength greater than 2 µm; this approach, together with GVD engineering, allowed for broad-band FWM generation with -38.6 dB efficiency and a bandwidth of 748 nm in a 1-cm-long SOI waveguide [30]. The pump was Continuous Wave (CW) and with a wavelength of 1950 nm. In terms of efficiency, the best result achieved in the C-band (1530 nm - 1565 nm), reported in Fig. 7, shows a peak conversion efficiency of +5.2 dB in a 2-cm-long SOI waveguide thanks to the pulsed nature of both the pump and signal waves which reduces the nonlinear losses [31]. Regarding the spectral conversion, the most notable outcome is a conversion from 2440 nm to 1620 nm [21], that was obtained exploiting the discrete band phase matching approach. This approach consists in achieving the phase matching between discrete sets of frequencies far away from the pump frequency [32]. In the case of the reported spectral conversion [21], the discrete band phase matching was achieved by balancing the anomalous second order dispersion β_2 with a small positive fourth order β_4.

Other studies focused on the reduction of the foot-print of the on chip system. A -9

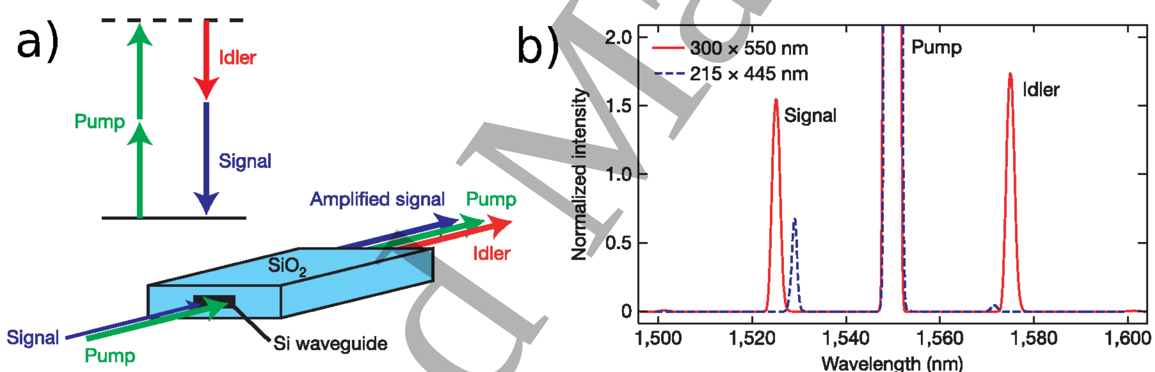

Figure 7. (a) Sketch of a FWM process in a waveguide. (b) Experimentally observed FWM spectra from waveguides exhibiting anomalous GVD (300 nm × 550 nm) and near-zero GVD (215 nm × 445 nm). Both plots are normalized to the signal power with the pump off. This figure is derived with permission from Ref. [31].

dB efficiency is shown by Photonic Crystal (PhC) silicon waveguides that are just 80 µm in length due to a slow c/30 group velocity [33]. Also slot waveguides are applied to enhance the efficiencies of FWM [34]. By inserting silicon nanocrystals (Si-nc) in the gap of a slot silicon waveguide it was possible to achieve a high nonlinear coefficient ($\gamma^{(3)}$=1100 W^{-1}m^{-1}), that led to -28.9 dB efficiency despite both the signal and pump sources were CW at about 1550 nm [34]. In addition, a graphene layer deposited on a hundreds-µm-long SOI waveguide boosts the efficiency to -20 dB [35]. Finally, hybrid plasmonic waveguides loaded with nonlinear organic materials combine the high optical confinement of these guiding structures with the high Kerr index and low refractive index of organic materials. This enhances the nonlinear response while lowers the linear losses with respect to non-hybrid silicon plasmonic waveguides, with simulated efficiencies of about -16 dB over 1 µm propagation length [36].

2.3.2. Third harmonic generation THG could be useful for applications in scanning microscopy [37, 38], light emission [39] , optical performance monitoring [40], material processing [41], and signal processing [42].
One of the first report of THG in silicon was in a PhC waveguide (Fig. 8). Here, the slow light properties ($v_g \simeq c/40$) were used to enhance the THG process. A spectral conversion from 1560 nm to 520 nm with an efficiency of -70 dB was obtained in a 80-µm-long PhC waveguide [39]. PhC waveguides guarantee an extremely high optical confinement together with the possibility to tailor the dispersion of the guiding structure to get phase matching.

A similar spectral conversion, from 1560 nm to 520 nm, but with a better efficiency,

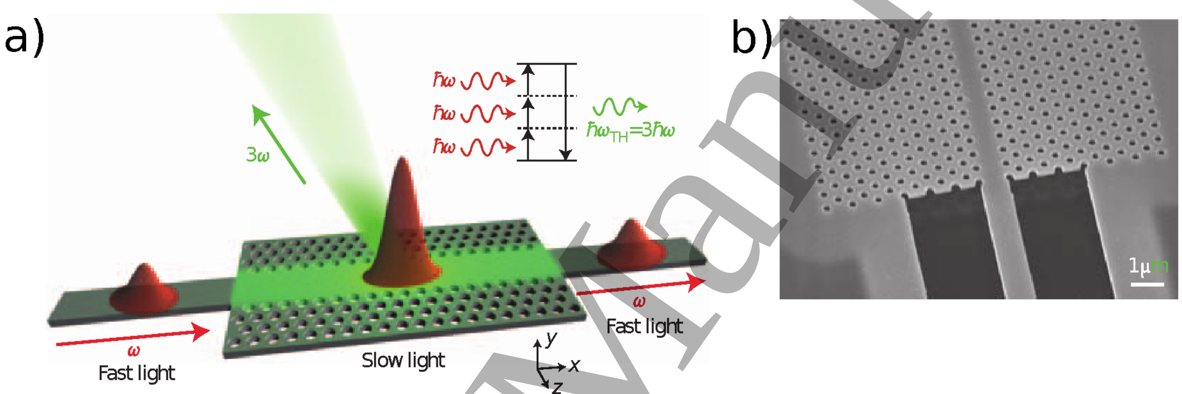

Figure 8. (a) Schematic representation of green light generation through THG in PhC waveguide. The pump light at frequency ω is coupled inside the PhC structure and the generated green light at 3ω is extracted out-of-plane with a specific angle off the vertical direction (b) Scanning-Electron-Microscopy image of the input of the PhC waveguide. This figure is derived with permission from Ref. [39].

about -50 dB, was achieved with a silicon loaded nanoplasmonic waveguide composed by a silicon core, with 95 nm width and 340 nm height, covered by a 60 nm thick gold cap that is only 5 µm long [43]. Other strategies for efficient THG have been studied, such as the application of silicon nitride Resonant Waveguide Gratings (RWGs) [44], where the low absorption of the UV light provided by silicon nitride and the strong optical confinement characterizing RWGs allowed the generation of light at UV wavelengths (355 nm) from a pump centered at 1064 nm with -120 dB efficiency. Also metal-clad plasmonic double-slot waveguides have been proposed for THG [45, 46], with theoretical efficiencies of about -50 dB for propagation lengths of few micrometers only.

2.3.3. Nonlinear multi-photon absorption The processes of electron-hole generation by Multi-Photon Absorption (MPA) [47] induces nonlinear losses in the silicon transparency region [48]. Two Photon Absorption (TPA) [49] or Three Photon Absorption (3PA) [50] efficiencies scale with the second and third power of the input optical intensity [4, 51]. When the photons involved in the MPA have the same wavelength, the process is said to be degenerate, as in Fig. 9(a), while if the absorption is mediated by photons with different frequencies the process is named non-degenerate, that is the case in Fig. 9(b). Typically, the non-degenerate TPA is achieved with the absorption of one photon coming from an intense pump that also triggers the absorption of a probe photon, belonging to a weaker light source; in this way, the absorption of the weaker field is ruled by the stronger one [52]. The excited electrons contribute to FCA and FCD.

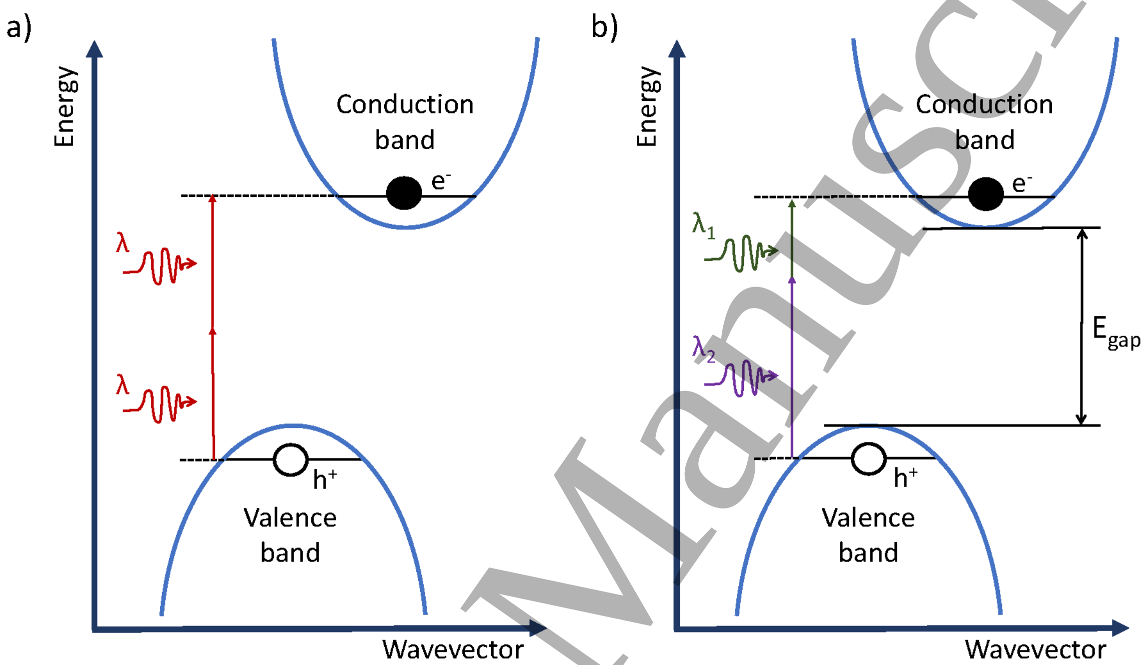

Figure 9. Illustration of TPA transition in silicon with the excitation of an electron–hole (e^-–h^+) pair. (a) Degenerate TPA, with the absorption of two identical photons. (b) Non-degenerate TPA, with the absorption of two photons with different wavelengths.

TPA, FCA and FCD are detrimental effects for optical applications of silicon waveguides requiring high optical powers [7, 47, 53]. In order to avoid TPA, whose nonlinear coefficient β_{TPA} falls between 0.44 cm GW^{-1} and 0.9 cm GW^{-1} at 1.55 µm wavelength, the best solution is to move the pump to wavelengths greater than 2.3 µm, where the silicon bandgap is too large for TPA [54]. More challenging is to face the free-carriers effects, whose relevance is strictly connected to their lifetime [55]. It has to be noticed that the impact of free-carrier effects is related also to the nature of the light-matter interaction, for example the slow light structures enhance these effects [56]. However, the main attention of current research is devoted to the reduction of the lifetime of free-carriers, since this parameter is critical for usual waveguides. The reduction of free-carriers lifetime can be accomplished with different strategies such as increasing the surface recombination velocity through the increase of the surface-to-volume ratio of the waveguide [57], or by introducing non-radiative centers through ion-implantation [58], or by surface chemistry modification [59] or by applying a reverse voltage bias through a p-i-n diode structure integrated in the waveguide to sweep out the TPA-generated carriers [60]. This last approach, shown in Fig. 10, is the most efficient approach and it was used to reduce the carrier lifetime from 3 ns to 12.2 ps [55]. Alternatively, pulsed lasers can be used with a repetition rate low enough to allow photo-generated carriers recombination between subsequent pulses [47].

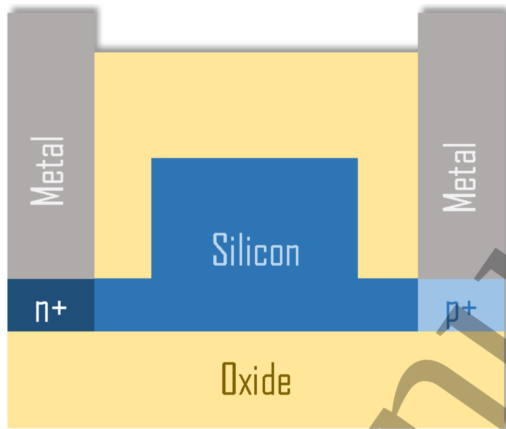

Figure 10. Schematic of the cross-section of the p-i-n waveguide device used in [55]. The system consists of a silicon rib waveguide on a SOI wafer. The waveguide has p+ and n+ type doping in the slab region on either side, and vias and contacts to the doped regions.

TPA, FCA and FCD have been also exploited in some applications. For example, FCD and FCA enable optical modulators [61, 62]. Free-carriers are used for optical pulse compression [63]. TPA is applied in photodetectors [64, 65], all optical logic gates [66, 67], ultrafast optical modulation and switching [68, 69] and in optical limiting [70].

2.3.4. Parametric amplification Parametric processes can also provide Optical Parametric Amplification (OPA), where an optical signal is amplified by means of a parametric nonlinearity [71, 72]. For a third order nonlinear material as silicon, OPA is mainly performed through FWM. Both the signal and the idler experience a parametric gain. The signal wave is amplified in the OPA process, while the idler is generated by the FWM process. The gain related to the idler generation by the signal wave is defined as the conversion gain. Due to the broad gain spectrum of OPA, multiple waves can be simultaneously amplified [31]. Moreover, OPA is also used in Optical Parametric Oscillators (OPOs) [73].

The efficiency of the OPA process is calculated as the off-chip gain, or as the on-chip gain, that is the equivalent to the off-chip gain but without considering the coupling losses.

Currently, the best result for OPA shows an off-chip gain of 20 dB that can reach values up to 30 dB through a Raman assisted FWM in a 2-cm-long waveguide, with a gain bandwidth around 200 nm [74]. This result, reported in Fig. 11, relies on a proper dispersion engineering of the waveguide together with the enhancement provided by the Raman effect [75]. These results benefit also of the absence of TPA losses, avoided by moving the pump wavelength to values greater than 2 µm, as shown in Fig. 3. In fact, the current best result for OPA near the common telecommunication wavelengths shows an off-chip peak gain of 2.9 dB between 1525 nm and 1540 nm in a 1.7-cm-long waveguide [31], that is definitely lower than the 20 dB gain observed with wavelengths greater than 2 µm.

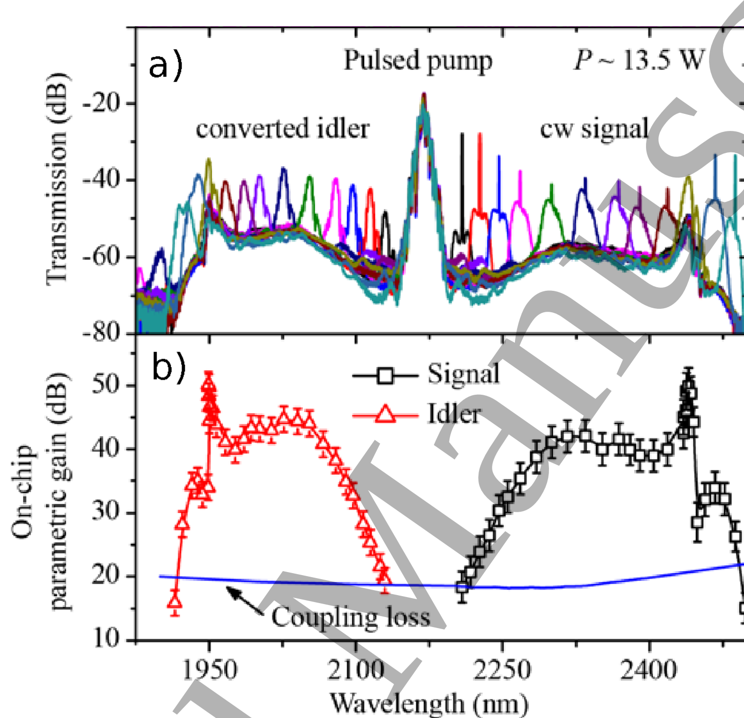

Figure 11. (a) Series of FWM spectra with the pulsed pump co-propagating with a CW mid-IR signal at various wavelengths. The pump is centered at $\lambda = 2173$ nm, and has a power around 13.5 W. (b) On-chip parametric signal gain (black squares) and idler conversion gain (red triangles). The coupling losses are shown with the blue line. To calculate the off-chip gain, the coupling losses have to be subtracted from the on-chip gain. This figure is derived with permission from Ref. [74].

2.3.5. Optical solitons A soliton is a self sustaining wave that maintains its shape while propagating with constant velocity through a medium. It results from the balance between nonlinear and dispersive effects [76]. Solitons can be spatial or temporal. A spatial soliton arises from the cancellation of diffraction effects with nonlinear processes, and it results in a self-confined wave propagating inside a nonlinear medium [77]. Temporal solitons, instead, propagate inside optical waveguides and fibers, and their pulse shape is preserved by the balance of nonlinear effects and dispersion [78]. The main contributions to the soliton formation come from SPM and from the anomalous GVD [8]. The formation of a fundamental soliton rather than a higher order one depends on the ratio between the nonlinear length associated to SPM chirping, L_{NL}, and the dispersion length related to the broadening induced by anomalous GVD, $L_D^{(2)}$. The order of the soliton is evaluated as $N = \sqrt{L_D^{(2)}/L_{NL}}$ [18]. To observe the fundamental soliton it is required that $L_{NL} = L_D^{(2)} << L$, with L the length of the waveguide. Since $L_D^{(2)} \propto 1/\beta_2$ and $L_{NL} \propto 1/\gamma^{(3)}$, the material should exhibit both large nonlinearity and dispersion . An experimental evidence of the formation of a fundamental soliton in a 5-mm-long silicon waveguide is shown in Fig. 12, which reports the results obtained in ref. [18]. The secant shape of the output pulse spectrum confirms the solitonic nature of the propagating light.

Light travelling as a soliton is applied in mode locking [79] and in frequency comb generation [80]. Higher order solitons are useful for applications such as pulse compression [81] and supercontinuum generation [82, 83].

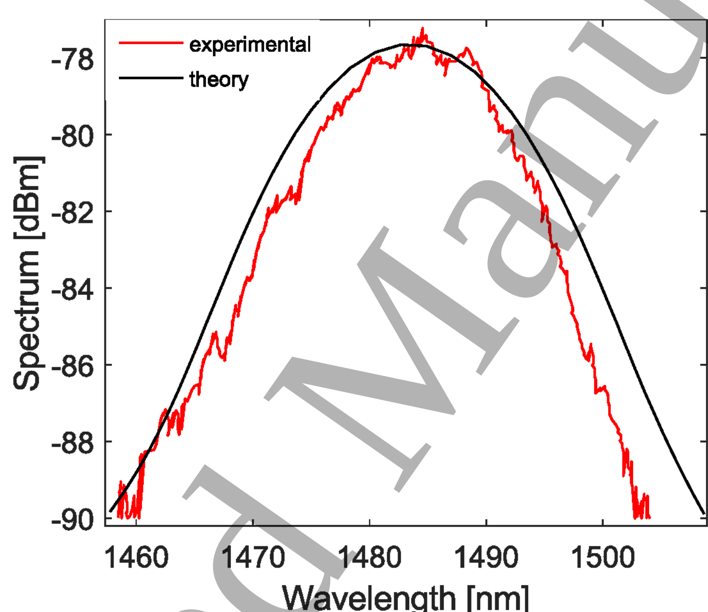

Figure 12. Comparison between experimental (blue) and numerically simulated (red) pulse spectra at the output of the waveguide. The secant shape confirms the formation of a fundamental soliton in the structure. Data reported in Ref. [18].

2.3.6. Supercontinuum generation Supercontinuum (SC) generation refers to the generation of a broad spectrum of coherent light by the simultaneous occurrence of several nonlinear optical processes. Depending on the temporal duration of the pump pulses, SC is distinguished in: short (femtosecond) and long (picosecond, nanosecond and continuous wave) pump pulses regimes. Within the short pulses regime SC is mainly dominated by the soliton dynamics and SPM, while in the long pulses regime FWM, Modulation Instability (MI) and Raman Scattering (RS) are the most important nonlinear processes [84]. The key parameters in order to achieve an efficient SC generation are a high nonlinear coefficient and a good control of the GVD. GVD is important because the broadest SC spectra are generated when the pump wavelength is placed in the anomalous GVD region, near to the Zero Dispersion Wavelength (ZDW), where $\beta_2 = 0$.

Integrated silicon SC sources guarantee small footprint together with ultra-broadband SC. In fact, while photonic crystal fibers, nonlinear optical fibers and silica waveguides require meters of nonlinear interaction [85], integrated silicon and silicon nitride waveguides exhibit better efficiency with lower nonlinear interaction lengths and pump powers [86, 87].It was demonstrated a supercontinuum generation in silicon wires spanning over 500 nm at telecommunication wavelengths [87], but better results were obtained in Si_3N_4 waveguides with a thickness of 1 μm, where SC was generated with a spectral broadening ranging from 470 nm to 2130 nm, as shown in Fig. 13 [88]. Another recent development involves the Silicon-On-Sapphire (SOS) platform which has lower propagation losses in the MIR than SOI. With SOS nanowires it was achieved a spectral broadening ranging from 1.9 μm to 6 μm [89], that is the widest spectral range ever achieved on a silicon platform.

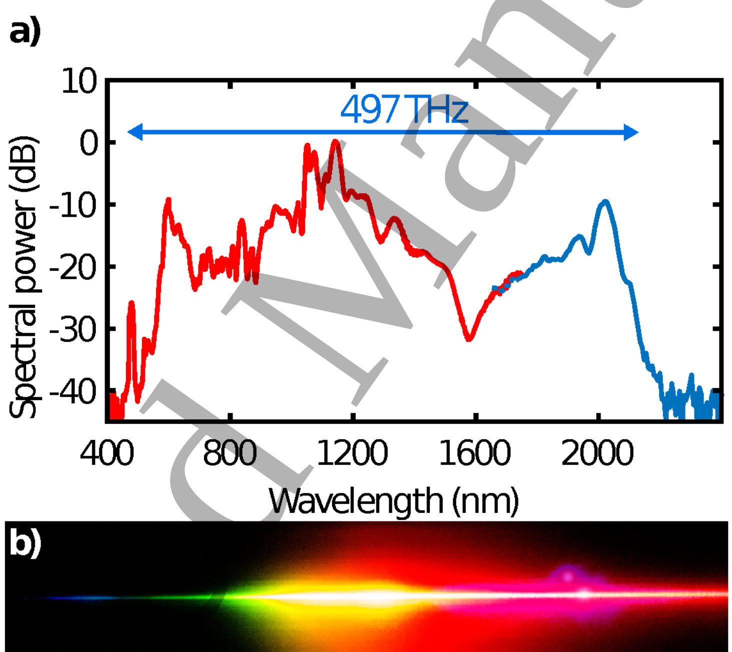

Figure 13. (a) Supercontinuum spectrum generated in a 5.5 mm long Si_3N_4 waveguide with 1.0 μm heigth and 0.8 μm width, and maximum coupled pulse energy 590 pJ. The spectrum is measured using an Optical Spectrum Analyzer (red) and a near-infrared spectrometer (blue). (b) Photograph of the spectrum after being dispersed by a diffraction grating. This figure is reprinted with permission from Ref. [88].

2.4. Pulse propagation in quadratic media

The theoretical description of the pulse propagation in waveguides under the effect of second order nonlinearities will be given in this section, considering the Pockels effect and the SHG processes.

2.4.1. The Pockels effect In the Pockels effect, an optical pulse propagates under the effect of a DC field $\mathbf{E}_{DC} = \mathbf{E}(\mathbf{r}, \omega = 0)$. The corresponding second order nonlinear polarization vector can be derived from Eq. (9), and is given by [90]:

$$\delta\mathbf{P}(\mathbf{r},\omega) = 2\varepsilon_0\chi^{(2)} : \mathbf{E}(\mathbf{r},\omega)\mathbf{E}_{DC}. \tag{50}$$

In this framework, using Eq. (19) and the definitions given in Eq. (16) and (17), the following equation can be derived:

$$\frac{du(z,\omega)}{dz} + i(\beta-\beta_0)u(z,\omega) = i\omega X u(z,\omega), \tag{51}$$

where we introduced the following parameter:

$$X = v_g \frac{\int_{A_\infty} \mathbf{e}^*(\mathbf{r}_\perp,\omega)\chi^{(2)} : \mathbf{e}(\mathbf{r}_\perp,\omega)\mathbf{E}_{DC}\ \ dA}{\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega)|^2\ \ dA}. \tag{52}$$

This equation can be solved analytically, obtaining that the propagation constant β can be written as $\beta = \beta_0 + \omega X$. This fact corresponds to a variation of the effective refractive index of the propagating pulse given by $\Delta n = cX$.

2.4.2. Second Harmonic Generation The simplest second order nonlinear generation process is SHG. In the SHG process, an equation for the pump pulse at frequency ω_p and an equation for the generated signal at frequency $\omega_{sh} = 2\omega_p$ must be written considering the proper polarization vector given in Eq. (9). As usual, Eq. (19) and the definitions given in Eq. (16) and (17) must be considered. By Fourier transforming, the following coupled equations describing the spatial and temporal evolution of the fields $u_p(z,t)$ and $u_{sh}(z,t)$ can be derived:

$$\frac{du_p}{dz} + \sum_{m\geq 1}\frac{(i)^{m-1}\beta_m}{m!}\frac{\partial^m u_p}{\partial t^m} = 2i\gamma_p^{(2)}\frac{P_p}{\sqrt{P_{sh}}}u_p^2 e^{i\Delta\beta z} \tag{53}$$

$$\frac{du_{sh}}{dz} + \sum_{m\geq 1}\frac{(i)^{m-1}\beta_m}{m!}\frac{\partial^m u_{sh}}{\partial t^m} = i\gamma_{sh}^{(2)*}\sqrt{P_{sh}}u_{sh}u_p^* e^{-i\Delta\beta z}. \tag{54}$$

Here we introduced the factor $\gamma_i^{(2)}$, which is defined as:

$$\gamma_i^{(2)} = \omega_i \frac{v_{g,p}\sqrt{v_{g,sh}}}{\sqrt{8A_0\varepsilon_0}}\Gamma^{(2)}, \tag{55}$$

while $\Gamma^{(2)}$ is given by:

$$\Gamma^{(2)} = \frac{\sqrt{A_0}\int_{A_\infty} \mathbf{e}^*(\mathbf{r}_\perp,\omega_p)\chi^{(2)} : \mathbf{e}(\mathbf{r}_\perp,\omega_{sh})\mathbf{e}^*(\mathbf{r}_\perp,\omega_p)\ \ dA}{\left[\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega_p)|^2\ \ dA\right]\left[\int_{A_\infty} n(\mathbf{r}_\perp)^2|\mathbf{e}(\mathbf{r}_\perp,\omega_{sh})|^2\ \ dA\right]^{1/2}}. \tag{56}$$

Analogously to the other generation processes, also in this case we introduced the phase mismatch parameter $\Delta\beta = 2\beta_p - \beta_{sh}$, which sets the maximum generation efficiency when $\Delta\beta = 0$. In this case, due to the energy relationship between the pump and the second harmonic signal, the phase matching condition simply requires the pump and the second harmonic generated signals to have the same effective refractive indexes ($n_p = n_{sh}$).

2.5. Second order processes in silicon waveguides

Inducing second order nonlinear effects in silicon waveguides requires the ability of destroying its centrosymmetric crystalline structure. In recent years, this goal has been achieved by using different techniques, which are mainly associated with the deposition of strain layers at the waveguide boundaries [13]. The experimental demonstration of Pockels effect and SHG in strained silicon waveguides is treated in this section. The large variation of the effective $\chi^{(2)}_{eff}$ values reported in the literature is also discussed, emphasizing the effect of the free-carriers on the reported results.

2.5.1. Pockels effect in silicon waveguides The first experimental observation of Pockels effect has been shown through the deposition of a stressing silicon nitride (Si_3N_4) layer on a silicon photonic crystal waveguide [91]. As it is sketched in Fig. 14(a), this layer induces an asymmetric strain on the silicon crystalline structure, determining a nonzero $\chi^{(2)}$ value. A phase modulator has been realized by means of a Mach-Zehnder interferometer (MZI). By applying a DC electric field to one of the MZI arms, the effective refractive index of the propagating mode is altered by means of the Pockels effect. An effective $\chi^{(2)}_{eff} \simeq 15$ pm/V has been determined. Recently, through the optimization of the waveguide geometry and the silicon nitride deposition, a $\chi^{(2)}_{eff} \simeq 190$ pm/V and the measure of the various elements of the $\chi^{(2)}$ tensor were reported [92, 93, 94]. Moreover, measurements of the spectral dependence of $\chi^{(2)}_{eff}$ show that it grows with wavelength, reaching values up to $\chi^{(2)}_{eff} \simeq 340$ pm/V at a wavelength of 1.63 μm [95]. Other works concerned the use of different methods to induce strain, such as the surface activation of second-order nonlinearity induced by a HBr dry etching process, through which an effective $\chi^{(2)}_{eff}$ value of (9 ± 1) pm/V has been detected [96]. However, recent studies raised many doubts about the interpretation of the $\chi^{(2)}$ measurements described so far [97, 98]. The authors showed that the phase-shift can be attributed to the accumulation of free-carriers inside the waveguide as a consequence of the applied DC field, together with the existence of a surface charge layer at the Si/Si_3N_4 interface [97]. In order to decouple the contributions to the effective $\chi^{(2)}_{eff}$ due to the Pockels effect from the ones related to free-carriers, the DC field has been substituted by an AC field in the GHz range, at a value larger than the maximum free-carrier modulation speed, which is limited to hundreds of MHz [98]. In this frequency region, only the contribution to the electro-optic modulation induced by bulk $\chi^{(2)}$ nonlinear effects is present, since the free-carriers are not expected to follow instantaneously the field variations. The experiment performed on a strained silicon racetrack resonator did not show any clear evidence of modulation due to Pockels effect. Considering the noise floor of the measurement setup, this sets an upper limit of (8 ± 3) pm/V to the effective $\chi^{(2)}_{eff}$ induced by an applied stress of -0.5 GPa. The result of this experiment is shown in Fig. 14(b).

2.5.2. SHG in silicon waveguides Strained silicon has been used to evidence SHG in silicon waveguides [99]. The study has been carried out on different waveguide geometries, providing the best performances with a (2×10.7) μm^2 cross section, which allowed to estimate a $\chi^{(2)}_{eff}$ of (40 ± 30) pm/V. This result has been achieved in a highly multimodal waveguide, without using any phase-matching mechanism. The pump wavelengths involved span from 2 μm to 2.3 μm, providing the corresponding generated signal at a halved wavelength, as it clear from the measured output spectrum shown in Fig. 14(c) and from the linear dependence between pump and SHG frequency shown in Fig. 14(d). Further investigations showed that not only strain contributes to the large $\chi^{(2)}_{eff}$, but also a contribution from a third order nonlinear process is present. This further effect is called Electric-Field-Induced-Second Harmonic Generation (EFISH). In this process, two photons at frequency ω and a DC field are coupled, giving rise to a variation of the polarization vector $\delta P = (3/4)\varepsilon_0\chi^{(3)}E_{DC}E^2_\omega e^{-i2\omega t}$ [100, 101, 102]. This results in the generation of a new signal at a doubled frequency 2ω, introducing a quasi-second-order nonlinearity $\chi^{(2)}_{EFISH} = \chi^{(3)}E_{DC}$. In the case of SHG process in strained waveguides, the DC field is ascribed to the charged layer which is created at the Si_3N_4 interface.

The EFISH process has been exploited for the achievement of SHG in an unstrained silicon waveguide [103]. In this work the DC field is induced by a series of lateral p-i-n junctions. The junctions are periodically separated along the waveguide propagation direction in order to provide a periodic DC field to quasi-phase-match the pump and the generated signal, which both propagate on the fundamental mode of the waveguide. Using this method, it has been possible to estimate an effective $\chi^{(2)}_{eff}$ of 41 pm/V.

Another recent work measures an effective $\chi^{(2)}_{eff}$ in an unstrained waveguide [104]. In this case the waveguide is realized in silicon nitride and the SHG process is achieved by means of multimodal phase-matching. Even if silicon nitride is an amorphous material, a second order bulk nonlinearity of the order of 10^{-1} pm/V is measured. By applying a voltage of 500 V across the waveguide, the second harmonic signal is increased by about $16 - 18\%$ thanks to EFISH.

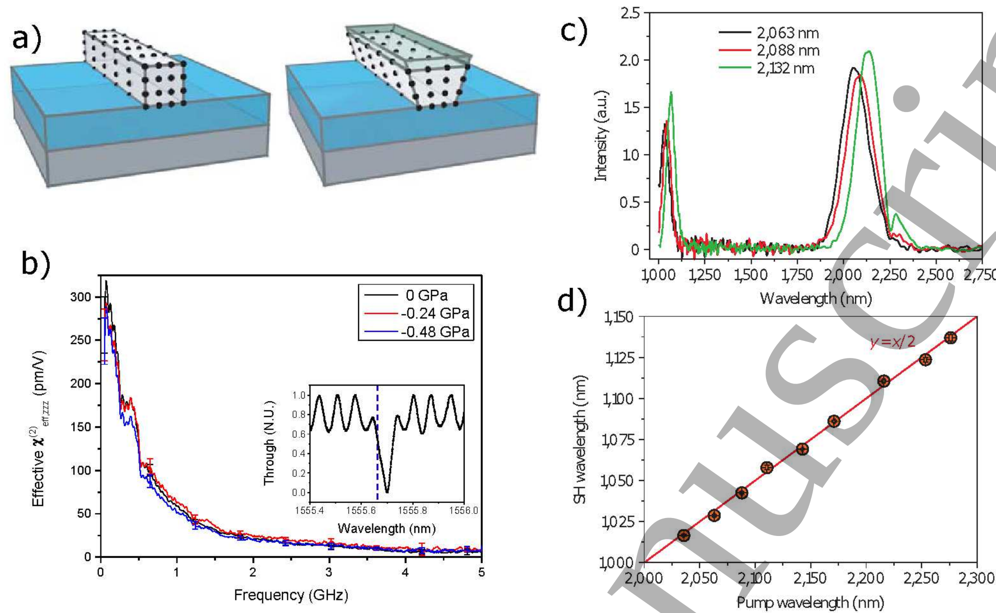

Figure 14. (a) On the left: sketch of a standard silicon waveguide. On the right: silicon waveguide covered by a layer of silicon nitride, which induces a strain in the crystalline structure that is sketched with the black points. (b) Effective $\chi^{(2)}_{eff}$ as a function of the electrical modulation frequency for three different stress values. The inset shows the trough port of the racetrack resonator, while the vertical blue dashed line shows the laser wavelength. (c) Waveguide transmission spectra for different pump wavelengths (d) Second-harmonic wavelength as a function of the pump wavelength. These figures are derived with permission from Ref. [91, 98, 99].

3. Nonlinear Cavities

3.1. Resonator basics

Optical microcavities are devices in which light is trapped in a closed path [105]. Energy confinement is achieved by TIR at the dielectric interfaces [106] or by exploiting photonic bandgaps in photonic crystal (PhC) nanocavities [107]. Over the years, several materials and geometries have been explored. These include silica micro-spheres [108], silica micro-toroids [109], suspended disks [110] or wedge resonators [111], crystalline Whispering Gallery Mode (WGM) resonators [112, 113], as well as monolithic devices like ring resonators realized in silicon, silicon nitride (Si_3N_4) or high index silica (Hydex) [114]. Some examples are shown in Fig. 15. Typically, microresonators are evanescently coupled to a waveguide (Fig. 15(b)). This waveguide is called exciting waveguide or bus waveguide and it is used to channel the input and output signal. Only certain wavelengths can be supported in an optical micro-cavity. These are associated with waves that, after a complete roundtrip, have accumulated a phase which is a multiple integer of 2π. This condition ensures that the wave continues to coherently-reinforce itself during the propagation, leading to a large energy storage inside the cavity. In travelling wave resonators, the resonance frequencies ω_m are given by the following

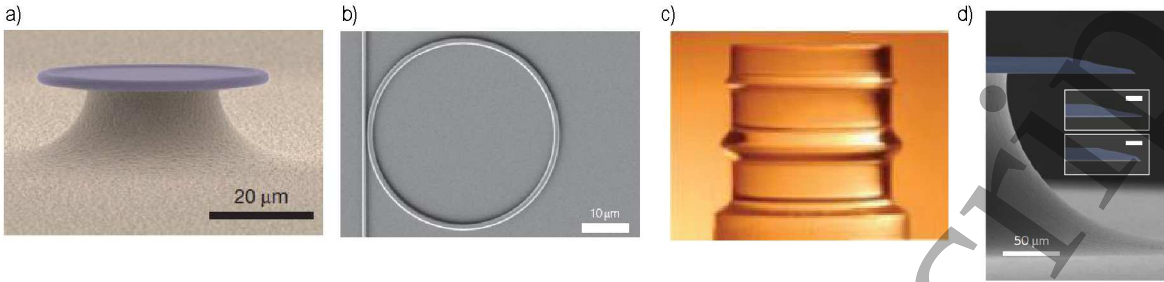

Figure 15. Examples of WGM microresonators realized in different platforms and geometries. (a) Silica micro-toroid resonator. (b) Silicon nitride planar ring resonator. (c) Crystalline Calcium Fluoride resonator. (d) Silica wedge resonator. These figures are derived with permission from Ref. [109, 115, 116, 117].

expression:

$$\omega_m = \frac{2\pi mc}{n_{eff}(\omega_m)L}, \tag{57}$$

where m is an integer number which expresses the resonance order, $n_{eff}(\omega_m)$ is the modal effective index at frequency ω_m, c is the speed of light and L is the resonator perimeter. From Eq. (57), the spectral distance between two adjacent resonance orders, which is called the resonator Free Spectral Range (FSR), is given by:

$$FSR_m = \frac{2\pi c}{n_g(\omega_m)L}, \tag{58}$$

where n_g is the group index. Since n_g is frequency dependent, the FSR is not constant but depends on the resonance order m. As a consequence, resonances are not equidistant in frequency. This fact is in general quantified by expressing the frequency distance $\omega_m - \omega_0$ (also called integrated dispersion D_{int} [118]), in which ω_0 is a reference eigenfrequency, as:

$$\omega_m - \omega_0 = D_1 m + \frac{1}{2}D_2 m^2 + ... \tag{59}$$

where $D_1 = FSR_0$ and D_2 corresponds to the difference of the two FSR adjacent to the expansion frequency ω_0 (Fig. 16). It can be demonstrated that D_2 is linked to the group velocity dispersion parameter β_2 through the relation $D_2 = -c/n_0 \cdot D_1^2\beta_2$, where n_0 is the material refractive index [119]. Each resonance has a finite linewidth, which is defined by its Full Width at Half Maximum (FWHM) $\Delta\omega_m$. The ratio between the resonance frequency and the FWHM is called the quality factor $Q = \frac{\omega_m}{\Delta\omega}$. It can be shown that the quality factor is directly proportional to the photon lifetime τ_m in the cavity, i.e., $Q_m = \frac{\omega_m \tau_m}{2}$ [120]. Resonators are appealing for nonlinear optics because they enhance stored optical power. The power enhancement factor F, defined as the ratio between the circulating power and the incoming one, is given by:

$$F_m = \frac{2\tau_m^2}{\tau_{rt,m}\tau_{l,m}}, \tag{60}$$

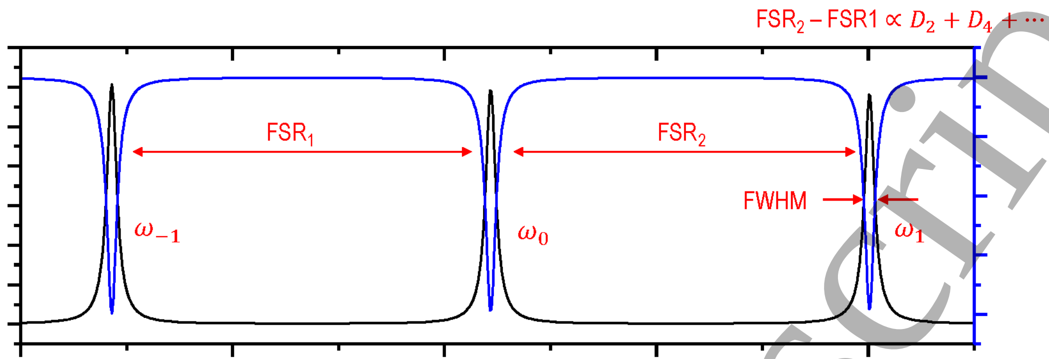

Figure 16. Simulated spectral response of a WGM microresonator in the Add-Drop filter configuration. Indicated are the resonance orders ω_m and the FSR. The simulation assumes a perimeter of 415 µm, a waveguide-resonator coupling rate of 2.5%, a cross-section of 400 × 250 nm and a linear loss rate of 5 dB/cm.

where $\tau_{rt,m}$ is the roundtrip time of a photon along the cavity and $\tau_{l,m}$ is the average time spent by a photon in the cavity before being scattered into the exciting waveguide. F can reach values up to ten thousands, which means that Watts of circulating power with sub-milliwat excitations can be achieved. It is worth to note that, from Eq. (60), the factor F results to be proportional to Q^2, which is the reason why high-Q cavities are ideal platforms for nonlinear optics. The total lifetime τ_m can be decomposed into two main contributions, i.e., $1/\tau_m = 1/\tau_{abs,m} + 1/\tau_{l,m}$, where $\tau_{abs,m}$ takes into account all the possible absorption and scattering mechanisms in the host material (linear and nonlinear absorption, Rayleigh-scattering induced by surface roughness [121]), while $\tau_{l,m}$ describes the energy leakage into the bus waveguide, whose amount is directly proportional to the coupling coefficient. From these quantities, one can define the intrinsic quality factor $Q_{int,m} = \frac{\omega}{2\tau_{abs,m}}$, and the extrinsic quality factor $Q_{ext,m} = \frac{\omega}{2\tau_{l,m}}$, which are related to the total quality factor $Q_{tot,m}$ through the relation $1/Q_{tot,m} = 1/Q_{int,m} + 1/Q_{ext,m}$. The ratio between the intrinsic and the extrinsic quality factor, hence the one between the intrinsic losses and the coupling coefficient, determines the regime under which the resonator operates. If $Q_{int}/Q_{ext} > 1$, the coupling rate exceeds the loss rate, and the resonator is said to be overcoupled. In the opposite situation, where $Q_{int}/Q_{ext} < 1$, the regime is called undercoupled. When $Q_{int} = Q_{ext}$, the coupling matches the losses, and the resonator is said to be critically coupled. It can be demonstrated that this condition coincides with the maximum field enhancement inside the resonator [122]. In a high-Q cavity, the time spent by a photon within the cavity before being absorbed, scattered or lost into the exciting waveguide should be made as long as possible. The latter is maximized by decreasing the resonator-waveguide coupling coefficient, which can be realized by controlling the distance between the resonator and the bus waveguide (named coupling gap). In monolithic devices, the fabrication process imposes that this distance (hence the coupling rate) can not be changed once the microresonator is fabricated. This is why more advanced coupling strategies, based on tunable Mach-Zehnder interferometers [123] or coupled resonators [124], have been proposed to achieve a full control on the coupling coefficient. On the other hand, to increase $\tau_{abs,m}$, all the loss mechanisms should be removed or minimized. For high-index contrast platforms like SOI, scattering at the sidewall roughness dominates. In Ref. [125], a etch-less fabrication process is used to fabricate micron-size silicon waveguides having the exceptionally low value of propagation loss of 0.7 dB/cm. The use of wedge resonators, in which angled edges are introduced to push the mode away from the edges to the inner regions, has also been proposed to decrease the overlap between the propagating mode and the surface scattering centers [126]. In devices realized in Si_3N_4, losses due to trapped hydrogen atoms which act as absorption centers, constitute the bottleneck for the realization of high-Q cavities. Multiple annealing cycles, in which the temperature is raised to 1200 °C for several hours, are used to outgas the hydrogen from the material. Using this technique, quality factors up to 1.0×10^6 have been reported for Si_3N_4 resonators with a radius of 230 µm [127]. Absorption induced by TPA generated carriers occurs when the photon energy is higher than half of the bandgap of the material. In this case, the waveguide is typically embedded in a reverse biased p-i-n junction to deplete the core from the generated carriers [125].

3.2. Dispersion engineering of optical micro-cavities

All the nonlinear light-matter interactions described in Section 1.2 for waveguides occur in resonators. However, there are two key aspects which differentiate the two geometries. The first is power enhancement. We can compare the expression for the idler power P_I^{wg} generated in the case of a phase matched FWM process in a straight waveguide of length L and nonlinear parameter γ (reported in Eq. (43)), to the one for a resonator of equal perimeter, which is given by $P_I^{res} = (\gamma P_p L)^2 F_p^2 F_s P_s$ [128]. In this expression, $F_p(P_p)$ and $F_s(P_s)$ are the enhancement factors (powers) of the pump and of the signal respectively. The ratio P_I^{res}/P_I^{wg} gives $F_p^2 F_s$, which tells that the signal conversion occurs with a much higher efficiency in the resonator with respect to the waveguide. This is because the pump and the signal powers get greatly amplified inside the cavity, with a consequent enhancement of the nonlinear interaction strength. The second feature is the phase matching relation. In a resonator, all the waves involved in a parametric process must be resonant with the cavity, otherwise they will undergo destructive interference. This implies that a new "energy matching"relationship must be satisfied, which states that the frequency of the pump and of all the generated waves must be coincident with the eigenfrequencies of the resonator (or at least fall within their FWHM). However, because of dispersion, the resonator eigenfrequencies are not equally spaced, whereas the comb lines have a fixed separation. So, it seems impossible to satisfy energy matching, especially for combs extending more than one octave. However, a closer look to Eq. (59) reveals that if all the D_i coefficients are made equal to zero for $i > 1$, the frequency distance between the eigenmodes becomes constant. This condition will yield phase matching for all $\chi^{(2)}$ and $\chi^{(3)}$ based parametric processes, since they all involve energy conservation relations of the form $\omega_1+\omega_2=\omega_3$ (for $\chi^{(2)}$-interactions) or $\omega_1+\omega_2=\omega_3+\omega_4$ (for $\chi^{(3)}$-interactions). In practical implementations, it is impossible to make all the D_i equal to zero, so one engineers the waveguide cross-section such that the GVD parameter D_2 is kept close to zero in a very broad spectral region. Since the material dispersion of silicon (or Si_3N_4) is normal ($\beta_2 < 0$) for wavelengths $< 6-7$ µm, the presence of waveguide dispersion is required to make β_2 approaching zero. This accounts in engineering the waveguide cross-section in such a way that its geometric dispersion is anomalous and cancels the material dispersion. Actually, for every practical application it is always preferred to work in a slightly anomalous regime ($D_2 > 0$) around the pump frequency, rather than having zero GVD. This is because the intense pump used for nonlinear frequency conversion induces a frequency dependent Kerr-shift of the resonance frequencies through SPM and XPM, which acts as an effective increase of the local GVD which compensates the anomalous dispersion. It is worth to note that it is through this mechanism that a comb can extend by more than one octave [129]. An example of dispersion engineering is shown in Fig. 17(b), in which the GVD of a Si_3N_4 waveguide of 730 nm thickness is tuned by changing its width. From this example, we

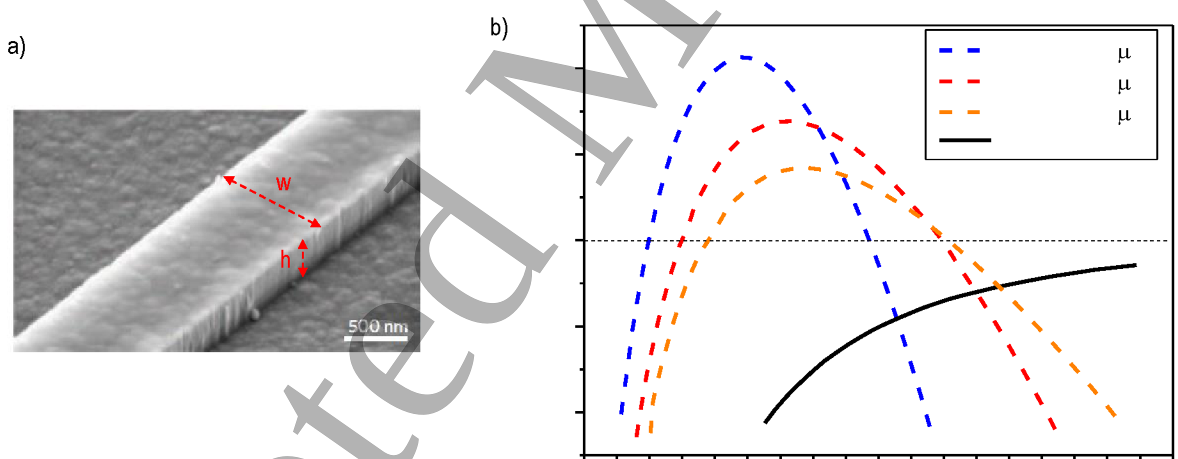

Figure 17. (a) Scanning Electron Microscope image of a Si_3N_4 channel waveguide, with indicated the width w and the height h. This figure is derived with permission from [114]. (b) Waveguide dispersion (dashed lines), defined as $-\frac{\omega^2}{2\pi c}\beta_2$, for the fundamental TE mode as a function of wavelength for different waveguide cross sections $w \times h$. The solid line refers to the dispersion of bulk Si_3N_4. Data from Ref. [130].

see that it is possible to optimize the waveguide cross-section to keep the dispersion slightly anomalous in a wavelength range which nearly spans one octave (i.e., a factor of two in frequency). It is relatively easy to engineer dispersion in SOI waveguides due to the tight confinement of the optical modes induced by the high index contrast with the silica or the air cladding materials. In case of Si_3N_4, the lower refractive index contrast requires thicker waveguides to successfully confine the optical mode, especially at Mid-

Infrared (MIR) wavelengths. It is a known issue that is difficult to grow thick (e.g. > 250 nm) layers of Si_3N_4 by Low Pressure Chemical Vapour Deposition (LPCVD) on the top of silica due the high tensile strains (> 800 MPa) which develops during the process, which can lead to crack formation [131, 127, 93]. To overcome this limitation, two new fabrication approaches emerged in the last years. One relies on the realization of crack isolation trenches into the silica cladding using a diamond scribe before Si_3N_4 deposition [127]. These trenches ensure that threading dislocations are terminated well before reaching the critical device area. Another fabrication process, known as the Photonic Damascene, has recently been implemented [131]. Here, the substrate is prepatterned prior to core material deposition, and once the necessary film thickness has been deposited, it is subsequently planarized. A checkboard prepatterning allows stress relaxation and avoid crack formation. Using these new techniques, film heights up to the record thickness of 1.35 µm have been successfully grown [131], with Q-factors reaching the exceptional value of 3.7×10^6. Even though the control of the waveguide cross section is the most easy and implemented tool for dispersion engineering, it would be of great interest to have access to more degrees of freedom to tailor the GVD at the user's will. Recently, a significant contribution towards this direction, which has been applied to free-standing silica wedge resonators but that can be straightforwardly extended to silicon-based planar devices, has been reported in Ref. [117]. In this work, they noticed that the GVD strongly depends on the wedge angle of the resonator. Furthermore, the centroid of the optical mode moves toward the inner of the resonator as the wavelength increases, as shown in Fig. 18(a). They merged these informations to form multi-wedge resonators. In such devices, as the wavelength increases, the optical mode senses a different wedge angle, so the GVD locally acquires its relative dispersion. It is then possible to engineer the wedge angles and their radial position to flatten the GVD and make it near zero in an ultra-broad spectral interval. Experimentally, they found that in quadrupole wedges, the GVD remains close to zero from 1.2 µm to 2.1 µm, a result which is in extremely good agreement with numerical simulations (Fig. 18(b)). More advanced dispersion control, which make use of mode-hybridization in coupled resonator Si_3N_4 structures, have been also proposed and experimentally investigated [124]. In some parametric processes, like SHG or THG, flattening the GVD near the pump frequency is of limited interest because the newly generated frequencies lie at very large spectral distance from the pump. In these cases, it is sufficient to match only the resonator modes involved in the process. For SHG(THG), this can be done by making the effective index of the doubled(tripled) frequency mode to be equal to the one of the pump mode m_p. In this way, through Eq. (57), it is possible to prove that phase matching will occur at the mode numbers m_p and $2(3)m_p$ for the pump and the SH(TH) mode respectively. Achieving index matching at spectral distances which spans hundreds of nanometers is made possible through multi-mode phase matching. Usually, the doubled (tripled) frequency propagates in a higher order waveguide mode, while the pump propagates into the fundamental one, since it is much easier to be selectively excited. For example, in Ref. [132], the SH mode lies in the 6^{th} transverse TE mode,

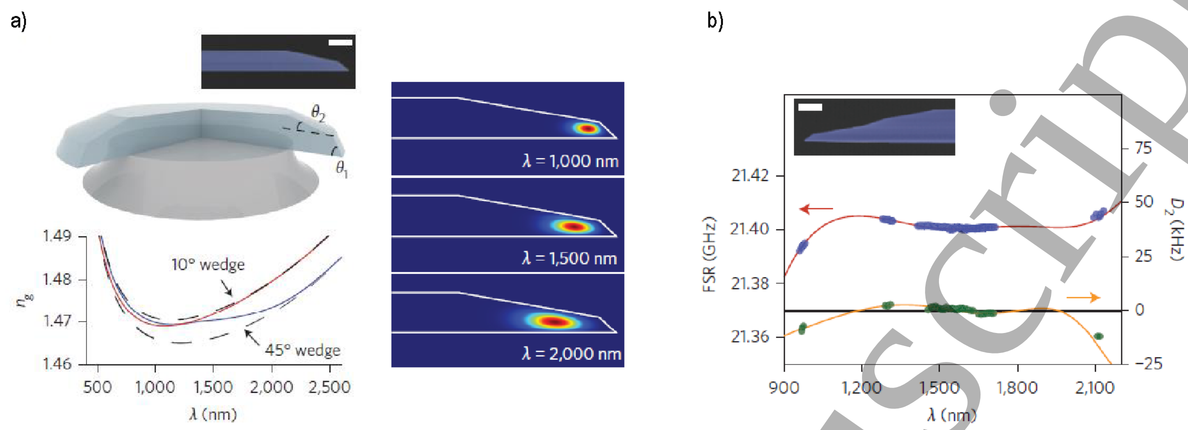

Figure 18. Broadband dispersion engineering of a suspended silica multi-wedge resonator. (a) 3D sketch of a double-wedge suspended wedge resonator. The outer angle is θ_1, while the inner angle is θ_2. The lower left panel shows the group index as a function of wavelength. The two dashed lines are relative to single-wedge resonators with the angle set either to θ_1 or to θ_2. The blue and red curves indicate the multi-wedge resonator where the angle ordering is red (45° (outer)→ 10° (inner)) and blue (10° (outer) → 45° (inner)). The right panel shows the intensity distribution of the modal profile for different wavelengths. (b) Experimental (dots) and simulated (solid lines) values for the FSR and for the GVD parameter D_2 as a function of wavelength for a quadrupole wedge resonator (sketch shown in the inset). This figure is derived with permission from Ref. [117].

while in Ref. [133] the TH mode will lie probably between the 18^{th} and the 20^{th} TE mode. Higher order modal index matching is typically realized by the engineering of the waveguide cross-section.

3.3. Optical frequency combs

Optical frequency combs perhaps represent the most widespread use of microresonators in nonlinear optics [116]. A frequency comb is constituted by a series of narrow-linewidth frequencies, equally spaced by the same amount Δf, which can be mutually phase-locked. Each comb line has a frequency f_n that can be expressed as $f_n = f_{ceo} + n\Delta f$, in which n is an integer number and f_{ceo} is an offset frequency which is called the "carrier envelope offset". A schematic representation of a frequency comb is shown in Fig. 19(a). Frequency combs are generated when an intense pump (which, for simplicity, will be assumed as CW, even if this is not strictly required) is resonantly coupled into a resonator in a frequency region where the GVD is slightly anomalous. Generally, the pump is initially slightly blue detuned from resonance, because this allows to use thermal lock [134] as a feedback mechanism for keeping constant the coupled pump power into the resonator. In the simplest picture, primary sidebands, symmetrically located with respect to the pump frequency f_p at $f_p \pm \Delta$, are generated by degenerate spontaneous FWM close to the resonances where the parametric gain exceeds the roundtrip losses (Fig. 19(a)). When this occurs, parametric oscillation can be sustained, which amplifies

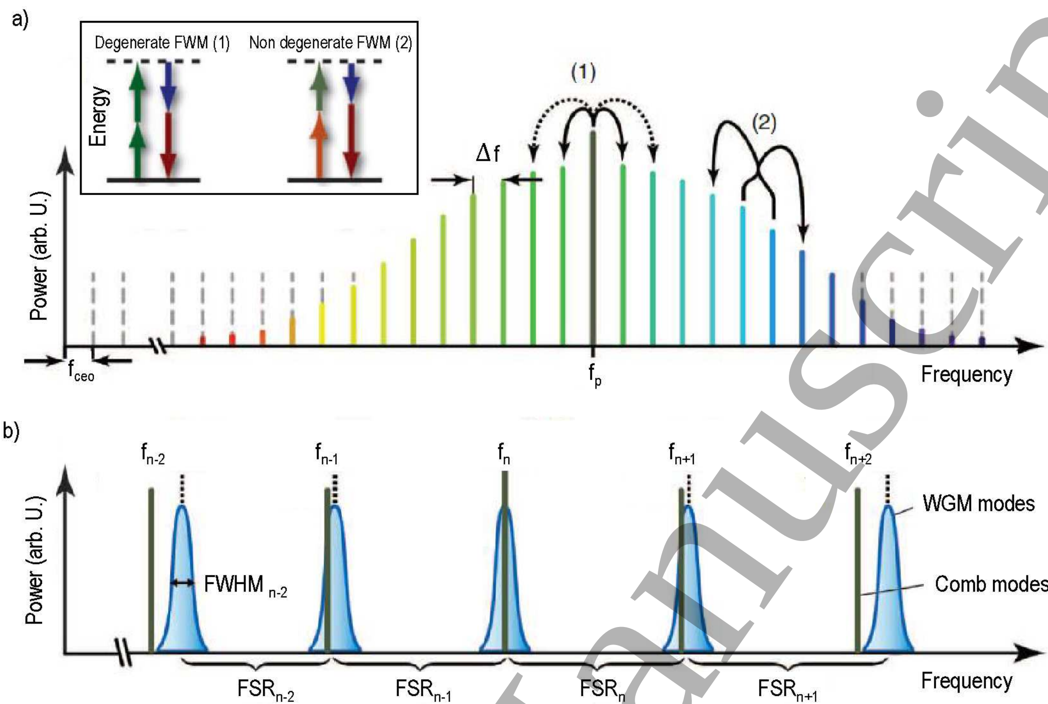

Figure 19. Schematic representation of a frequency comb. (a) Each comb line has a frequency $f_n = f_{ceo} + n\Delta f$. In this simple picture, the comb is initiated by spontaneous degenerate FWM(1), in which two pump photons (green arrows in the upper left inset) at frequency f_p are annihilated to produce a photon pair at frequencies $f_p + \Delta f$ (red arrow) and $f_p - \Delta f$ (blue arrow). Non degenerate FWM (2) between the generated sidebands extends the comb. (b) Dispersion causes the resonator FSR to be frequency dependent, in such a way that the WGM resonances are not equally distant. After some FSRs, the comb lines do not more overlap with the resonator modes, a fact that limits the bandwidth of the generation process. This figure is derived with permission from Ref. [116].

the newly generated frequencies. These can be separated by two or more FSRs, depending on the GVD. However, this picture is greatly oversimplified, and many other factors, such as the pump detuning with respect to the cavity eigenmode, are involved in the generation of the sidebands. Depending on the relative magnitude between the GVD, the initial pump detuning, and the pump power, very little incoherent noise (which can be classical as well as induced by vacuum fluctuations [135]) superimposed the pump field can break the CW regime and generate pulse trains, giving raise to a phenomenon called Modulation Instability (MI). In the frequency domain, these pulses are represented by sidebands which are symmetrically located with respect to the pump frequency. The complex dynamics of comb formation through MI lies outside the scope of this review, and we refer to more specific papers for further reading [136, 137]. Once the main sidebands are generated, non-degenerate FWM occurs between these and the pump, and new lines are created with a spectral separation Δ. This is a direct consequence of energy conservation. At this stage, mini-combs can be generated by degenerate FWM or non-degenerate FWM around the primary and secondary sidebands (it must be pointed out that secondary combs have a tooth separation which is different from the separation of the primary comb teeth, see, e.g., Ref.[138]), at spectral separations which nearly coincide with the mode spacing. The process of multiple cascaded FWM densely fills with comb lines all the resonator modes. The bandwidth is limited to the region in which the GVD remains slightly anomalous. Outside this range, the resonance positions start to deviate from the frequencies of the comb lines by more than a FWHM, so parametric oscillation ceases to occur (Fig. 19(b)). There are some routes of comb formation in which all the comb lines maintain a fixed phase relationship, i.e., the comb is fully-coherent. When viewed in the time domain, this is a fundamental requirement for the generation of pulse trains with a repetition rate Δf. In some cases, the phase relations randomly changes in time but with still some degree of correlation, and the comb is said to be partially coherent. When the phases are completely uncorrelated, the comb is incoherent, and no pulses can be observed at the output. The degree of coherence of the comb [139] depends on its formation dynamics, and will discussed later in this review. First combs where observed in silica microtoroids [109] and crystalline calcium flouride resonators [113]. Fig. 20 offers an overview of the recent achievements in this field. In Ref. [130], one octave spanning fully coherent frequency comb from 1170 nm

Material	Q-factor	Pump power	Spectral range	Reference
Si_3N_4	7.7×10^5	~ 2 W	[1300 – 2000] nm	[118]
Si_3N_4	~ 10^6	~ 400 mW	[1170 – 2350] nm	[130]
Si_3N_4	3.0×10^5	~ 300 mW	[1350 – 2150] nm	[118]
Si	2.2×10^5	~ 150 mW	[2100 – 3500] nm	[125]
Si_3N_4	1.5×10^6	~ 700 mW	[765 – 776] nm	[141]
Si_3N_4	1.3×10^6	~ 100 mW	[502 – 580] nm	[133]
Si_3N_4	~ 10^6	~ 500 mW	[2300 – 3500] nm	[127]

Figure 20. Recent comb observations in different microresonator devices.

to 2350 nm is generated in a dispersion engineered Si_3N_4 resonator of 200 µm diameter, $Q = 3 \times 10^6$, and 400 mW of input power in the waveguide. The related spectrum is shown in Fig. 21(a). In another work [118], the formation of frequency combs covering 2/3 of one octave, from 1300 nm to 2000 nm, in a 238 µm diameter Si_3N_4 monolithic resonator has been reported. The peculiar *sech*^{2} envelope of the comb lines suggested that these are associated with single or multiple temporal dissipative soliton states inside the resonator [140, 80]. It is also shown that the comb can extend in a region of normal GVD if a dispersive wave associated to Cherenkov radiation emission from the soliton is generated [118]. This wave has been demonstrated to be fully coherent with the other comb teeth. The recent achievements in the deposition of thick Si_3N_4 layers allowed to extend GVD engineering to MIR wavelengths, and combs spanning 1.2 µm, from 2.3 µm to 3.5 µm, have been demonstrated in resonators with $Q = 1.0 \times 10^6$ at a wavelength of 2.6 µm [127], which is, up to now, the highest Q reported at this wavelength for an

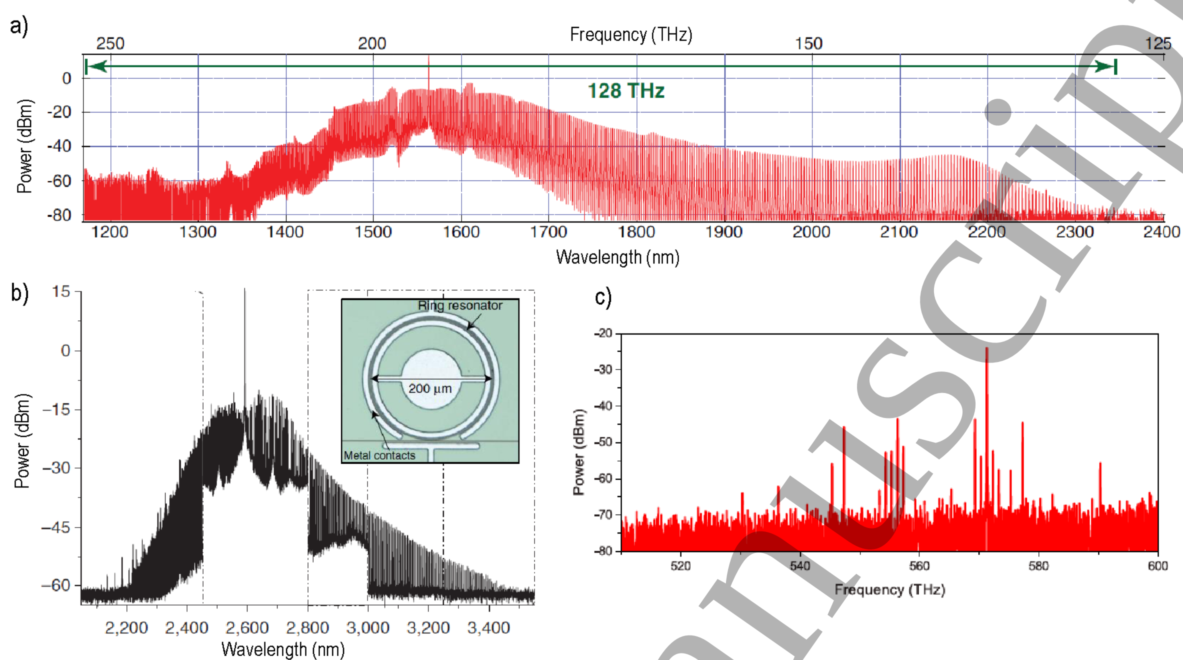

Figure 21. (a) Spectrum of the octave spanning frequency comb in the NIR realized in a dispersion engineered Si_3N_4 resonator of 200 µm diameter and $Q = 3 \times 10^6$. The input power was set to 400 mW. (b) MIR frequency comb using a silicon ring resonator embedded in a p-i-n junction (shown in the inset). (c) Green frequency comb by a Si_3N_4 ring resonator, achieved by THG and SFG of a NIR comb spanning 2/3 of one octave. Data from Ref. [133]. These figures are derived with permission from Ref. [130, 125]

on chip resonator. A MIR comb extending from 2.1 µm to 3.5 µm has been realized also in ultra-low loss silicon resonators which use etchless waveguides embedded in a reverse biased p-i-n junction (Fig. 21(b)) [125]. The application of a reverse bias allows to deplete the waveguide from the carriers generated by Three Photon Absorption, thus reducing the amount of free carrier absorption and leading to a remarkable increase of the number of comb lines . Combs in the visible are difficult to achieve since the material GVD is strongly normal in this spectral range, and usually too large to be compensated by waveguide dispersion. Moreover, losses induced by Rayleigh scattering are increased [12], making the realization of high-Q cavities more challenging. These difficulties can be overcome by generating frequency combs in the NIR, where anomalous dispersion is easily achieved, and then by up converting them into the visible range using SHG, THG or SFG (or a combination of them). Using this approach, combs in the red (765 nm) [141] and in the green (502 nm, Fig. 21(c)) [133] have been demonstrated in Si_3N_4 resonators, even if still limited to a few number of comb lines. Recently, pure $\chi^{(2)}$-based frequency combs have been experimentally observed in bulk quadratic cavities [142] and have been theoretically described in optical microresonators [143, 144].

3.4. Coherence and stabilization of frequency combs

Different methods have been implemented to monitor the phase and the amplitude noise in frequency combs, some of them are sketched in Fig. 22 [139]. The most

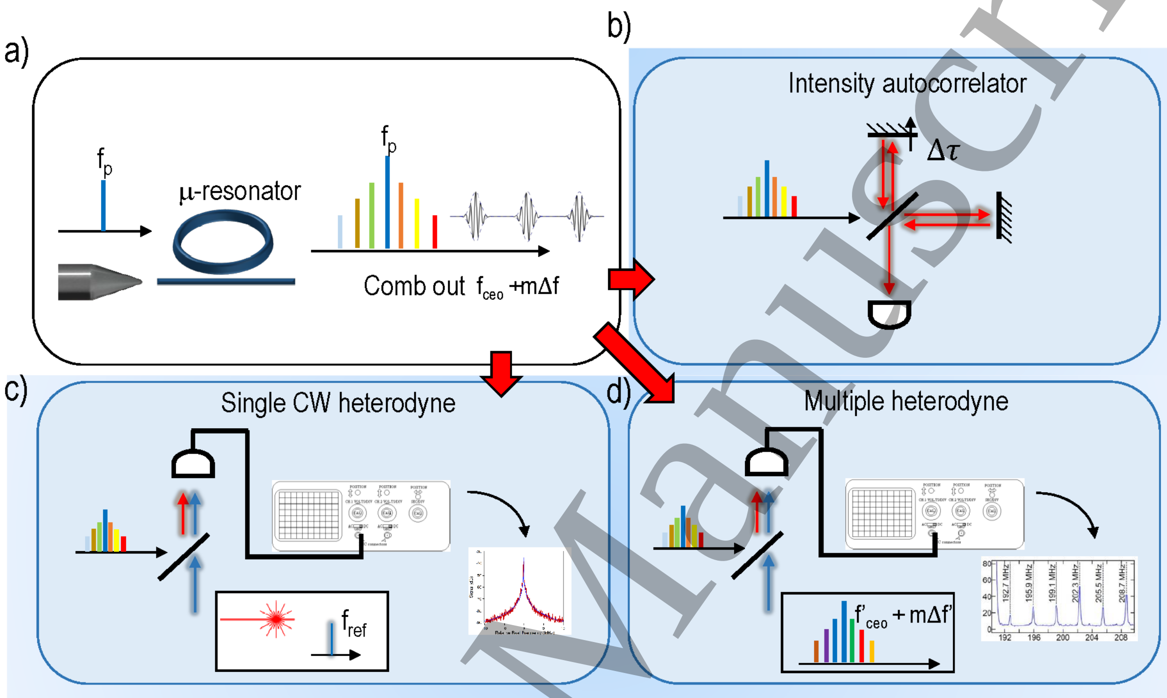

Figure 22. (a) The spectra at the output of the microresonator consist of a comb of lines, whose frequencies are determined by the relation $f_{ceo} + m\Delta f$. The comb is generated by a pump laser at frequency f_p, which is part of the comb. In the time domain, the comb can be constituted by a periodic train of pulses, separated by a time delay $1/\Delta f$. (b) In the simplest implementation, an intensity autocorrelator is constituted by an interferometer in which the pulse interferes with a delayed copy of itself. (c) In a technique which implements a single heterodyne, the comb is combined with a reference CW laser at frequency f_{ref}, and the RF beatnote at $|f_m - f_{ref}|$ is recorded. (d) In a multiple heterodyne technique, the comb is mixed with a reference comb $f'_m = f'_{ceo} + m\Delta f'$, and the multiple RF beatnotes $|f_m - f'_m|$ are recorded.

direct approach is to send the output light from the microresonator to an intensity autocorrelator (Fig. 22(a)) [112]. If the comb state corresponds to a pulse train, this is marked by narrow peaks in the autocorrelation trace, equally separated by a time interval which corresponds to the inverse of the repetition rate of the comb teeth. From the width of these peaks, it is possible to extract the pulse width. Another technique relies in the single heterodyne measurement of the Radio Frequency (RF) beatnote between the comb lines and a reference CW laser (Fig. 22(b)) [145], or in the multi-heterodyne beatnotes of several comb lines with a reference comb produced by an external mode locked fiber laser (Fig. 22(d)) [146]. Low phase noise states are characterized by a single and narrow CW heterodyne beatnote, or by multiple-heterodyne beatnotes of equal spacing corresponding to the offset frequency between the reference comb and the microresonator one [138, 147, 112]. In Ref.[147], spontaneous transitions to phase-locked states, corresponding to sub-200 fs pulse generation with extremely high extinction ratio, has been observed and suggested to be linked to the creation of soliton states, a fact that has been independently supported by later works [80, 118]. A comprehensive treatement of universal comb formation dynamics is given in Ref. [138]. In this work, platform and geometry independent routes of comb generation are analysed and directly linked to single-heterodyne features which creates in the RF spectrum as the comb evolves. A significant contribution to the understanding of comb dynamics has been given in Ref.[140], in which the phase and the amplitude of a variety of different combs have been measured and used as an input for numerical simulations [148]. Some of these comb states, together with their reconstructed time traces, are shown in Fig. 23. By comparing experimental results with simulations, the authors

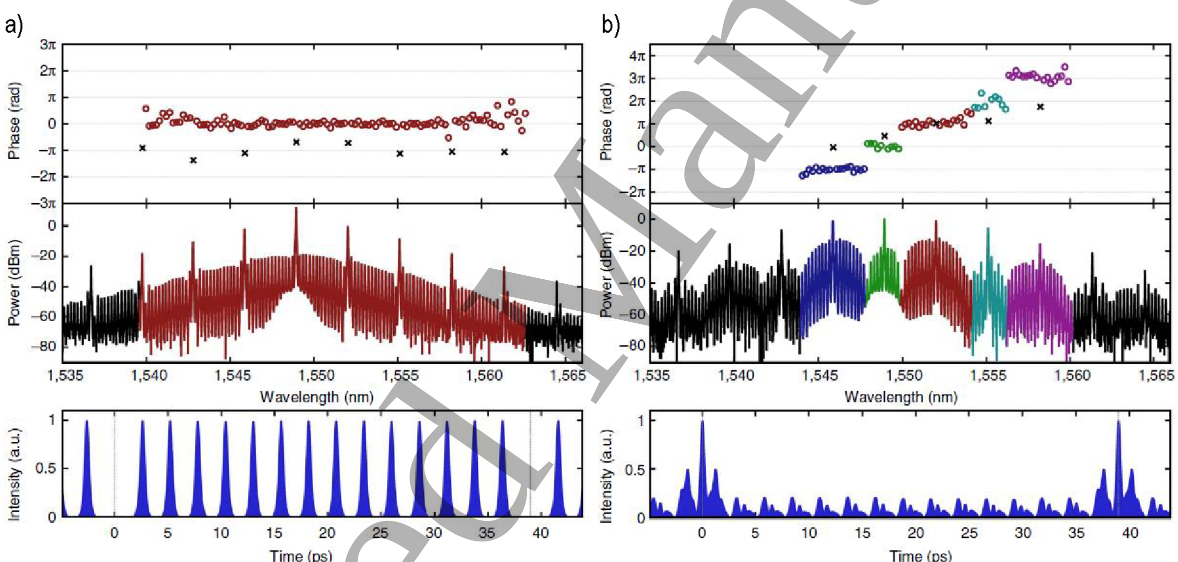

Figure 23. Measured phases, spectra and reconstructucted time traces for two different comb states achieved in a fused silica microtoroid resonator of $Q = 1.5 \times 10^8$. (a) Top panel: measured phases relative to each comb line. Middle panel: frequency comb spectra Lower panel: time trace reconstructed starting from the measured amplitude and phases shown in the middle and upper panel respectively. (b) Same as for panel (a), but relative to a different comb state. This figure is reprinted with permission from Ref. [140].

found that the observed comb states are compatible with multiple solitons circulating in the resonators, which are referred as Turing patterns [149]. In parallel to the study of coherence and formation dynamics, lots of efforts have been dedicated to comb referencing, stabilization and control. For the implementation in absolute frequency metrology, the carrier envelope offset frequency f_{ceo} and the mode spacing Δf must be known with high accuracy. The common method for measuring f_{ceo} makes use of self referencing, which requires one octave spanning frequency comb [150]. The offset frequency can be finely tuned by changing the pump laser frequency relative to the pump resonance [145]. The determination of the mode spacing Δf can be done by measuring the multi heterodyne beatnotes between the comb lines and the modes of a reference comb laser [146]. An alternative technique, which eliminates the need of a reference laser, called Parametric Comb Folding, has been introduced in Ref. [146]. In this approach, each mode f_m on the blue side of the pump is converted by a transfer laser at frequency $f_p+\Delta\omega$ (f_p is the pump frequency) to a new line at $f'_{-m} = f_{-m}+2\Delta\omega$. Then, the beatnote between f'_{-m} and f_m is recorded. If the comb is symmetric with respect to the pump frequency, each pair of lines should exhibit the same beatnote frequency $\Delta\omega$. Foster showed that the average deviation from this value was less than 0.5 Hz in a wavelength range of 115 nm. Recently, a new technique which allows to measure the detuning of the comb lines to the resonator modes, even when the comb has been already created and the resonator is "hot"(internal powers sufficiently high to activate nonlinear phase Kerr phase shifts), has been introduced [140]. This relies in measuring the beatnote between the back reflected light of the resonator at the input of the waveguide and a co-propagating External Cavity Diode Laser (ECDL). The precision of this method has been shown to be comparable to the other aforementioned techniques, and in parallel it allowed to show that the temporal dynamics of the comb is almost totally governed by the "cold" resonator GVD. The mode spacing is affected both by the pump frequency f_p and by the pump power P_p through the Kerr and the thermal phase shifts [145]. By using an active control of f_p and P_p based on a Phase Locked Loop (PLL) between f_{ceo} , $\Delta\nu$ and two reference RF signals, full stabilization of the microresonator frequency comb has been achieved [145]. Within the PLL, both the mode spacing and the offset frequency are stable within the mHz range (10^{-16} relative to the optical frequency).

3.5. Dynamical instabilities in resonators induced by thermal and free carrier effects

Kerr induced frequency combs are not the only source of time dynamics in microresonators. At 1.55 µm, the nonlinear dynamics of light inside a silicon resonator is governed by thermal and free carrier effects, rather than by the Kerr nonlinearity. Carriers are generated by TPA, which induce a blue shift of the resonance wavelength. At the same time, they introduce FCA, which raises the temperature of the resonator, determining a red shift of the resonace wavelength. These competing shifts act at two different timescales, which lie typically in the ps-ns range for free carriers [151], and in the μs range for temperature variations [121]. The interplay of these effects can lead to dynamical instabilities, and the resonator can show a self-pulsing behaviour. Intuitively, this can be understood by considering the thermal and the free carrier nonlinearities as two springs which "pull" the resonance wavelength in opposite directions, so as the resonance oscillates with respect to its equilibrium position ("cold" cavity condition) around the laser wavelength, modulating the transmitted intensity. This phenomenon has been intensively studied both theoretically and experimentally [121, 152, 153, 154]. Cavity self-pulsing is a threshold activated phenomenon. It has been demonstrated that it is possible to turn the resonator state from stationary to time varying by applying a small overshoot of optical power over a continuous bias background [155]. It has been suggested that this behaviour can mimic the excitability of a spiking neuron, so the resonator can be used as the basic building block of more complex neural networks [155, 156]. In Ref. [155], it has been studied how selective excitability can be addressed in a chain of resonators, and how this property can be cascaded from one resonator to another. There have been some theoretical works in which the transition from a periodic self-pulsing to a chaotic regime has been investigated. These make use of an exotic non-instantaneous Kerr effect introduced by liquids and metals in silicon [157], or the intra-cavity optical feedback in a CROW geometry [158] or in an in-line sequence of photonic crystal nanocavities [159]. An experimental demonstration of chaos in a passive chain of resonators, organized in the Side Coupled Integrated Spaced Sequence of Resonators (SCISSOR) geometry, has been also reported in Ref. [160]. As shown in Fig. 24, chaos is seen to arise when a collective self-pulsing of resonators is activated. The complexity of the output waveforms scales with the number of resonators involved in the extended oscillation. These can be recognized in Fig. 24(a) as the ones which scatter the largest amount of radiation. It has been suggested that the randomness in the light fluctuations can be exploited for the realization of a CMOS compatible, all passive random number generator.

3.6. Harmonic generation using resonators

The use of resonators allows to generate higher harmonics with a reduced pump power with respect to a bare waveguide. SHG with a peak efficiency of -35 dB has been reported in a Si_3N_4 resonator using 300 mW of input power at 1554 nm, but evidences of wavelength conversion are present with input powers as low as 3 mW [132]. The scattered red light at 777.1 nm from the resonator is visible in Fig. 25(a). The enhanced light-matter interaction induced by the resonator is crucial, since the magnitude of the $\chi^{(2)}$ induced by surface effects is of the order of 10^{-2} pm/V, i.e., ten thousand times smaller than the one measured by SHG in strained silicon waveguides [132]. THG in the green, has also been reported in this structure, as shown in Fig. 23(b). In centrosymmetric crystals, an effective second order nonlinearity can be obtained through the EFISH effect, which has been introduced in Section 43. In Ref. [161], a periodic poling of the dressed $\chi^{(2)}$ is obtained by continuously reversing the sign of the electric field which is applied across a p-i-n junction embedded in a silicon ridge waveguide. The engineering of the quasi phase matching allowed to obtain a SHG efficiency of $(13 \pm 0.5\%\mathrm{W}^{-1})$ at a wavelength of 2.28 µm, and a corresponding EFISH induced $\chi^{(2)}$ of (41 ± 1.5) pm/V. A proposal of a resonator based device has been given in Ref. [162]. In this work, a periodic EFISH-poling realized by 64 electrodes of alternate polarity, placed around the resonator perimeter, is introduced to realize quasi-phase matching. A theoretical SHG efficiency of -21.67 dB is derived with 60 mW of input power and 30 V of applied voltage. SHG and SFG in a Si_3N_4 resonator of $Q = 1.5 \times 10^6$ (input power of 0.7 W) has been exploited to up-convert a frequency comb in the NIR to the

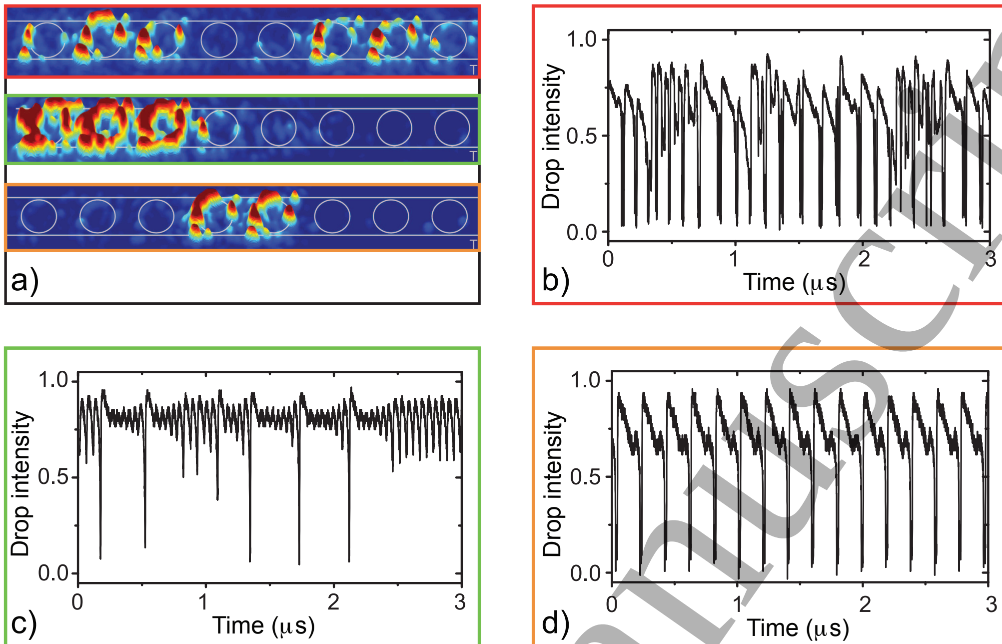

Figure 24. (a) Scattered radiation from each resonator, recorded using an infra-red camera placed at the top of the sample. The position of the rings in the sequence is indicated with gray lines. The pattern with a red border is associated to the waveform in panel (b), the one with the green border to waveform (c) and the one with the orange border to panel (d). (b) Output waveform in the Drop port of the SCISSOR corresponding to a chaotic regime. (c) Output waveform in the Drop port of the SCISSOR corresponding to a transition from a periodic self-pulsing to a chaotic regime. (d) Output waveform in the Drop port of the SCISSOR corresponding to a periodic self-pulsing regime. This figure is reprinted with permission from Ref. [160]

red spectral interval covering [765 − 776] nm [141]. The reported efficiencies are −29 dB for SHG and −20 dB for SFG. In both cases, higher order modal phase matching has been exploited. A similar approach has been adopted in Ref. [133] to up convert a NIR frequency comb covering 2/3 of one octave to the green ([502 − 580] nm). A spectrum of the green comb is shown in Fig. 19(c). The results of this work constitute, up to now, the broadest comb ever reported in this wavelength region for a on-chip resonator.

3.7. Photonic crystal nanocavities

All the resonators geometries considered so far are based on travelling wave WGM. Resonant structures can be realized as well by using photonic crystals. There are several ways to confine light in PhC, which essentially are based on point defects [163], line defects [164], or mode gap (or hetero-structure) nanocavities [165]. These are reported in Fig. 26. In the first (Fig. 26(a)), the periodicity of the crystal lattice is broken by introducing a vacancy [163], or a substitutional site [166] in the periodic arrangements of dielectric elements. The surrounding elements of the point defect are often adiabatically shifted from their lattice position as they get far from the defect center to tailor the

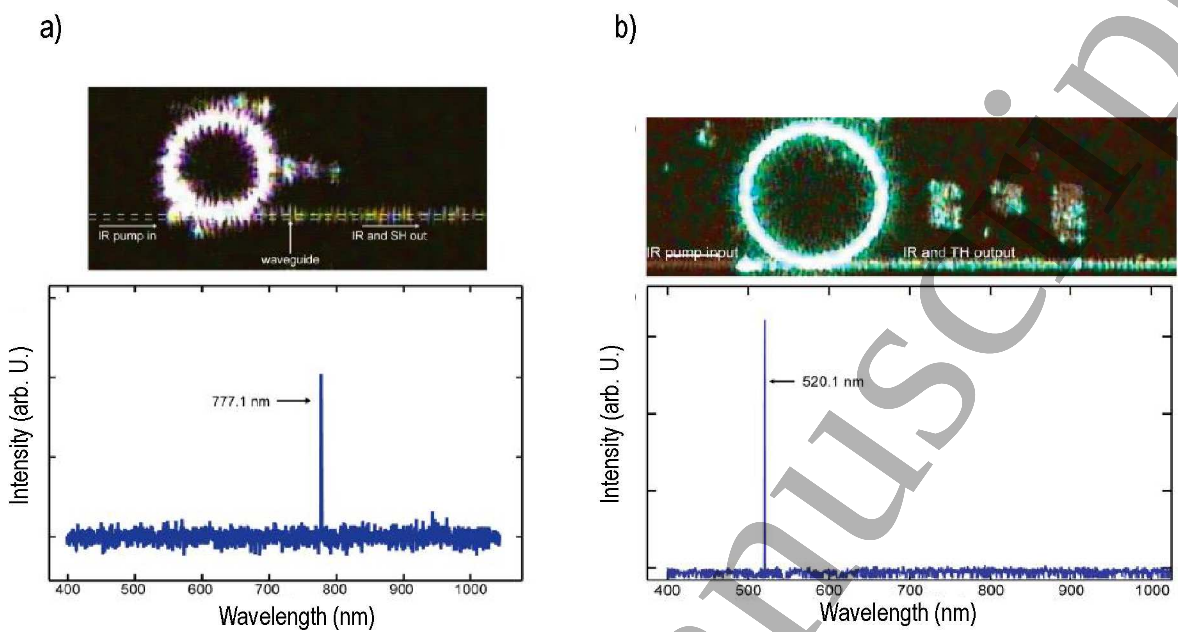

Figure 25. SHG and THG from a monolithic Si_3N_4 ring resonator. (a) Upper panel: an image of the red light scattered (generated by SHG) from the top of the ring resonator, acquired with a visible CCD camera. The input infra-red light (not detected by the camera) is injected from the left into the waveguide. Lower panel: spectra of the light transmitted by the resonator, restricted to the wavelength interval where the second harmonic is generarated. (b) Same as for panel (a), but relative to THG. In this case, the higher order armonic is emitted in the green. This figure is derived with permission from Ref. [132]

quality factor or the coupling with an external waveguide or with an incoming laser beam [167]. If one or more adjacent point defects are introduced (Fig. 26(b)), a line defect cavity is formed [164]. Mode gaps resonators are based on a series of different line defects (Fig. 26(c)) [165]. A portion of a PhC line defect waveguide, supporting one or more optical modes, is sandwiched between two other PhC line defect waveguides in which those modes are not supported. The external waveguides behave then as mirrors which confine light in the central waveguide, realizing a resonant structure. Photonic crystal resonators (especially the ones with highest Q) are in general not CMOS compatible, since they require the formation of frangible air-bridges and the complete removal of the sacrificial silica layer by isotropic etching. Moreover, the high precision in hole positioning requires the use of electron beam lithography, which is a very accurate but time consuming technique. Nevertheless, recently there have been some important efforts toward the realization of high-Q PhC by standard photolithography, which made possible to produce heterostructure resonators with Q of 2.2×10^5 with a mode volume of $1.7\left(\frac{\lambda}{n}\right)^3$ µm^3 [168]. There are several features in which PhC resonators are superior to WGM ones. High-Q travelling waves resonators possess often a large diameter (especially for low index contrast materials as Si_3N_4), which increases the modal volume and, at the same time, decreases the FSR. On the contrary, PhC cavities

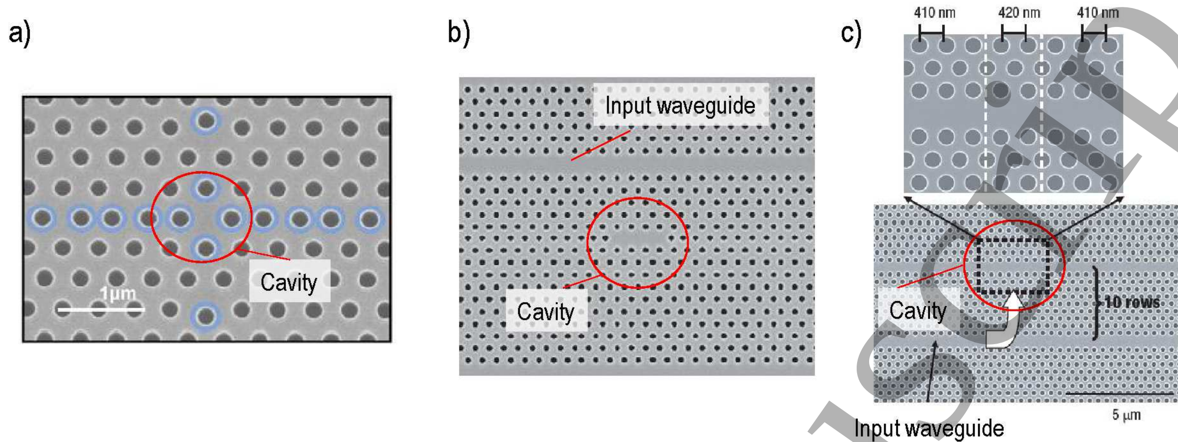

Figure 26. SEM images of different types of photonic crystal resonators. (a) Point defect, constituted by a missing hole in the center of the periodic array. The blue highlighted holes are slightly shift to their original position in the lattice to tailor the Q and the mode profile of the resonance. (b) Line defect, constituted by a series of missing holes in the periodic array. The cavity is evanescently coupled to a line defect waveguide to provide mode excitation. (c) Mode gap, constituted by a series of three line defect waveguides, each one having a different lattice constant (shown in the inset). These figures are derived with permission from Ref. [163, 164, 165]

are characterized by ultra-small mode volumes, well below $(\frac{\lambda}{n})^3$ μm^3, quality factors that can reach values of 9×10^6 [169], and large FSR, of the order of tens of nanometers [164]. Moreover, the high surface to volume ratio of these structures decreases the effective free carrier lifetime by introducing a large number of recombination centers, thus lowering the FCA induced losses. With respect to channel waveguides, which have a group index that hardly exceeds a value of $n_g \sim 4 \div 4.5$, PhC waveguides can be engineered to have values of $n_g \sim 40 \div 80$, i.e., they are suited for slow light propagation [170]. This feature increases the nonlinear interaction strength, since as shown by Eq. (23), the nonlinear parameter γ is proportional to n_g^2. Combined with a ultra-small effective area of less than 0.05 μm^2, a Coupled Resonator Optical Waveguide (CROW) consting of more than 200 coupled cavities has been demonstrated to have a nonlinear parameter $\gamma > 10000\,\mathrm{W^{-1}m^{-1}}$, which is the highest reported so far for a silicon waveguide [170]. As a matter of comparison, a typical channel waveguide having similar effective area has a γ of the order of $\gamma \sim 300\,\mathrm{W^{-1}m^{-1}}$ [171], while a slow-light PhC waveguide of comparable cross-section (but without the CROW) has a γ value of the order of $\gamma = 3000 \div 7000\,\mathrm{W^{-1}m^{-1}}$ [172, 33]. The high nonlinear parameter achieved by using CROW enhances the FWM efficiency with respect to a bare slow-light waveguide of the same effective area by more than one order of magnitude, and nearly of about three orders of magnitude with respect to a channel waveguide. This performance is comparable to a WGM based CROW, in which a FWM enhancement up to 30 dB has been reported, but with a lower bandwidth (30 GHz) with respect to the PhC-CROW (> 100 GHz), mainly caused by the lower number (8) of cascaded resonators

[173]. The extremely high Q/V_{eff} has been exploited for lowering the threshold of TPA-induced free carrier bistability down to 15 µW [163]. Up to now, this value is the lowest reported for a on chip silicon resonator, greatly outperforming WGM ring resonators [151, 174]. This makes PhC resonators particularly attractive for realizing ultra-low power consumption all optical memories or switches. An aggregate bandwidth of 42 Tb/s/mm^2 has been demonstrated by the simultaneous implementation of 25 cavities, mainly limited by the free carrier lifetime ($\sim$ 300 ps). Following this work, but with an increased number of 125 cavities that can be individually addressed by different wavelengths, it has been recently demonstrated the largest scale all-optical memory ever realized on a silicon chip [164]. A sketch of this device is shown in Fig. 27(a). Each cell of the dynamic memory can be written and reset by a CW bias ($79 \div 400$ µW power) and read using a single pulse ($100 \div 200$ fJ), showing an extinction ratio between the "1" and the "0" state of $10 \div 20$ dB. The memory stores 105 bit in a footprint area of 20 µm^2 per channel wavelength. The switching performance of this device is illustrated in Fig. 27(b,c).

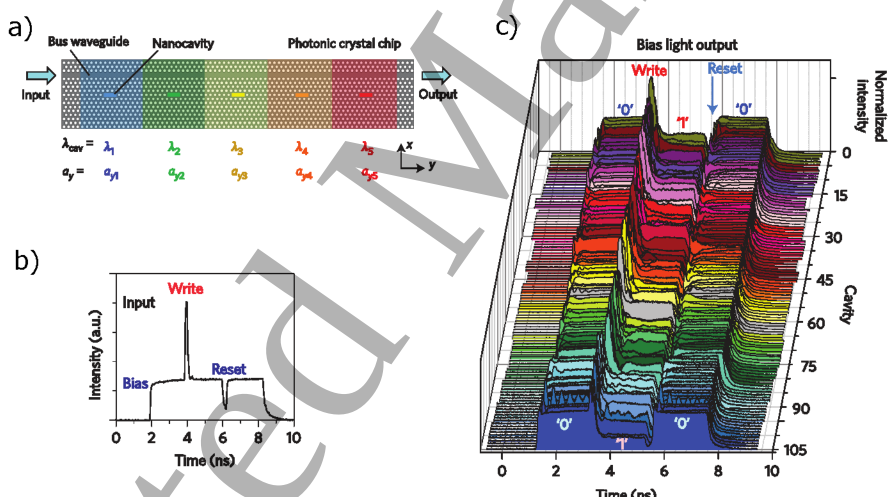

Figure 27. (a) Photonic crystal nanocavity based all-optical memory. The resonant wavelength of each cavity λ_{cav} is tuned by locally modulating the lattice constant a. (b) Input optical waveform. (c) Measured temporal waveform for 105 output channels. This figure is derived with permission from Ref. [164].

4. All-Optical Nonlinear Devices and Applications

4.1. Temporal imaging and time lenses

An interesting application of nonlinear optics to manipulate optical signals is offered by temporal imaging. This technique is based on the space-time analogy between paraxial diffraction and the propagation of pulses in dispersive media. A time-domain imaging system requires the realization of the temporal equivalent of a spatial lens, which is referred as a time lens. In analogy with a spatial lens, a time lens imparts a quadratic phase to the input signal [175, 176]. In this approach, a chirped pump pulse transfers its quadratic phase to an input signal through a wave mixing process. Some examples of this technique have been recently shown in silicon waveguides by means of FWM process [177, 178, 179, 180].

A schematic representation of a time lens system based on FWM is shown in Fig. 28. The input signal passes through a dispersive element characterized by a Group-Delay Dispersion (GDD) $\phi_1^{''} = \beta_2^1 L_1$. Within the time lens the input signal is mixed with a chirped pump signal, which has experienced a GDD of $\phi_p^{''} = \beta_2^p L_p$. By means of FWM, the output signal is generated at a frequency $\omega_2 = 2\omega_p - \omega_1$, passing then through a dispersive element characterized by $\phi_2^{''} = \beta_2^2 L_2$. It is possible to define the focal GDD of the lens $\phi_f^{''}$. In this scenario, an equation analogue to the one valid for the spatial lenses can be written:

$$\frac{1}{\phi_1^{''}} + \frac{1}{\phi_2^{''}} = \frac{1}{\phi_f^{''}}. \tag{61}$$

Analogously to spatial lenses, temporal magnification can be defined as $M = -\phi_2^{''}/\phi_1^{''}$. When dealing with FWM, the focal GDD is given by $\phi_f^{''} = -\phi_p^{''}/2$.

The time lens idea can be better understood from Fig. 29, where the temporal ray diagram is sketched for the case of negative magnification. The horizontal axis represents the accumulated GDD. The vertical axis represents the time τ relative to the center of the waveform propagating at the group velocity of the carrier frequency ω_0, and is then given by $\tau = (t - t_0) - (z - z_0)v_g(\omega_0)$, being z_0 and t_0 the references in space and time. The magnification scheme shown in Fig. 29 has been experimentally demonstrated in a silicon waveguide [177, 178], showing a magnification factor larger than 500. Through this approach, an optical sampling at rates larger than 1−tera-sample per-second has been demonstrated using a 1−GHz oscilloscope.

A similar approach has been used for the temporal compression of an optical pulse using a time-domain telescope [179]. In this work a maximum compressing factor of 27 has been demonstrated. Using a standard 10−GHz-bandwidth electro-optic modulator, the waveform has been compressed to a bandwidth greater than 270 GHz. The original and the compressed signal are shown in Fig. 30.

An object placed at the front optical plane of a spatial lens produces the Fourier transform of the object at the back focal plane. This fact can be used to get a frequency-to-time conversion. In Ref. [180] an optical spectrometer is used for measuring the FWM-generated spectrum, from which the temporal profile of the input signal is determined. Using this method, a signal larger than 100 ps with a 220 fs resolution has been recorded, with a record-length-to-resolution ratio larger than 450.

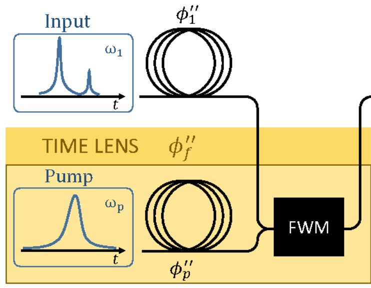

Figure 28. Scheme of a temporal imaging system based on FWM described in Ref. [177].

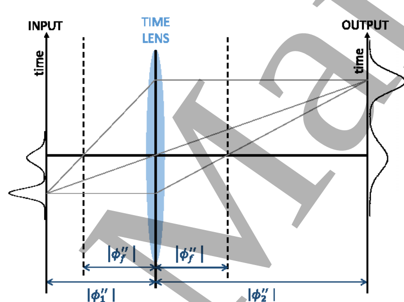

Figure 29. Temporal ray diagram for a time lens with $M < 0$. On the horizontal axis the accumulated GDD is reported, while on the vertical axis it is shown the time relative to the center of the waveform propagating at the group velocity of the carrier frequency.

4.2. Frequency comb-based applications

Many interesting applications are offered by frequency combs. These include the generation of ultrashort pulses, the waveform shaping, spectroscopy and the synthesis of microwave and radiofrequency signals. Combs realized by silicon photonics can guarantee the realization of small footprint and mass manufacturable optical devices, with repetition rates larger than 10 GHz or spectral bandwidths $\Delta\lambda/\lambda > 10\%$ [114, 116, 181].

4.2.1. Ultra-short pulse generation and telecommunication applications The generation of pulses of the order of 100 fs has been demonstrated [147, 182]. Recently, the transmission of a comb-generated 1.44 Tbit/s data stream was demonstrated by multiplexing 20 channels, each operating at a bit rate of 72 Gbit/s [183]. The proposal for a future chip-scale terabit-per-second transmitter is shown in Fig. 31.

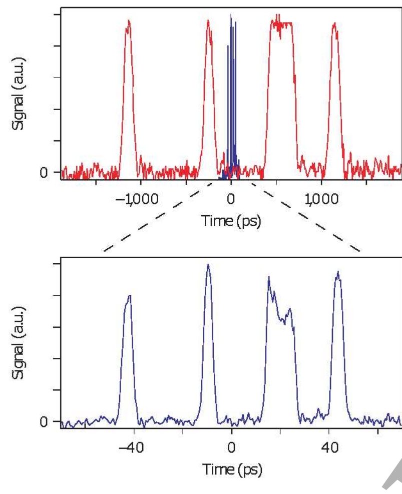

Figure 30. Data packet encoded at 10 Gb/s via standard modulation technology and compressed to 270 Gb/s. In the top row the original packets (red) and compressed packets (blue) are shown. An expanded view of the compressed packets is shown in the bottom row. The figure is derived with permission from Ref. [179].

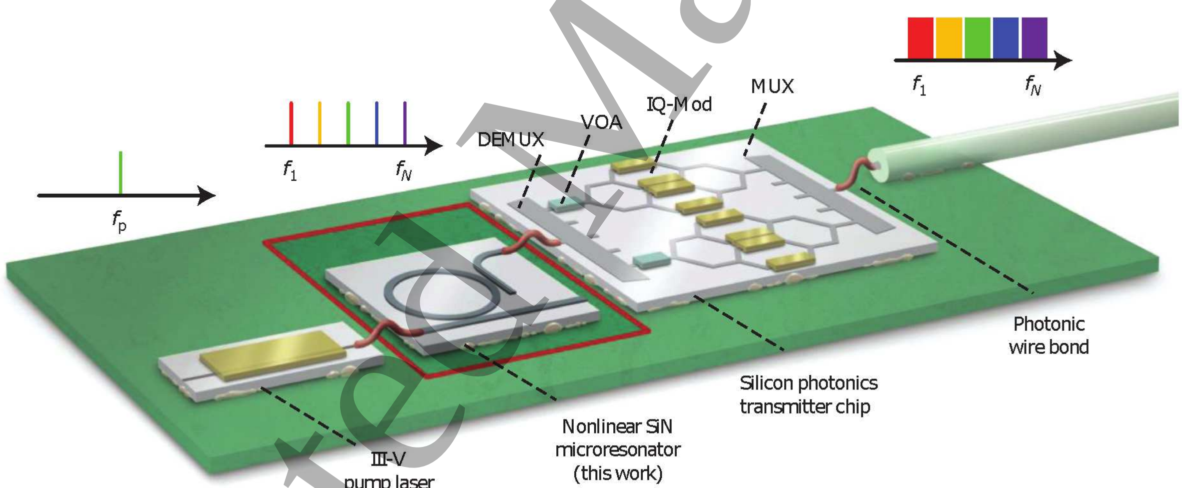

Figure 31. Schematic representation of the chip-scale terabit-per-second transmitter proposed in [183]. The generated comb is demultiplexed, modulated and then again multiplexed for the fiber transmission. The figure is derived with permission from Ref. [183].

4.2.2. Optical arbitrary waveform generation The large spectral spacing of frequency combs facilitates the pulse shaping of optical arbitrary waveform generation. This is achieved by coupling a frequency comb to a Fourier transform pulse shaper, which incorporates a modulator array able to control independently the phase and the intensity of each spectral line [184]. In Fig. 32(a) the spectrum of the comb after the pulse shaper is reported. The black dots show the phase applied to each spectral line. This phase is optimized for maximizing the zero delay autocorrelation measurement of the output signal. In Fig. 32(b) the autocorrelation signal without the phase compensation (blue) is compared with the one obtained maximizing the zero delay autocorrelation (red). The pulse is unmodulated in time when the phase is not controlled, while a pulse appears if the phase is set properly, demonstrating the compression of the pulse and the arbitrary waveform generation. Another example is shown in Fig. 32(c), where a π-phase step is applied to one half of the spectrum determining the formation of a so called odd pulse.

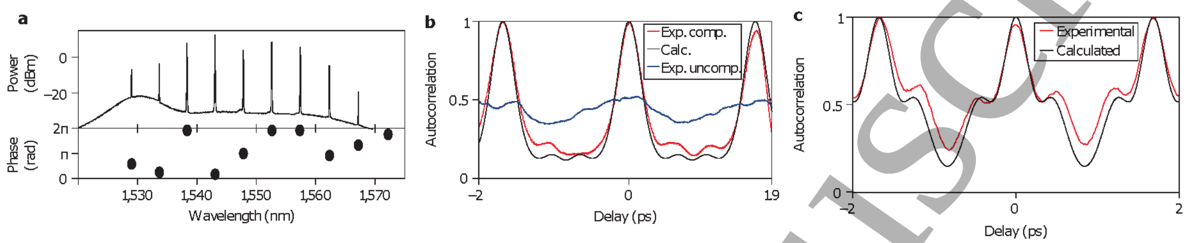

Figure 32. (a) On the top: spectrum of the generated comb after the pulse shaper. On the bottom: phase applied to the spectral lines. (b) Autocorrelation trace for the uncompressed pulse (blue) and for the compressed one (red). The black line is obtained from simulations. (c) Experimental (red) and calculated (black) autocorrelation signal where a π-phase step is applied to one half of the spectrum. The figure is derived with permission from Ref. [184].

4.2.3. Frequency comb spectroscopy Frequency combs are also spectroscopic sources for absorption spectroscopy of atoms and molecules [116, 181]. Particularly interesting are the applications of frequency combs generated in the MIR, where typical fingerprints of different gases exist [125, 185, 186].

4.3. All-optical wavelength conversion

4.3.1. FWM based frequency conversion Spectral translation of MIR radiation to NIR radiation was demonstrated by using coherent upconversion processes in Silicon-On-Shappire (SOS) waveguides [187]. The nonlinear process occurring in the SOS waveguide was a FWM based Modulation Instability (MI). MI is a $\chi^{(3)}$ nonlinear process, which can be interpreted in terms of a FWM process that is phase-matched by XPM [5]. The MI allows frequency band conversion of a MIR signal to 1.55 µm. A comparison of the performances of state-of-the-art MIR detectors (PbSe, HgCdTe, InSb) with a module formed by a SOS waveguide coupled to an InGaAs photodetector was reported. The module improved the Signal-to-Noise Ratio (SNR) up to 40 dB, compared to commercial MIR detectors at 0.1 µW of input MIR power.
Another scheme which deals with FWM based MI in silicon nanophotonics wires was proposed in [21]. The underlying phase matching mechanism is known as discrete-band phase matching, which has been already discussed in section 2. As reported in Fig. 33, the 2-cm-long silicon waveguide is coiled into a compact spiral, with cross sectional dimensions of (900×220) nm^2. The upper cladding is composed simply by air, while the lower cladding is a 2 µm buried oxide. Spectral translation of a signal at 2440 nm to 1620 nm, across a span of 62 THz was demonstrated.

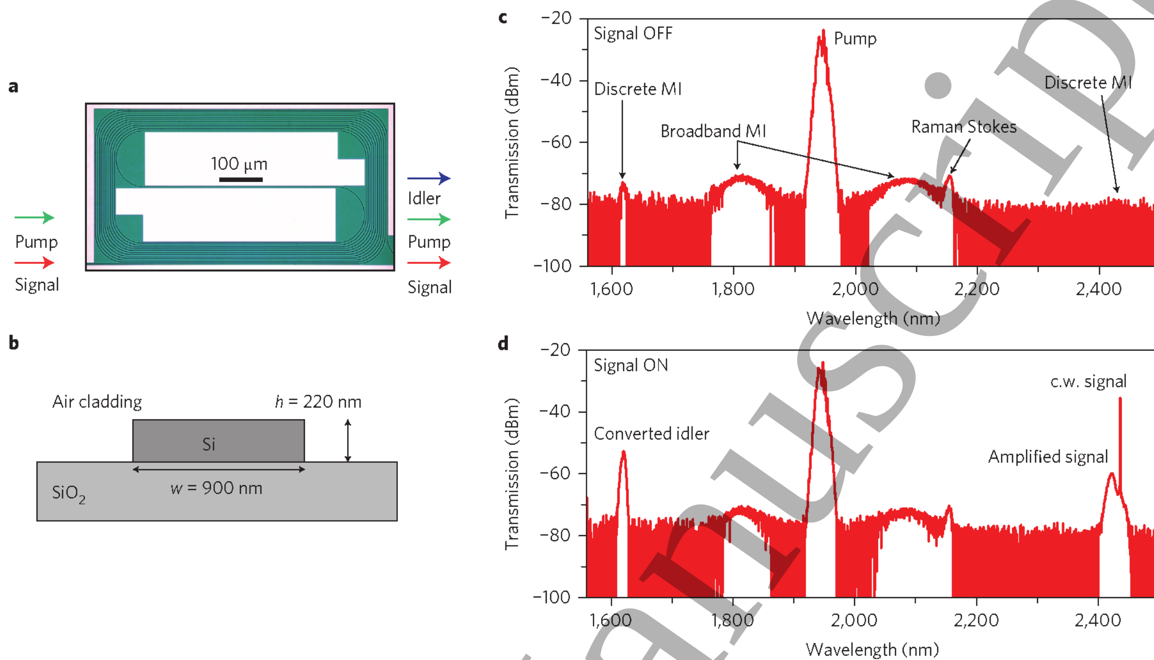

Figure 33. Waveguide design and transmission characteristics of the wavelength converter device. This figure is reprinted with permission from Ref. [21].

4.3.2. Single photon frequency conversion In the traditional wavelength conversion processes based on FWM which have been described so far, pairs of pump photons are destroyed, generating pairs of sideband photons (the signal and the idler) [21, 187]. This mechanism provides, on one side, the amplification of the signal and, on the other side, the generation of an idler pulse. However, this process involves also spontaneous FWM process due to vacuum fluctuations, which sets a theoretical limit to the maximum SNR for the converted idler, preventing the possibility of realizing single photon frequency conversion.

To overcome this problem, a FWM Bragg Scattering (FWMBS) process was used in Ref. [188]. As it is shown in Fig. 34, in this mechanism for each generated idler photon a signal photon is destroyed, determining a full power transfer from the signal to the idler. Since the total power in the sidebands is constant, the vacuum fluctuations are not amplified and no excess noise is produced [188, 189].

4.4. Photon-phonon interactions

4.4.1. Stimulated Raman scattering Following measurement of spontaneous Raman scattering from silicon waveguides [190], the first evidence of Stimulated Raman Scattering (SRS) has been reported in Ref. [191]. A Stokes signal $\lambda = 1542.3$ nm was amplified by 0.25 dB with a pump signal at $\lambda = 1427$ nm. The frequency difference between pump and signal was 15.6 THz, which is the frequency of optical phonons in

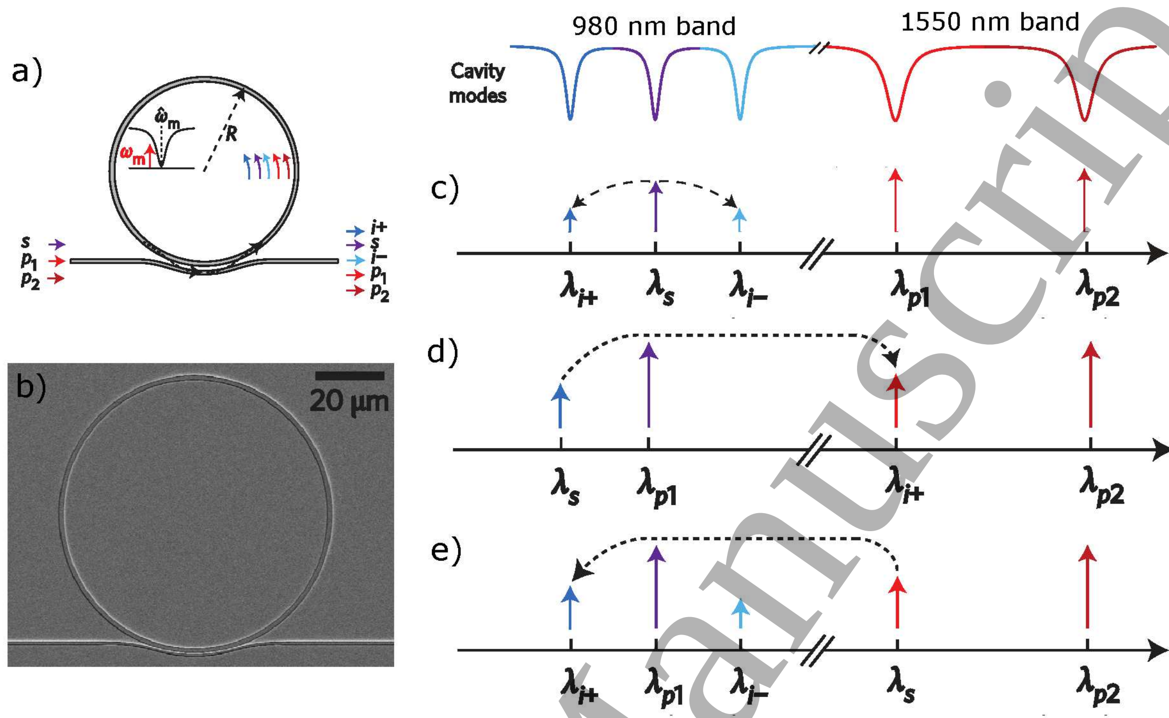

Figure 34. (a) Schematic of the microresonator geometry, where an input signal s is converted to idlers $i+$ and $i-$ by FWMBS process provided by two pump fields $p1$ and $p2$. (b) Scanning electron micrograph of the silicon nitride ring resonator. (c-d-e) Schematic of the FWMBS process for the intraband, upconversion and downconversion processes demonstrated in [188]. The figure is derived with permission from Ref. [188].

silicon. Even if SRS can only occur on a narrow bandwidth at 15.6 THz from the pump wavelength, it is a fully tunable process which is limited only by the available pump wavelengths [15]. Therefore, it has been possible to demonstrate also a MIR silicon Raman amplifier [192]. One of the main problems affecting the SRS amplification efficiency is the TPA-induced FCA, which has been faced introducing pulsed pump with a repetition rate shorter than the free-carrier recombination lifetime [193, 194].

Starting from these results, the first Raman silicon laser has been demonstrated in an optical fiber loop cavity incorporating a silicon waveguide [195]. The integration of Raman lasing in an all-silicon Fabry-Perot waveguide cavity has been also demonstrated [196]. In this design, a p-i-n diode has been introduced across the waveguide in order to sweep out the TPA-generated free-carriers when a reverse bias is applied. This mechanism for reducing FCA opened the door for the realization of the first continuous-wave Raman silicon laser, characterized by a 200 mW lasing threshold and a 4.3% slope efficiency [60]. A great improvement has been obtained using a ring resonator cavity configuration, setting the lasing threshold to 20 mW and the slope efficiency to 28% [197]. A micrometer-scale Raman laser (Fig. 35), with 1 µW threshold, has been demonstrated by using a PhC resonator. This sustains two eigenmodes of opposite parity whose spectral separation coincides with the silicon Raman shift [198]. The precise tuning of the two eigenfrequencies allowed to measure the silicon Raman shift with an accuracy of $\sim 10^{-3}$ THz, improving the previous estimations by more than one order of magnitude [199].

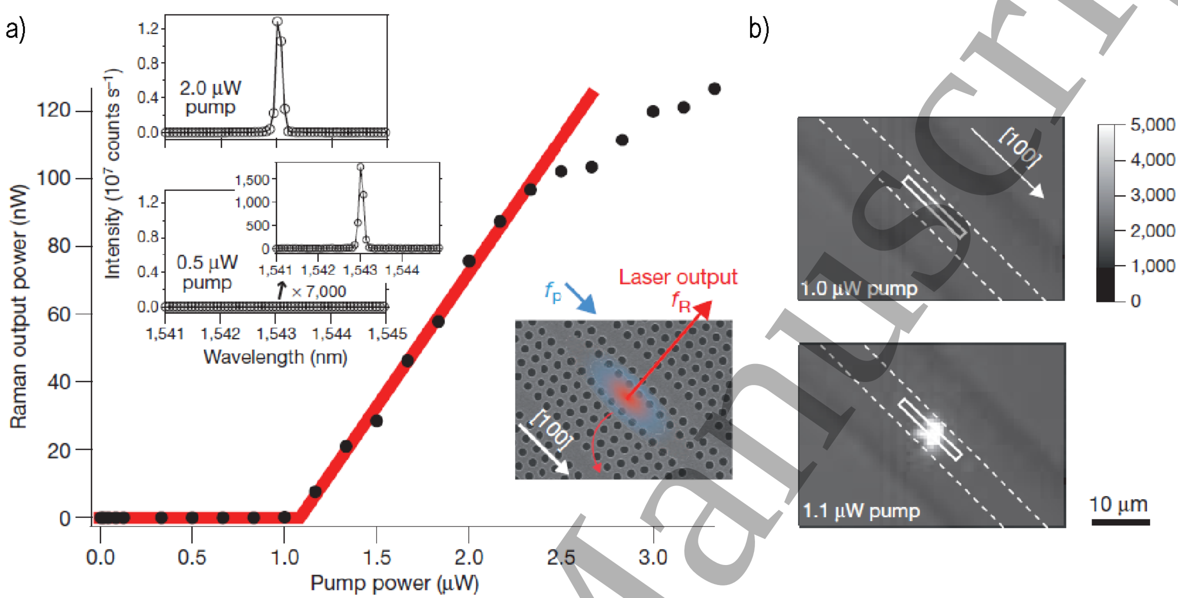

Figure 35. (a) The laser output power (black scatters) as a function of the pump power in the PhC nanocavity. The red line is a fit of the experimental data, showing the typical threshold (near 1 μW) behaviour which marks the onset of the laser oscillation. The insets show the spectra of the light transmitted to the output waveguide when the pump power is set below and above threshold respectively. (b) Near-infrared camera images of the nanocavity Raman laser below and above the threshold. A long-pass filter is placed in front of the camera to eliminate the pump light. Dotted lines represent the waveguides. Rectangles indicate the nanocavity position. The mode-gap cavity is fabricated along the [100] crystallographic direction to maximize the Raman gain. The figure is derived with permission from Ref. [198].

4.4.2. Coherent anti-Stokes Raman scattering In the Coherent Anti-Stokes Raman Scattering (CARS) process two photons of the same pump laser at frequency ω_p interact with the Stokes signal photon ω_s generating an output photon at frequency $\omega_{CARS} = 2\omega_p - \omega_s$. The efficiency of this process is strongly related to the phase-matching condition. If phase-matching is satisfied CARS process is dominant, otherwise SRS dominates [8, 15]. On these basis, CARS processes in a silicon waveguide were observed [200]. With a pump at 1427 nm, a signal at 1542.3 nm was converted to 1328.8 nm with an efficiency of −50 dB. With respect to wavelength conversion processes based on FWM this process is quite interesting, since it is mediated by the Raman susceptibility $\chi_R^{(3)}$ whose strength is about 44 times larger than standard electronic $\chi_e^{(3)}$ [8]. However, one of the main limitations of this process is the lack of tunability once the signal wavelength is fixed.

4.4.3. Stimulated Brillouin scattering Two different Stimulated Brillouin Scattering (SBS) processes can occur in silicon waveguides. In the Backward SBS (BSBS) the optical pulses propagate in the opposite direction, requiring the participation of a phonon propagating in the waveguide propagation direction. Co-propagating optical pulses give rise to Forward SBS (FSBS), involving acoustic phonons propagating orthogonally to the waveguide axis. The difficulty for the excitation of SBS phenomena in silicon waveguides has been ascribed to many effects, such as the acoustic mode leakage, the small elasto-optic coefficient and the large acoustic loss in SOI devices [201].

FSBS process has been recently demonstrated, making use of suspended nanoscale silicon devices for the realization of hybrid photonic-phononic waveguides [202, 203, 204]. One of these hybrid structures is shown in Fig. 36, where a suspended silicon waveguide is embedded within a silicon nitride membrane with etched air slots realized to guide acoustic modes. In these structures, the combined effect of electrostrictive forces and radiation pressures allowed to evaluate a SBS gain value up to 1000 times larger than the ones previously demonstrated on different platforms [202, 203]. The use of these hybrid structures is highly favorable, since it allows to have a separate control of the optical and the acoustic modes.

The achievement of SBS in silicon waveguides offers interesting perspectives for practical applications. As an example, a mechanism based on photon/phonon transduction for coherent information processing has been demonstrated, creating a radiofrequency photonic filter with frequency selectivity, narrow-linewidth and high power-handling [205]. Other examples refer to the realization of SBS-based slow-light, Brillouin dynamic gratings, narrow-line-width laser sources and nonreciprocal devices [201].

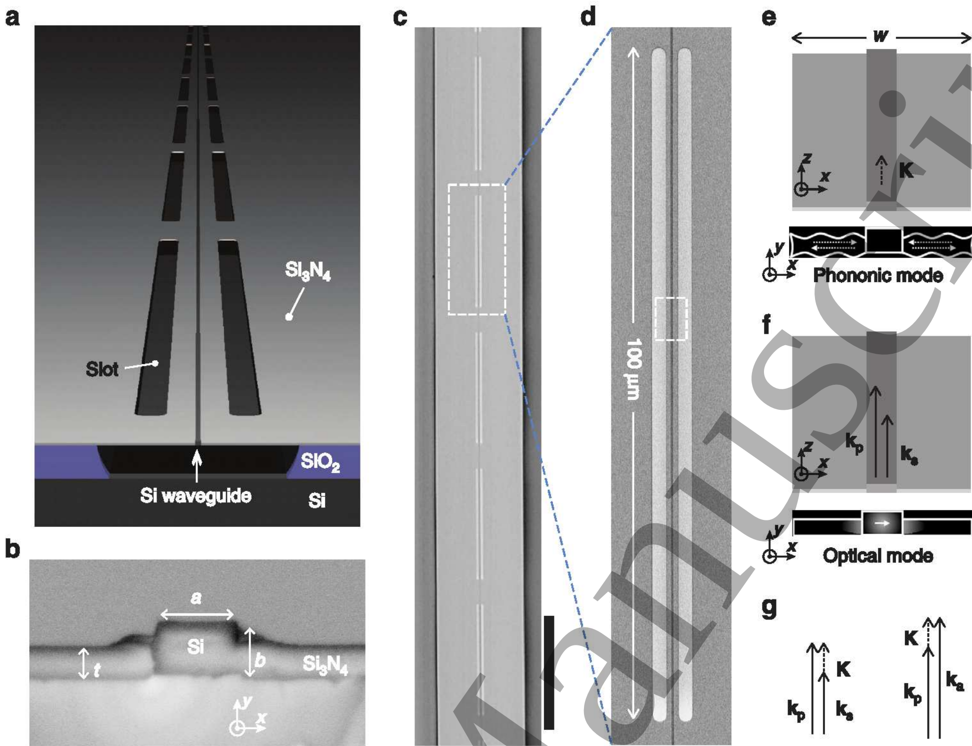

Figure 36. Hybrid photonic/phononic waveguide. (a) Geometric view of the photonic/phononic waveguide. (b) SEM view of the cross-section of the silicon waveguide core. (c-d) Top-side view of the waveguide showing the air slot sections. (e-f) Propagation of the guided phononic and optical modes. (g) Phase matching diagrams for Stokes and anti-Stokes FSBS. This picture is licensed by CC.3.0, and reprinted from Ref. [202].

5. Nonlinear wave interactions in quantum photonics

5.1. Deterministic vs probabilistic sources

In the last years, there has been a growing interest in the production and manipulation of non-classical states of light, because of applications in quantum information and quantum computation [23, 206, 207, 208, 209, 210]. Indeed, photons interact weakly with the environment over long distances and can be manipulated with linear optics. Among all the techniques to generate non-classical states of light (entangled photons or single photon states), one may distinguish between *deterministic* sources and *probabilistic* ones. Let us remark that in general an entangled state is defined as a quantum state in the composite system of two or more subsystems, which cannot be factored as a product of states of its local constituents [211].

The deterministic sources mainly deal with the generation of single photons. An ideal single-photon source would be the one for which: a single photon can be generated at any arbitrary time defined by the user (on-demand generation), there is 100% probability of generating a single photon and the repetition rate is arbitrary fast. Examples of deterministic sources are: semiconductor quantum dots [212], single atoms [213] or molecules, ions [214] and color centers [215].

The probabilistic sources generate correlated photon pairs at random times. Typically these kinds of sources involve a laser excitation of a nonlinear optical material. The nonlinear processes involved in the emission of entangled photon pairs are the Spontaneous Parametric Down Conversion (SPDC) and the Spontaneous Four Wave Mixing (SFWM).

SPDC is a $\chi^{(2)}$ process, whereas SFWM is a $\chi^{(3)}$ process. The classical framework developed in Chapter 2 fails to provide a full description of both SPDC and SFWM processes due to the fact that they are of *spontaneous* nature. Both of them are stimulated by random vacuum fluctuations and they happen without the coupling of any additional weak field. As it will be discussed in the following, this will imply a generally very low generation efficiency, with a typical value of -100 dB.

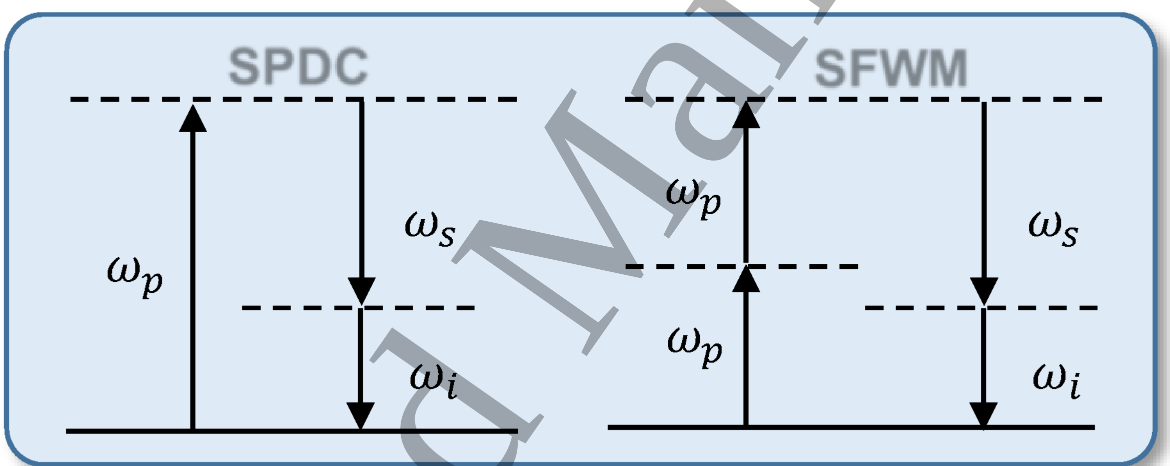

Figure 37. Sketch of the energy diagram of the SPDC and SFWM processes.

As can be seen in Fig. 37, the essential feature of SPDC is that a single pump photon passing through a nonlinear optical material can decay into two daughter photons, called signal and idler. These daughter photons will be correlated in momentum, energy and time (in some configurations also in polarization). In Fig. 37 it is also sketched the energy diagram for the SFWM process. It is an higher-order nonlinear effect compared to SPDC, where two pump photons are converted into two correlated photons.

The most important nonlinear platforms are:

- Silicon-based platforms, which include silicon (Si) as well as the silicon nitride (SiN)[216, 217, 218].
- Platforms based on III-V compound semiconductor, including indium phosphide (InP), gallium arsenide (GaAS), and gallium nitride (GaN) [219, 220]. III-V materials typically have a strong $\chi^{(2)}$ coefficient, which allows for compact and efficient SPDC sources or frequency conversion devices. These platforms have also the advantage to allow laser emission due to their direct bandgap, but suffer from

the fact that they are not CMOS compatible. However, full system integration, including electrically pumped lasers, may be conceived with this technology [221].

- Nonlinear optical dielectric materials, in particular lithium niobate ($LiNbO_3$) and potassium titanyl phosphate (KTP) [222, 223]. These materials present a high $\chi^{(2)}$ nonlinearity which makes them ideal for SPDC. Their fabrication is well studied and controlled and, in order to increase the nonlinear generation efficiency, they are periodically poled. In periodically poled crystals, the sign of the $\chi^{(2)}$ coefficient is periodically reversed along the crystal length, achieving the so called quasi-phase matching condition [224]. However, due to the relatively large size of devices made by these platforms, they are less favourable for the development of scalable systems and are instead often employed for proof-of-concept experiments.
- Platforms based on glass waveguides, such silica on glass [225], or direct femtosecond laser writing in glass [226], have been recently developed and used for quantum photonics experiments. The advantages of these platforms are the low propagation losses at telecom wavelengths and the excellent mode-matching with standard optical fibers for the input/output chip coupling. The drawbacks are the low $\chi^{(3)}$ non-linear coefficient as compared to silicon and that they are not CMOS compatible.

From the above discussion, it emerges that the next quantum photonics integrated devices will most likely adopt hybrid integration technology. The most promising one is the merging of the CMOS compatible SOI platform, with the III-V platform, so that active devices can be fabricated.

5.2. Generation of photons with spatial and temporal entanglement in silicon

The most important way to generate entangled photon pairs in silicon is to employ the SFWM process (Fig. 38) in waveguides or ring resonators. An example of entangled photon source is reported in Fig. 38(f) [227]. It is a compact integrated high quality ring resonator (5 μm radius), which is evanescently coupled to a straight waveguide.

The standard technique for detecting photon pairs is the *timing coincidence* [228]. Let us discuss the case of CW laser excitation, which is the most intuitive one. The entangled photon pairs under CW excitation are emitted at random times. The detection of one photon of the generated entangled photon pair *heralds* the existence of the other, as a consequence of the strong correlation which characterizes the twin photons. Therefore, a marked correlation peak appears at nearly zero time delay in a coincidence measurement, due to the fact that the arrival times of the twin photons are correlated. This is what is observed in Fig.38(b)-(c). A pair generation rate of 0.2 MHz with a 200 µW of coupled continuous wave laser excitation, working at telecom band around 1550 nm was obtained in [227]. It corresponds to a net SFWM generation efficiency of about -98.8 dB (without taking into account propagation and insertion losses) and to a coincidence-to-accidental ratio as high as 250.

The use of the heralding property to develop a single-photon source has been

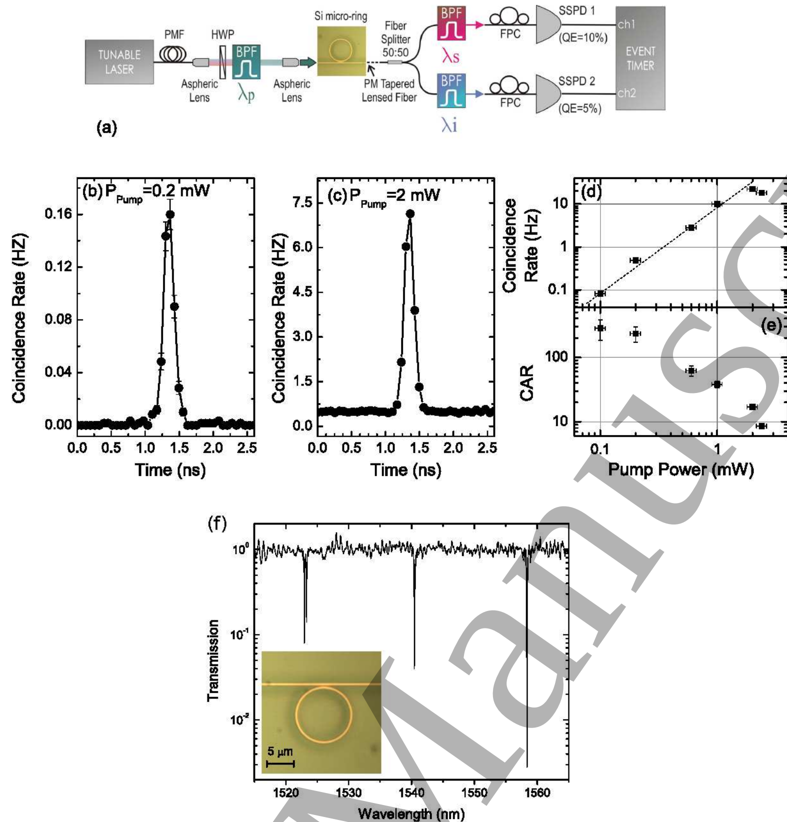

Figure 38. (a) Sketch of the experimental set-up. (b)-(c) Coincidence measurements as a function of the pump power. (d)-(e) Coincidence rate and coincidence to accidental ratio (CAR), as a function of the pump power. (f) Transmission spectrum of a waveguide evanescently coupled with a ring resonator. The inset shows an optical microscope image of the ring resonator. This figure is derived with permission from Ref. [227].

demonstrated in a Coupled-Resonator-Optical-Waveguide (CROW) [229]. This is characterized by antibunching in the second order correlation function ($g^{(2)}$), which measures the joint probability to have coincidence detections as a function of the relative time delay between two signals. Observation of antibunching indicates sub-Poissonian photon statistics. This is usually performed in a Hanbury Brown-Twiss interferometer set-up [230], in which the emitted light is split with a 50/50 beam-splitter onto two single photon detectors. The observation of a dip in the coincidence number around relative zero delay, reveals that the source is dominantly composed of single photons.
In [229] after the manipulation of the generated photon pairs, where the signal (1529.5 nm) photons are heralded by the detection of the idler (1570.5 nm), an antibunching measurement ($g^{(2)} = 0.19 \pm 0.03$) was performed. In this experiment, triple coincidences between signal and idler photons were observed after the signal photons were split by a 50/50 coupler.
The standard technique to create time-bin entangled states is to inject a photon pair

source in a Franson interferometer [231], sketched in Fig. 39.

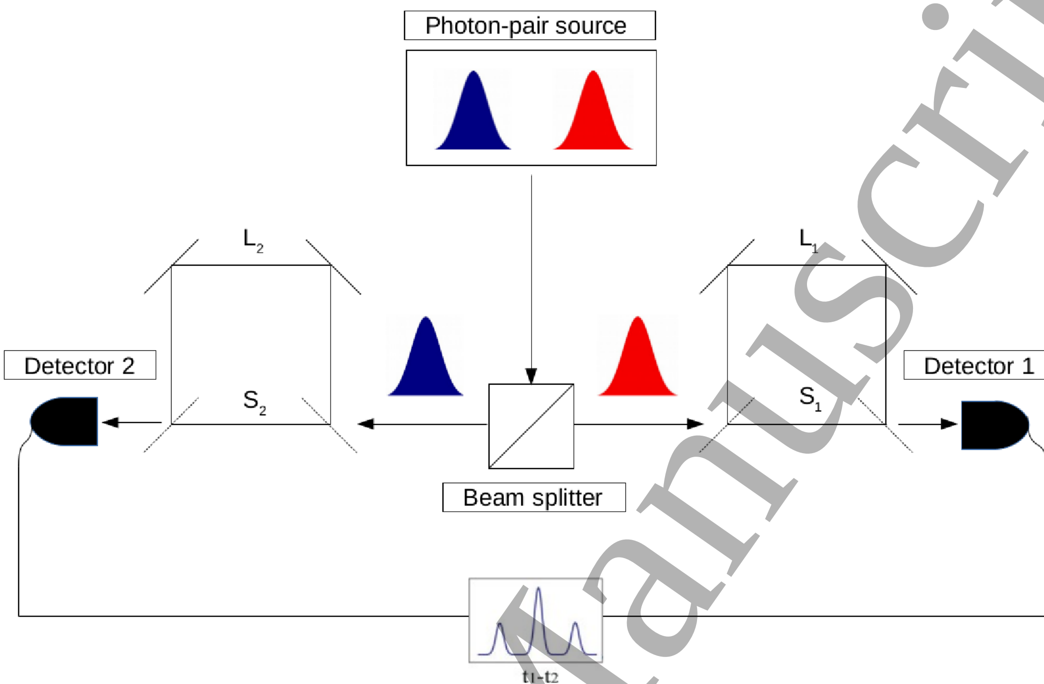

Figure 39. Sketch of the Franson interferometer.

Under CW excitation, SPDC or SFWM produce photon-pairs at random times. Signal and idler photons are separated with a beamsplitter, for example sensitive to the polarization or the wavelength, and both are directed to an unbalanced Mach-Zehnder Interferometer (MZI). Each photon may take the long path (L_1 and L_2) or the short path (S_1 and S_2) of the MZI. To ensure that there is no first-order interference at the output of the MZI, the path length difference between the long path and the short path is made much larger than the coherence time $L_c \ll \Delta L_{1,2} = L_{1,2} - S_{1,2}$. The photons are then detected by two single-photon detectors and the arrival times are cross correlated, with $\tau = t_1 - t_2$.

Actually there could be four probability amplitudes for the joint detection of photon pairs: $|S_1\rangle\,|S_2\rangle$, $|S_1\rangle\,|L_2\rangle$, $|L_1\rangle\,|S_2\rangle$, $|L_1\rangle\,|L_2\rangle$. This will give rise to three different peaks in the coincidence graph. The left (right) peak is due to $|S_1\rangle\,|L_2\rangle$ ($|L_1\rangle\,|S_2\rangle$), where one photon of the pair takes the shortest path and the other the longest one. The central peak, instead, comes from both $|S_1\rangle\,|S_2\rangle$ and $|L_1\rangle\,|L_2\rangle$, i.e. both photons take either the shortest or longest path. The creation time uncertainty is given by the coherence time of the pump laser L_p that excites the photon pair creation, which could be SPDC or SFWM. Therefore, we have to impose the general condition $L_c \ll \Delta L \ll L_p$. By post-selecting only the central peak of the coincidence graph through a narrow coincidence window, one gets a time-bin entangled state of the form:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\Big(|S_1\rangle\,|S_2\rangle + e^{i\phi}\,|L_1\rangle\,|L_2\rangle\Big) \tag{62}$$

where ϕ is the phase difference between $|S_1\rangle\,|S_2\rangle$ and $|L_1\rangle\,|L_2\rangle$, which can be tuned by setting $\Delta L_{1,2}$.
Therefore, it is possible to create time-bin entangled states with photon pairs of the form given by Eq. (62). These states have applications in Quantum Key Distribution (QKD) [232, 233].
The first silicon-based time-bin entanglement generation was demonstrated in [234]. Here, SFWM in a 1.09 cm-long silicon wire waveguide was used to generate 1.55 μm time-bin entangled photons. Then, non-classical two-photon interference fringes with $> 73\%$ visibility were observed.
Also microring resonators allowed to generate time-energy entangled photons in a Franson interferometer [235].
Another class of entangled states, which involves multiple photons, is the $N00N$ states, or path-entangled states. These states are such that N photons are in a superposition of all being in one location and all being in a second location. These states have applications in interferometry [236, 237].

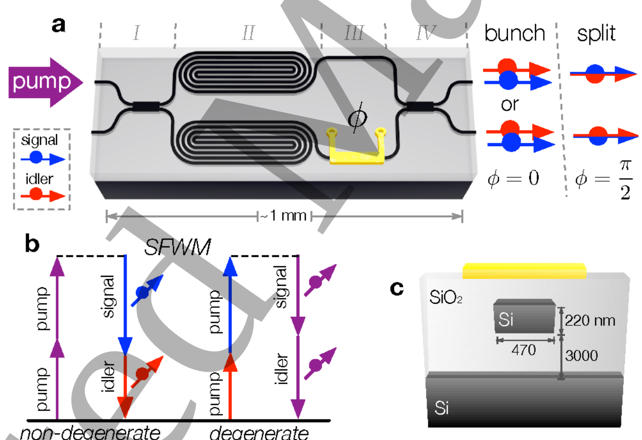

Figure 40. (a) Sketch of the operation of the device. (b) SFWM energy diagram. (c) Cross section of the silicon on insulator photonics device. This figure is reprinted with permission from Ref. [23]

Although $N00N$ refers to large N, most of the experimental demonstrations are limited to $N = 2$. A pioneering work reported on a SOI photonics device, where two spiralled waveguides 5.2 mm long, produced entangled photon pairs through SFWM [23]. To produce a 2002 path-entangled state (Fig. 40) a pump beam was equally split by a 2 X 2 Multimode Interference couplers (MMI), so that the pump excited a path entangled state $\frac{1}{\sqrt{2}}\Big(|20\rangle - |02\rangle\Big)$ in the two spiralled waveguides. Note that the system was operated in the weak pump regime. Therefore, only one pair was likely to be generated, creating a path entangled state where there is the superposition of states composed by 2 photons in one spiralled waveguide and 0 photons in the other one and viceversa. The state was manipulated by a thermal phase shifter, and a 100.0 ± 0.4 % visibility of the

on-chip two-photon quantum interference and off-chip Hong-Ou-Mandel [238] quantum interference with 95 ± 4 % visibility were achieved. On-chip two photon quantum interference refers to the fact that the photon pairs are thermo-optically phase shifted in a reconfigurable manner and made to interfere within the chip. Off-chip Hong-Ou-Mandel quantum interference was performed firstly by on-chip manipulating the state to obtain pair splitting, with one photon in each mode. Then, the photon pairs are let to interfere on an off-chip fibre beamsplitter and coincidence events are monitored as a function of a free-space implemented displacement. In both cases the pulsed pump laser source and the single photon detectors are off-chip, i.e. not integrated on the SOI chip.
A follow-up of this work was published in [239], where the path entangled generation was achieved through SFWM within two microring resonators.

5.3. Quantum-classical correspondence in nonlinear wave mixing

In silicon FWM can be used to generate entangled photon pairs in many configurations. In the stimulated FWM, the signal triggers parametrically the generation of a new wavelength, such that the energy conservation is fulfilled (see Fig. 2). It is worth to remark that, in this case, a classical treatment is enough to properly discuss the physics underlying FWM (see Section 2). Actually this is not the case of spontaneous FWM (SFWM), where a quantum optical approach is needed to describe the process. Let us consider the following questions:

- Is there a link between the stimulated and spontaneous FWM?
- Is it possible to infer the SFWM efficiency process by knowing the generation efficiency in the stimulated FWM?
- Is it possible to access the SFWM quantum wavefunction (*biphoton wavefunction* [240]), by means of a classical measurement?

These questions were answered in [241] by the derivation of the relationship between SPDC and the stimulated counterpart Difference Frequency Generation (DFG). Similar arguments hold also for the comparison between spontaneous and stimulated FWM. It was shown that the biphoton wavefunction, which represents the quantum state of the correlated photon pair, is accessible by stimulated DFG measurements. Tuning the properties of the input signal allows mapping the biphoton wavefunction with a high fidelity. This method, named Stimulated Emission Tomography (SET), outperforms the quantum state tomography approach, which aims to reconstruct the density matrix of a quantum state.
The idea was applied in [242] to a 3 mm long, 220×460 nm^2 cross section, silicon nanowire with an effective nonlinearity of $\gamma \sim 236$ W^{-1}m^{-1}. By measuring the Joint Spectral Intensity (JSI), i.e. the square modulus of the biphoton wavefunction which represents the probability distribution of the generated energy-time entangled photon pairs in the frequency space, was possible to determine experimentally the relation

between stimulated and spontaneous FWM. Three different JSI were measured: one via coincidence measurements from SFWM, and two via stimulated FWM, using different detection methods. The experimental results are reported in Fig. 41. Since the pump

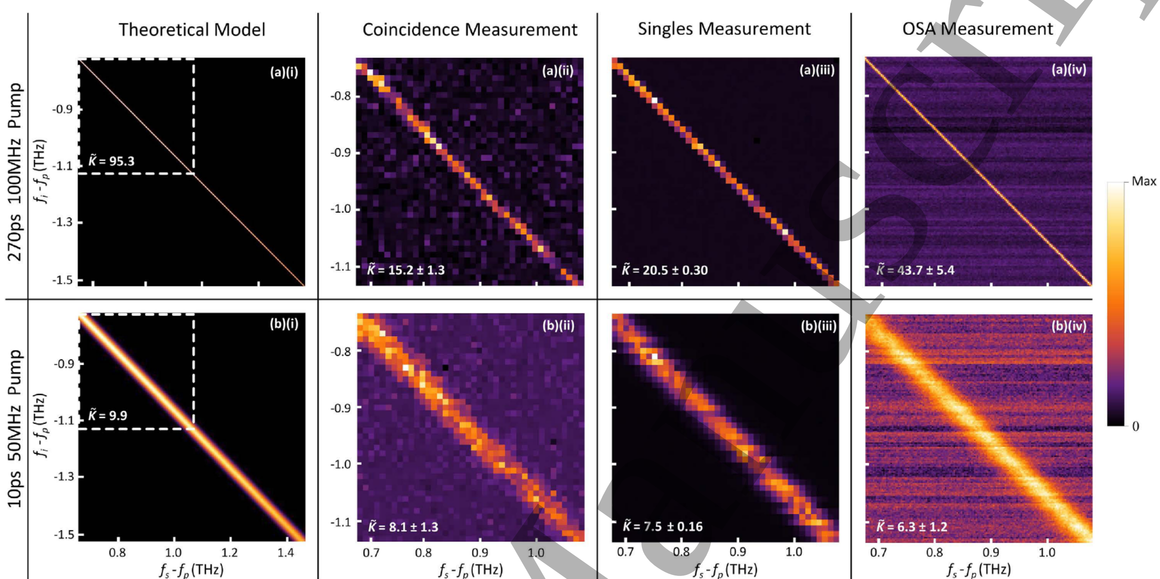

Figure 41. Theoretical and experimental JSI, for (a) the 270 ps and (b) 10 ps pump laser pulses. Schmidt Number Lower Bound (SNLB), $\tilde{K}$, are reported for each plot. This figure is licensed by CC.4.0, and derived from Ref. [242].

pulse duration influences the degree of spectral correlation, measurements were taken for 270 ps and 10 ps long pulses. The usefulness of the method is evidenced by considering that coincidence-based measurements require an integration time of approximately 36 hrs while the classical OSA measurements were completed within 2 hours with 16 times higher resolution. Still, moving to the stimulated measurement with the single-photon counter results in a significant decrease in the required integration time (1.5 hours) when compared to the coincidence measurement, while providing comparable Schmidt Number Lower Bound (SNLB), $\tilde{K}$ [243]. $\tilde{K}$ gives actually an estimation of the degree of entanglement of the source and can be derived from the theoretical and experimental JSI measurements reported in Fig. 41.

SNLB is a singular value decomposition applied to the modulus of the biphoton wavefunction, i.e. the square root of the measured JSI [243]. Since photon detection is proportional to the light intensity, experimentally one has direct access only to the square modulus of the biphoton wavefunction, with an inherent loss of phase information. The biphoton wavefunction can only be retrieved via an interferometric-based approach. Once the biphoton wavefunction has been measured, the standard Schmidt decomposition can be used to fully characterize the quantum mechanical state by means of the Schmidt number K.

Another work [244] reconstructed the joint spectral density (JDS) of photons pairs generated by spontaneous FWM by SET techniques. A silicon ring resonator, with a quality factor of Q=40000 and radius R = 15 μm, was side coupled to a 500 × 220 nm^2 silicon bus waveguide in a critical coupling condition .

By measuring the square modulus of the biphoton wavefunction the Joint Spectral

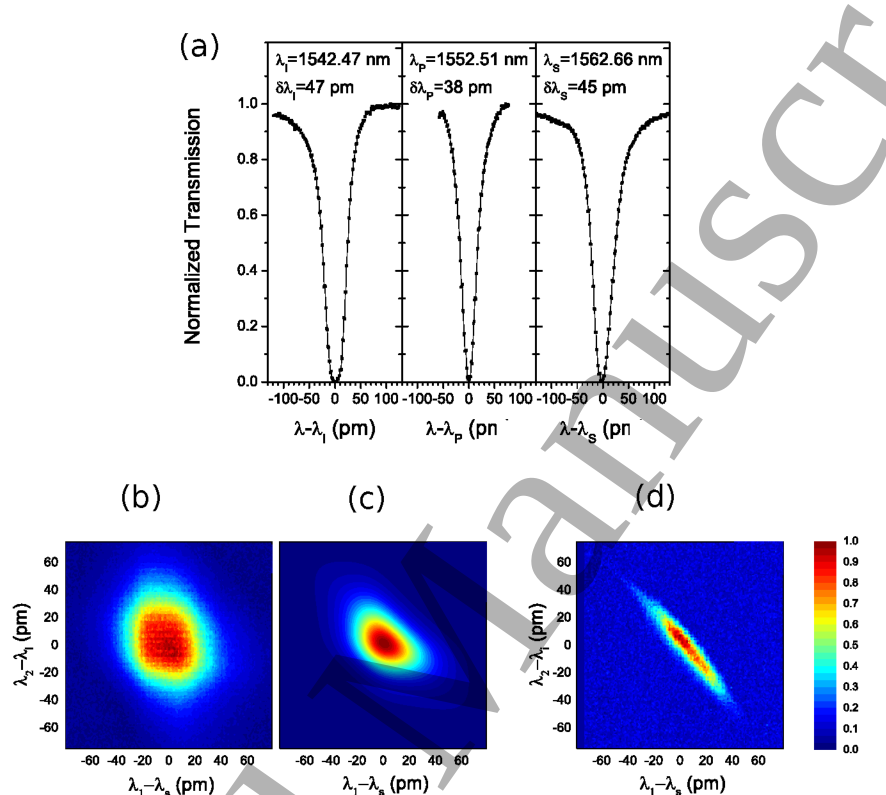

Figure 42. (a) The transmission spectra of the idler, pump and signal resonances are reported (from left to right respectively). (b) Experimental JSD under pulsed excitation, which has to be compared with the calculated one in (c). (d) JSD under cw excitation. This figure is licensed by CC.4.0, and reprinted from Ref. [244].

Density (JSD) was extracted. Fig. 42 shows the measurements performed on the ring. Two experimental JSDs (b),(d) and one theoretical JSD (c) for a pulsed pump source are reported. A theoretical estimate of the Schmidt number $K = 1.09$ indicates nearly uncorrelated photons ($K = 1$ corresponds to a separable state, while $K > 1$ indicates an entangled state), which is expected when the pump pulse duration is comparable to the lifetime of the photons in the ring resonator [128]. From the experimental JSD measurement reported in Fig. 42(b), it is possible to derive a lower bound for the Schmidt number $\tilde{K} = 1.03 \pm 0.01$, which well matches the theoretical estimation. In the CW experiment, reported in Fig. 42(d), the elongated shape of the JSD qualitatively manifests spectral correlations. The lower bound of the Schmidt number in this case is $\tilde{K} = 3.93$, which is limited by the experimental resolution. Therefore, it was demonstrated that an integrated silicon-based microring resonator, is able to generate nearly uncorrelated or time-energy entangled photon pairs depending on the pump characteristics.

5.4. The quantum photonic chip

Thanks to quantum photonics, huge improvement has been experienced in quantum enhanced sensing [237], quantum metrology [245], quantum cryptography [246], quantum computing [247] and quantum litography [248].

Therefore, strong efforts are nowadays spent to realize a Quantum Photonics Chip (QPC). QPC refers to an integrated device incorporating all the components required for a full handling of the quantum properties of light. The availability of such a device would make possible the large scale diffusion of quantum technologies [249].

In Fig. 43 a possible configuration is shown for a silicon chip including some of the building blocks of a QPC, the source of quantum states, the pump filtering stage and the manipulation stage.

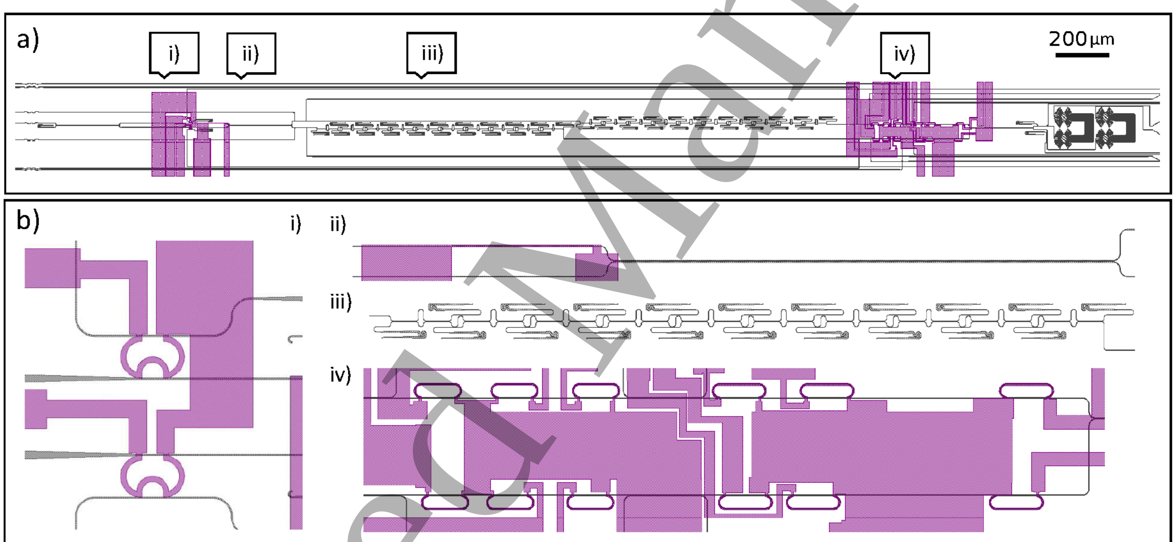

Figure 43. a) Design of a quantum chip able to generate and manipulate N00N states. b) Details on the different parts of the chip: i) source of entangled pairs, ii) deterministic quantum state preparation stage, iii) pump filtering stage, iv) manipulation stage.
The black lines refer to waveguides while the purple lines refer to metallic wires used to thermally tune the photonic components. Figure derived from [20].

5.4.1. Integrated laser source In order to induce the strong nonlinearities required for entangled pairs generation, high optical powers are needed. Until now, nonlinear processes in integrated devices have always been induced with an off-chip pump source. However, the integration of the laser source is a fundamental step forward towards the fabrication of a QPC. The problem with the scalability of the laser source on a silicon platform is silicon itself, because this material cannot emit laser light due to its indirect band gap. This obstacle can be overcame by integrating on the silicon platform a gain medium, III-V materials typically, that is coupled to the silicon circuit [250]. There are several techniques for the integration of the gain medium on the silicon platform, but one of the most interesting and viable approaches is reported in Fig. 44. This hybrid silicon laser involves a single ring resonator integrated inside a laser cavity that is composed by two Distributed Bragg Reflectors (DBRs) and a III-V gain medium [251]. Light coming from the silicon waveguide is coupled to the gain medium through an adiabatic mode converter, which consists of an adiabatic taper in both the III-V medium and silicon waveguide. This solution for the integrated laser source offers 10 dBm maximum power with 100 mA current. Even better results should be possible with a pulsed source of coherent light, that would reach extremely higher peak powers. Because of this, great effort is currently focused on pulsed III-V/SOI lasers. In particular, an improvement of the device of Fig. 44 has been proposed in [252]. Here a mode-locked hybrid silicon laser was fabricated by introducing a saturable absorber section in the laser cavity. In this way, the output mean power was 0.3 mW with about 120 mA, with about 4.8 GHz repetition rate and 1.5 ps pulse duration.

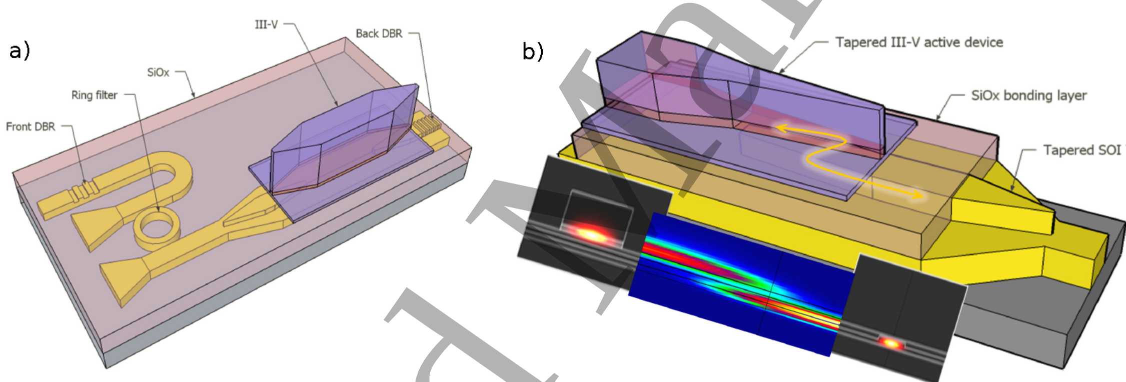

Figure 44. (a) Schematic view of the laser structure. (b) Coupling structure in the gain section with mode profiles in two cross-sections. This figure is derived with permission from Ref. [251].

5.4.2. On-chip generation of quantum states of light A chip where quantum N00N states are generated was proposed in [239]. There, it was reported a SOI device in which the path entangled generation was achieved through SFWM in two microring resonators, as shown in Fig. 45. Narrow-band photon sources, spectral demultiplexers and reconfigurable optics were integrated into a single device, resulting in one of the best examples of integration on the same chip of the generation and manipulation stages. The entangled pairs generated from the microring resonators were made highly indistinguishable by means of thermal tuning of the rings' resonances, achieving $95.8 \pm 2.1\%$ visibility.

An ideal source of single photons should deterministically, i.e. on demand, emit indistinguishable single-photons [253]. For the QPC, this is achieved by heralded photon generation through SFWM in silicon wires. Still, there are two main problems:

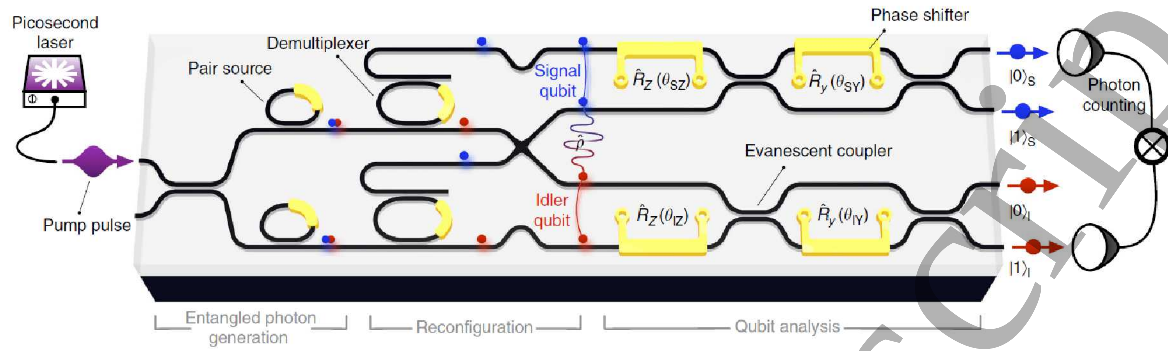

Figure 45. Generation of N00N states via SFWM in ring resonators. The chip is composed by a stage of entangled photons generation, a reconfiguration stage and a stage of analysis of the generated qubits. The path entanglement is separated into signal (blue) and idler (red) path qubits. This picture is licensed by CC.3.0, and reprinted from Ref. [239].

the occurrence of multipair generation and the non-deterministic nature of SFWM [254]. One mitigation action is to reduce the pump power to minimize multi-photon noise, and then increase the heralded photon generation rate through multiplexing. Both spatial [255] and temporal multiplexing have been studied, with the second most promising in terms of multiphoton suppression and near-deterministic single photon emission [256, 257]. Another challenge for the heralded single photon source is the simultaneous generation of many independent single photons at different wavelengths for quantum communication protocols. This was achieved in [258], where a ring resonator generates multiple, simultaneous and independent photon pairs at different wavelengths in a frequency comb centered at 1556 nm (Fig. 46).

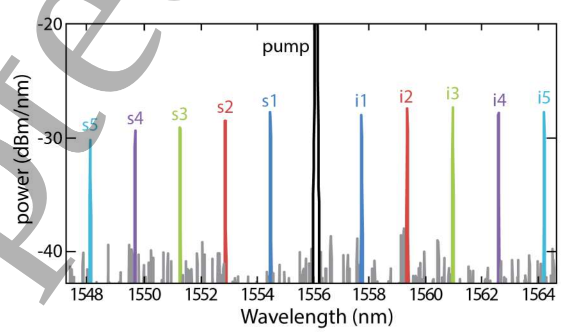

Figure 46. Spectrum of the frequency comb showing the five different signal/idler (s5-s1 and i1-i5, respectively) couples detected. The black curve is the pump, while the grey traces are due to the noise of the optical spectrum analyzer. This figure is derived with permission from Ref. [258].

5.4.3. Integrated filters for pump extinction To generate correlated pairs in silicon, a large optical power is required. Since the generation efficiency is roughly -100 dB, very efficient pump filters are needed. This means that for QPC also on-chip filters have to be developed. An example is reported in [218], where a device based on a DBR to filter the pump by more than 95 dB was described. The system is shown in Fig. 47. The photon pairs are generated in a ring resonator via SFWM. A first pump filtering is performed through a 2.576-mm-long DBR consisting of a grating. After the DBR stage, two add-drop ring resonators demultiplex the signal and idler photons, with a further filtering of the pump, whose total rejection amounts to 95 dB.

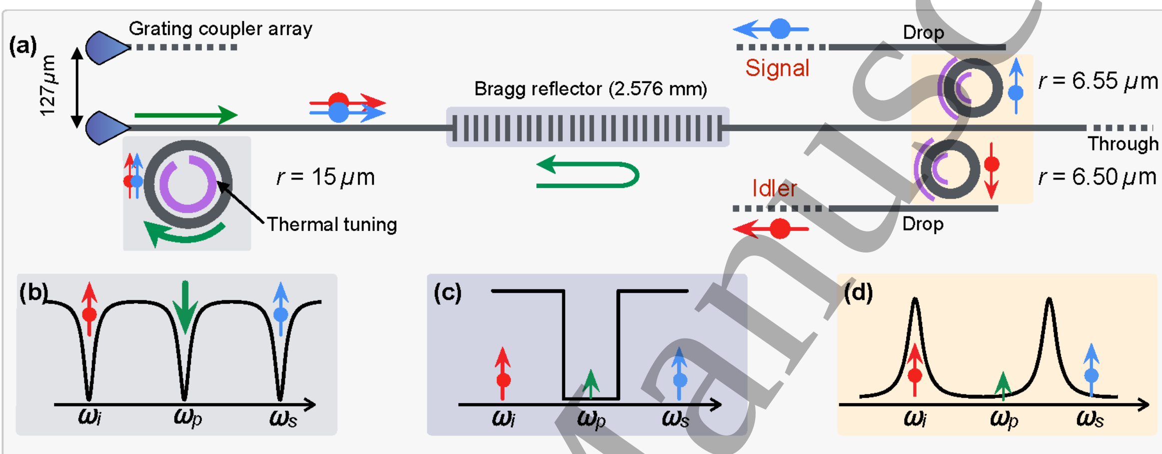

Figure 47. (a) Schematic layout of the photonic integrated circuit composed of ring for pair generation by SFWM, followed by a DBR for pump rejection and the add-drop ring-resonator filters for the demultiplexing of signal and idler photons. Optical coupling to a single-mode polarization-maintaining fiber array is achieved via focusing grating couplers separated by a 127-µm pitch. (b) Schematic transmission spectrum of the first ring around the pump wavelength ω_p. (c) Schematic transmission spectrum of the DBR with the stop band overlapping with the pump wavelength ω_p. (d) Add-drop filter spectrum tuned to route idler photons to the drop port. This picture is licensed by CC.3.0, and reprinted from Ref. [218].

5.4.4. On-chip manipulation of quantum states of light Since quantum applications involve a low number of photons, manipulation and processing of the quantum states of light occur at low photon number levels. In this context, optical nonlinearities would be useful. However, in silicon, the nonlinear response associated with individual photons is negligible [259]. Indeed single photon nonlinear interaction has been achieved only with atomic transitions of single atoms in cavities [213]. A recent promising approach involves a quantum dot embedded in a PhC waveguide [260]. Despite this, the "No-go theorem" [261] states that linear optics is not sufficient for qbit processing. A mitigation to this was introduced in [262] with a probabilistic scheme based on effective nonlinearities of linear optics. It was suggested that the high fidelity and reconfigurability needed for quantum light manipulation, can be achieved by a high control of many linear optical elements (MZIs, phase shifters, beam splitters) integrated together on the same chip. Indeed, a single reprogrammable optical circuit to implement many linear optical protocols was demonstrated in [263]. As shown in Fig. 48, the QPC is coupled to an off-chip source of quantum photon states and to a bulk photon counting stage constituted by Single Photon Avalanche Diodes (SPADs). The phase shifters are tuned through an electronic controlling stage. With this system, several quantum optics experiments were performed on the chip.

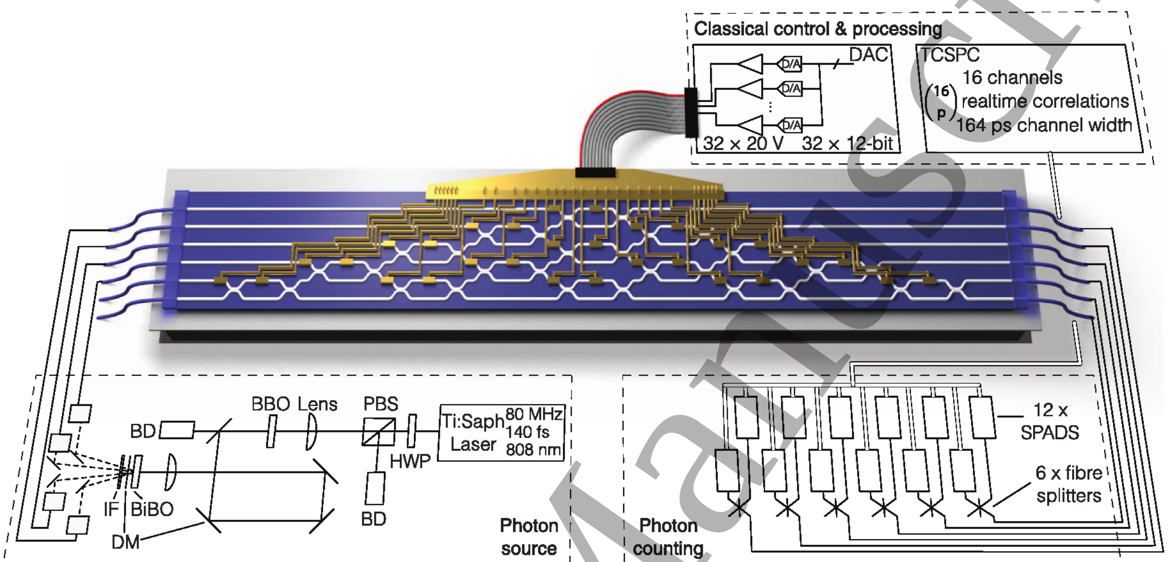

Figure 48. Multiphoton states are generated via SPDC. Photons are collected into polarization-maintaining fibers and delivered to the LPU. The processor is constituted by a cascade of 15 MZIs, controlled with 30 thermo-optic phase shifters, set with a digital-to-analog converter (DAC), and actively cooled with a Peltier cooling unit. Photons are then out-coupled and sent to SPADs and counted. This figure is derived with permission from Ref. [263].

In this section it has been shown that the main elements required for the fabrication of a complete QPC are already existing and ready for the integration on the silicon platform. Several experiments reported successfully the integration of some of the building blocks required for a QPC [218, 239, 264]. The challenge for the future is to improve the performance of the QPC building blocks reaching those efficiencies that are needed for quantum state manipulation, and, at the same time, to find a viable integration strategy for those functional blocks that are still missing (single photon detectors and pump laser).

Acknowledgments

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